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SPECIALIZED PROJECT (CO4029)

Project

IELTS Mastermind

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DECLARATION

We hereby declare that, apart from the results referenced from other works related and clearly cited in this thesis, the contents presented in this thesis are our own work, and no part of it has been submitted to obtain a degree at any other institution.

Ho Chi Minh City, July 10, 2021.



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Although we made every effort to carry out the topic with strong determination, shortcomings are inevitable. We look forward to receiving feedback and contributions from our teachers so that our thesis can be improved and made more complete.



ABSTRACT

In recent years, the demand for IELTS preparation has increased significantly due to requirements for education, employment, and international migration. However, many learners face difficulties in self-study because of the lack of personalized guidance, challenges in accurately assessing their own proficiency, and the fact that existing platforms often do not fully support all four IELTS skills, especially Writing and Speaking. These limitations highlight the need for a more comprehensive and learner-centered IELTS preparation platform.

To address these issues, this project proposes IELTS Mastermind, an online IELTS learning platform that supports all four skills: Listening, Reading, Writing, and Speaking. The system provides exam-format practice tasks with official IELTS timing, along with result evaluation, learning feedback, and progress tracking. Within the scope of this Specialized Project, the focus is on requirement analysis, system architecture design, data modeling, user interface design, and partial implementation of core features. The result is a well-structured and feasible platform that can be further developed and expanded in the Graduation Project to include advanced features and intelligent learning support services.



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1 Introduction

1.1 System Description

The IELTS Mastermind is an online learning platform designed to provide comprehensive preparation for the IELTS exam. The system integrates practice modules for all four skills, including Listening, Reading, Writing, and Speaking, aligned with the official IELTS test format. The system aims to simulate the benefits of one-on-one tutoring by assessing learner's knowledge, identifying strengths and weaknesses, and adapting the learning path accordingly.

1.2 Problem Statement

Preparing for the IELTS exam is often challenging due to the lack of personalized guidance, limited access to qualified tutors, and the high costs of traditional preparation courses. Learners frequently struggle with self-study because they cannot accurately assess their own performance or identify the specific areas that need improvement. Standard practice materials typically provide static feedback, which does not adapt to individual learning needs.

Moreover, the IELTS exam requires balanced proficiency across four distinct skills. Many existing platforms focus mainly on Reading and Listening, leaving Writing and Speaking insufficiently supported. As a result, learners may feel unprepared in some areas and fail to achieve their target band score, especially when aiming for major goals such as studying abroad, immigration, or career advancement.

1.3 Challenges

There are several problems with traditional IELTS preparation that IELTS Mastermind aims to address.

1. Learners can't easily find materials. They struggle to browse and search practice content in one place.
2. Learners have trouble managing time. Study sessions and test simulations don't provide enough timing support.
3. Learners don't get enough feedback. They lack guidance on how to improve.
4. Learners don't get personalized practice. Everyone follows the same path.
5. Learners can't see their progress clearly. It's hard to track performance and measure improvement over time.
6. Preparation is expensive. Private tutoring and courses cost a lot and limit access for many learners.

1.4 Reasons for Implementation

Implementing IELTS Mastermind is necessary to address the challenges identified in Section 1.3. We aim to achieve two major outcomes:



- Improved band scores with reduced study time per learner.
- Reduced cost of effective preparation relative to tutoring or offline courses.

Altogether, IELTS Mastermind is a scalable, cost-effective solution that directly solves the mentioned challenges and improves learners outcomes.

1.5 Scope of Implementation

IELTS Mastermind focuses on the process of preparing for the IELTS exam. The goal is to make preparation simple, adaptive, and accessible. The system lets learners browse practice content, take a quick evaluation, and receive a personalized study plan. Learners can practice any of the four skills, complete timed mock tests, and get automated feedback, especially for Writing and Speaking. It also gives administrators tools to manage users, roles, and practice content.

This implementation delivers the following capabilities:

- Authentication and Authorization
- User Profile Management
- Practice Content Access
- Personalization and Learning Paths
- Feedback and Band Score
- Progress Tracking
- User and Role Management
- Practice Content Management

2 System Overview

2.1 Survey of Related Systems

There are some existing websites that offer similar functions to ours. Each of them has its own strengths and weaknesses. Our goal is to learn from their good features and avoid their drawbacks. Below are two well-developed websites and their advantages and disadvantages.

2.1.1 YouReady

Advantages

- The interface is clean and user-friendly.
- Accessible with desktop and mobile devices.
- Offers practice for all four IELTS skills.
- Follows the official IELTS test format and timing.
- Provides performance statistics for all four skills.
- Integrates an AI-powered chatbox to assist learners.
- Applies AI to evaluate Speaking and Writing.
- Includes a blog featuring IELTS updates, tips, stories, and learning materials.

Disadvantages

- Writing and Speaking modules are not fully implemented.
- Performance statistics are limited, displaying only scores and results by question type.
- Lacks a notification system to encourage consistent practice.
- The blog content is minimal, with only a few articles available.

YouReady is a typical and accessible platform that supports IELTS preparation. However, the website appears to be incomplete, as the Writing and Speaking features remain under development. Additionally, the platform feels somewhat simplistic and could benefit from more advanced analytics, more content, and stronger engagement features. These are areas where our project can improve.

2.1.2 IELTSOnlineTests.com

Advantages

- Features a professional yet user-friendly interface.
- Offers a comprehensive library of IELTS exam materials.

- Provides IELTS tips and learning resources.
- Supports all four IELTS skills.
- Offers live lessons and courses taught by certified tutors.
- Includes information and guidance for studying abroad.
- Developed by a community of IELTS teachers, examiners, and experienced test takers.
- Provides an advanced analytic system with detailed statistics.
- Grants free access to a some IELTS tests.

Disadvantages

- Most premium features and lessons require payment.
- Pricing is relatively high compared to the average income level in Vietnam.

IELTSONlineTests.com is a comprehensive and professional IELTS learning platform that covers every aspect of test preparation. Its paid features may limit accessibility for certain users, but its quality and completeness set a strong standard. Our website can draw inspiration from its structure, analytics, and professional design while focusing on affordability to better serve local learners.

2.1.3 Development Opportunities

Based on the strengths and weaknesses of YouReady and IELTSONlineTests.com, IELTS Mastermind can improve in these areas:

- **Improve Writing and Speaking support:** YouReady's Writing/Speaking features are not complete, and IELTSONlineTests.com often requires payment for advanced learning. IELTS Mastermind should build strong Writing and Speaking practice with clear tasks and helpful feedback.
- **Provide better progress tracking:** YouReady's statistics are quite basic. IELTS Mastermind can offer clearer and more detailed results (by skill and question type), so learners know exactly what they need to improve.
- **Add motivation features:** YouReady lacks a notification system. IELTS Mastermind can add reminders, study schedules, goals, and streaks to help learners practice regularly.
- *Offer richer learning content at a more affordable level:* YouReady has limited blog content, while IELTSONlineTests.com may be expensive. IELTS Mastermind can provide more learning resources while staying affordable for students.

IELTS Mastermind should focus on complete skill coverage (especially Writing/Speaking), better feedback and analytics, study support tools, and affordable learning resources.

2.2 Objectives

Based on the challenges identified in Section 1.3, the main objectives of the IELTS Mastermind are:

1. Develop a system that allows learners to browse, search, filter, and manage practice content easily.
2. Develop a system that offers timed mock tests, and evaluation tests.
3. Develop a system that generates estimated band scores and delivers feedback for improvement.
4. Develop a system that generates study plans and adaptive recommendations for each learner.
5. Develop a system that shows dashboards, attempt history, and reports showing trends, skill progress, and time on task.
6. Develop a system that provides affordable access to high-quality IELTS preparation.

2.3 Stakeholders

2.3.1 Internal Project Team

- Project Owner: Sets the vision and budget, chooses priorities, and approves major milestones.
- Engineering Team: Builds and runs the website, APIs, databases, and third-party integrations.
- UX/UI Designer: Designs easy-to-use screens that work well for all learners.
- Testers: Find bugs and confirm that features work correctly on different devices before release.
- Content Authors: Write exercises, mock tests, evaluation tests, model answers, and rubrics that match the official format.

2.3.2 External Partners

- Speech-to-Text Service Provider: Converts learners' spoken answers into text for analysis.
- AI Provider: Estimates band scores and generates feedback.

2.4 Users

Learners and Administrators are the two primary users of IELTS Mastermind. Here are their user stories:

2.4.1 For Users

- As a user, I want to register with email/phone/Google/Facebook so that I can create an account.
- As a user, I want to log in, log out, and reset my password so that my account stays secure.

2.4.2 For Learners

- As a learner, I want to browse, search, and filter practice content so that I can quickly find what I need.
- As a learner, I want to bookmark and unbookmark practice content and see my bookmarked list so that I can save practice content for later.
- As a learner, I want to preview a practice content's metadata so that I know what I'm about to attempt.
- As a learner, I want to start practice and submit answers manually or automatically so that I can finish tasks my way.
- As a learner, I want to receive an estimated band score and automated feedback after each attempt so that I know how to improve.
- As a learner, I want to resume in-progress practice so that I can study in short sessions.
- As a learner, I want to view my attempt history so that I can review what I've done before.
- As a learner, I want to take an evaluation test so that the system creates my skill profile.
- As a learner, I want to receive a skill profile so that I understand my strengths and weaknesses.
- As a learner, I want the system to use skill profile to generate a study plan for me.
- As a learner, I want to set a target band score and my test date so that the study plan fits my goal.
- As a learner, I want the system to create a study plan and goals for me so that I always know what to do next.
- As a learner, I want adaptive recommendations for next practice content so that practice matches my weaknesses.
- As a learner, I want AI to analyze my Writing so that I get feedback and an estimated band.
- As a learner, I want my Speaking to be transcribed and analyzed so that I know what to fix.
- As a learner, I want all scores and feedback to be saved and shown so that I can review them anytime.
- As a learner, I want a skill progress dashboard so that I can see trends over time.
- As a learner, I want an overall performance dashboard so that I can spot gaps.
- As a learner, I want to see time spent per skill and total study time so that I can manage my schedule.
- As a learner, I want to generate a progress report so that I can keep a record of my learning.



2.4.3 For Administrators

- As an admin, I want to create and edit user accounts so that people can access the platform.
- As an admin, I want to deactivate accounts so that I can remove access when needed.
- As an admin, I want to assign roles and manage access levels so that permissions stay secure and appropriate.
- As an admin, I want to add new practice content with metadata so that the catalog grows in an organized way.
- As an admin, I want to edit content and metadata so that practice content stays accurate and up to date.
- As an admin, I want to remove content so that outdated or low-quality practice content doesn't confuse learners.
- As an admin, I want to categorize content by skill so that search work well.

3 System Requirements

3.1 Functional Requirements

3.1.1 For Users

- The system shall allow a user to register via email, phone, Google, or Facebook.
- The system shall verify new accounts created via email and phone.
- The system shall allow a user to log in with registered credentials or Google/Facebook.
- The system shall allow a user to log out from the current session.
- The system shall allow a user to reset a forgotten password via email or phone verification.

3.1.2 For Learners

- The system shall allow learners to browse the catalog of practice content.
- The system shall provide search across practice content.
- The system shall allow learners to filter practice content.
- The system shall display metadata for a practice content before starting.
- The system shall let learners bookmark and unbookmark practice content.
- The system shall provide a bookmarked practice content list.
- The system shall allow learners to take a practice content.
- The system shall allow learners to submit answers manually or automatically.
- The system shall allow learners to resume in-progress practice attempts.
- The system shall store and display a practice attempt history.
- The system shall allow learners to take an evaluation test.
- The system shall generate a skill profile from evaluation test results.
- The system shall allow learners to set a target band score and test date.
- The system shall generate a study plan based on the learner's skill profile, target band, and test date.
- The system shall create learning goals from the study plan.
- The system shall produce adaptive recommendations for the next practice content.
- The system shall use AI to analyze responses and return feedback and an estimated band score.
- The system shall transcribe audio responses.

- The system shall present scores and feedback to the learner.
- The system shall save all scores and feedback.
- The system shall provide a skill progress dashboard.
- The system shall provide an overall performance dashboard.
- The system shall track and display time spent per skill and total study time.
- The system shall allow learners to generate a progress report.

3.1.3 For Admin

- The system shall allow an administrator to create user accounts.
- The system shall allow an administrator to edit user account.
- The system shall allow an administrator to deactivate and reactivate user accounts.
- The system shall allow an administrator to assign roles.
- The system shall allow an administrator to revoke roles.
- The system shall allow an administrator to manage access levels
- The system shall allow an administrator to add new practice content.
- The system shall allow an administrator to edit practice content.
- The system shall allow an administrator to remove practice content.
- The system shall allow an administrator to categorize content by skill.

3.2 Non-Functional Requirements

The three most critical architectural characteristics for IELTS Mastermind are, in order of priority: Responsiveness, Reliability, and Overall Cost. The prioritization is explained as follows:

- **Responsiveness:** If pages, practice items, or AI feedback feel slow, learners drop off and do not complete tasks. Quick loads and fast feedback keep study flow, motivation, and learning quality high.
- **Reliability:** Learners often practice close to exam dates and depend on the platform to work consistently. If the system is unstable or data is lost, users lose trust and cannot continue learning effectively.
- **Overall Cost:** Since IELTS Mastermind is still in the development phase, the available budget is limited. Therefore, the system must be designed to keep the overall cost as low as possible while still meeting functional and performance requirements.

3.2.1 Responsiveness Requirements

- The system must load practice content catalog within 4 seconds for 95% of requests.
- The system must load the search or filter results within 3 seconds for 95% of requests.
- The system must open a practice item within 3 seconds for 95% of requests.
- The system must submit an answer within 3 second for 95% of requests.
- The system must return AI feedback and an estimated band score within 15 seconds for 90% of requests.

3.2.2 Reliability Requirements

- The system must maintain at least 95% uptime per month, measured over a calendar month.
- The system must ensure that no more than 5 minutes of data is lost after an incident.
- The system must restore core services within 60 minutes after an incident.

3.2.3 Overall Cost Requirements

- The system must be deployable using low-cost services with a total infrastructure cost less than 4,000,000 VND per month.
- The system must limit AI feedback usage so that the total AI-related cost does not exceed 1,000,000 VND per month.

4 System Analysis

4.1 Use Case Diagrams

4.1.1 Authentication and Authorization Module

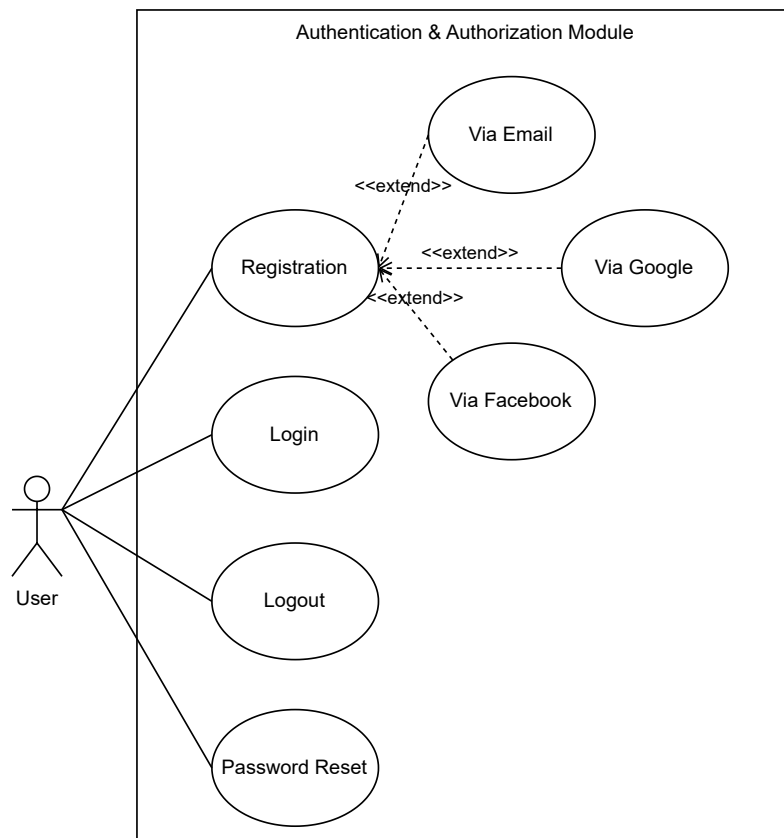


Figure 1: *Use Case Diagram of Authentication and Authorization Module*

The Use Case Diagram in Figure 1 for the Authentication and Authorization Module shows how users access and manage their accounts in the IELTS Mastermind system. The User can perform key actions such as Registration, Login, Logout, and Password Reset. The registration process supports multiple methods, including Email, Google, and Facebook.

4.1.2 Practice Content Access Module



Figure 2: Use Case Diagram of Practice Content Access Module

The Use Case Diagram in Figure 2 for the Practice Content Access Module describes how a Learner interacts with practice materials in the IELTS Mastermind system. Learners can browse, search, and filter practice content to find suitable exercises. They are able to preview content, view metadata, and start practice sessions, as well as bookmark materials for later

review and manage their bookmark list. During practice, learners can submit answers manually or automatically, receive band scores and automated feedback, and track their learning through progress lists and attempt history.

4.1.3 Personalization and Learning Paths Module

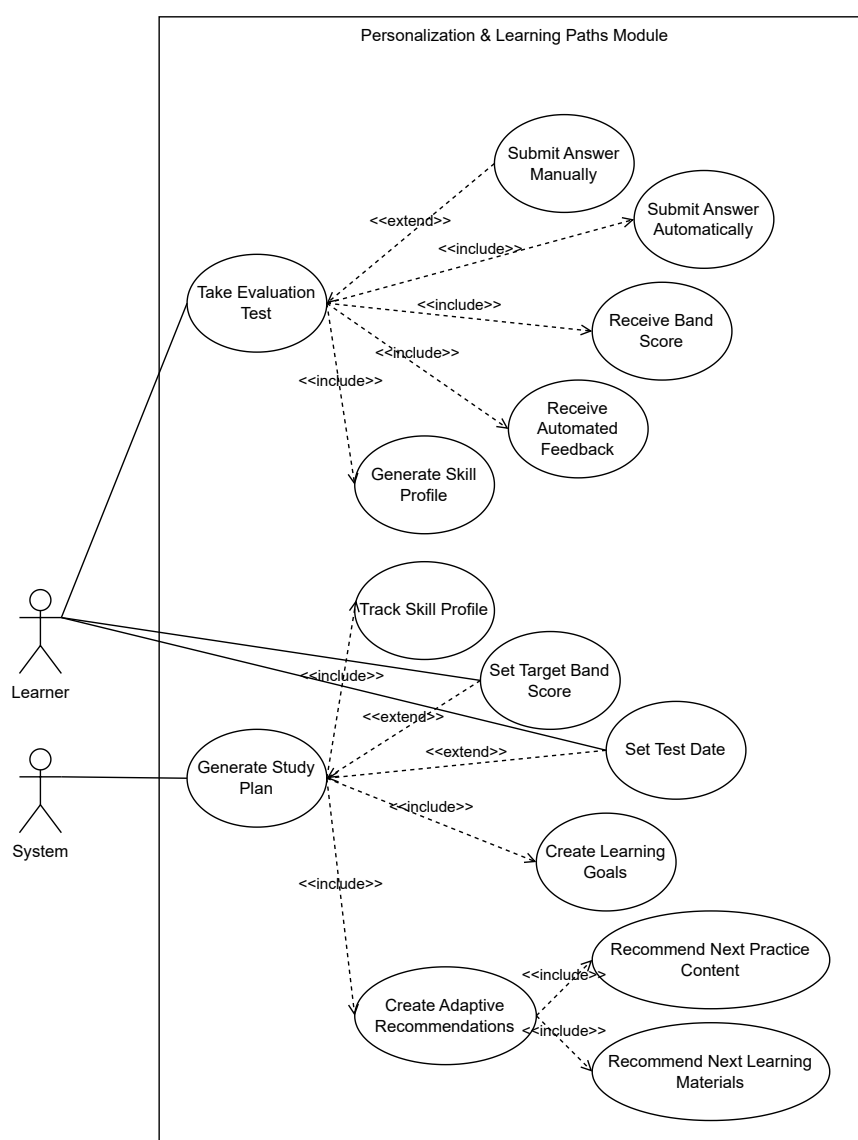


Figure 3: Use Case Diagram of Personalization and Learning Paths Module

The Use Case Diagram in Figure 3 for the Personalization and Learning Paths Module shows how the system supports personalized learning for each Learner. Learners can take an evaluation test by submitting answers manually or automatically, after which the system provides band scores and automated feedback to generate a skill profile. Based on this profile, learners can track their skills, set target band scores, define test dates, and create learning goals. The system then uses this information to generate a study plan and provide adaptive recommendations, such as suggesting the next practice content and learning materials.

4.1.4 Feedback and Band Score Module

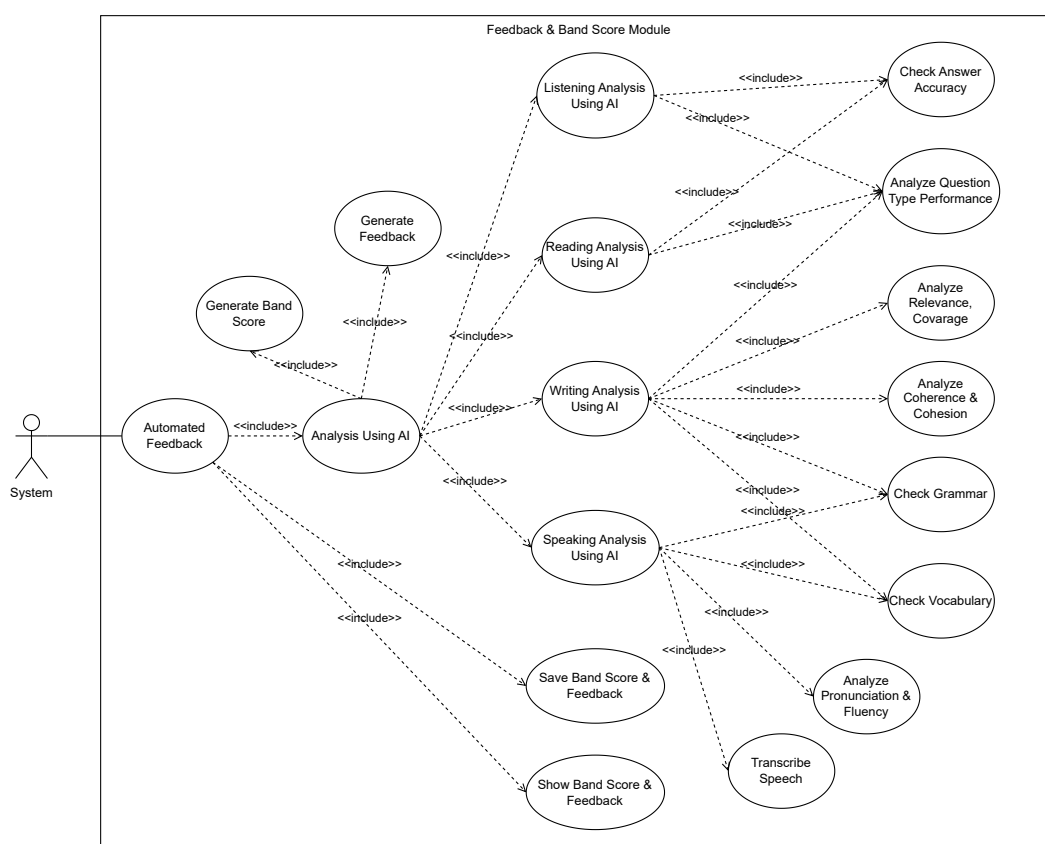


Figure 4: Use Case Diagram of Feedback and Band Score Module

The Use Case Diagram in Figure 4 for the Feedback and Band Score Module illustrates how the system automatically evaluates learner responses and provides detailed feedback. The System analyzes answers for all four IELTS skills, including Listening, Reading, Writing, and Speaking, using AI-based analysis. For objective skills such as Listening and Reading, the system checks answer accuracy and performance, while for Writing and Speaking, it evaluates aspects such as grammar, vocabulary, coherence, task response, and pronunciation. Based on these analyses, the system generates band scores, creates feedback, and stores the results so learners can view their

scores and feedback later.

4.1.5 Progress Tracking Module

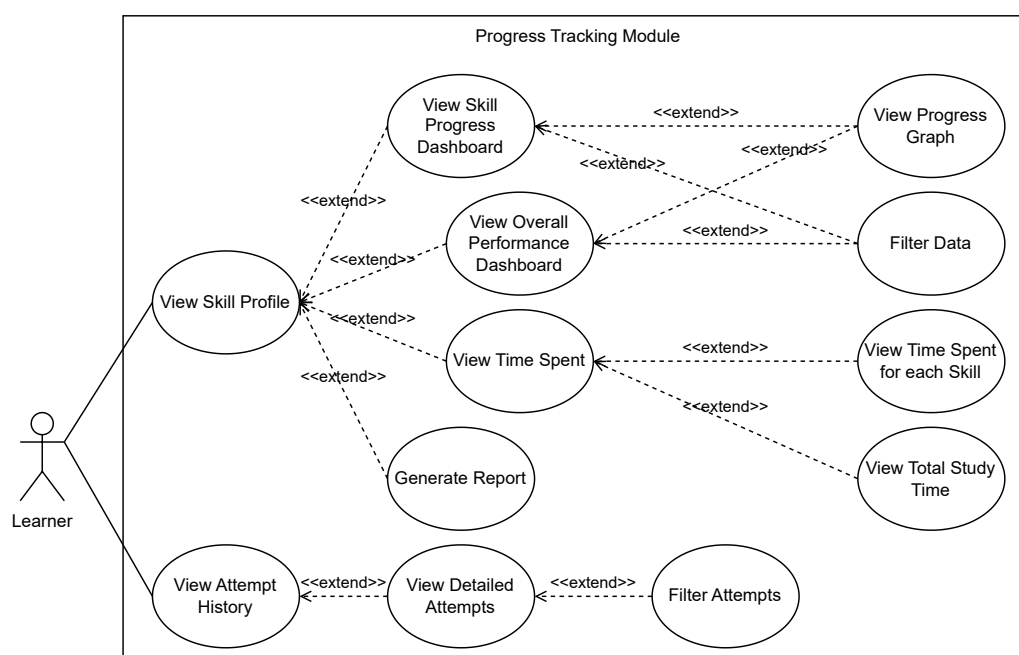


Figure 5: Use Case Diagram of Progress Tracking Module

The Use Case Diagram in Figure 5 for the Progress Tracking Module shows how learners monitor their learning progress in the IELTS Mastermind system. Learners can view their skill profile, attempt history, and access different dashboards to see skill progress, overall performance, and time spent on practice. The system allows learners to view progress through graphs, filter data, and check detailed statistics such as time spent for each skill and total study time. Learners can also review detailed attempts and generate reports to better understand their learning patterns.

4.1.6 User and Role Management Module

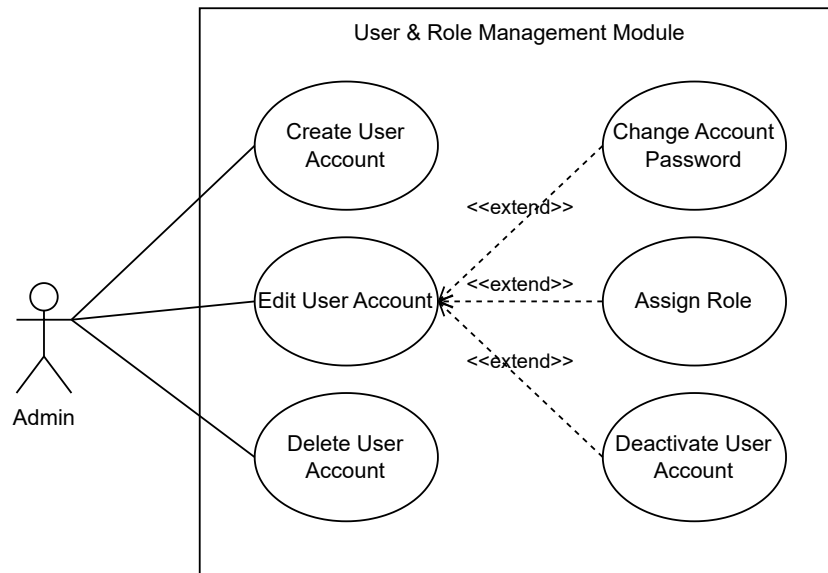


Figure 6: *Use Case Diagram of User and Role Management Module*

The Use Case Diagram in Figure 6 for the User and Role Management Module illustrates how an Admin manages user accounts in the IELTS Mastermind system. The admin can create, edit, and delete user accounts to maintain the system. When editing a user account, the admin can perform additional actions such as changing passwords, assigning roles, and deactivating accounts, which are shown as extended use cases.

4.1.7 Practice Content Management Module

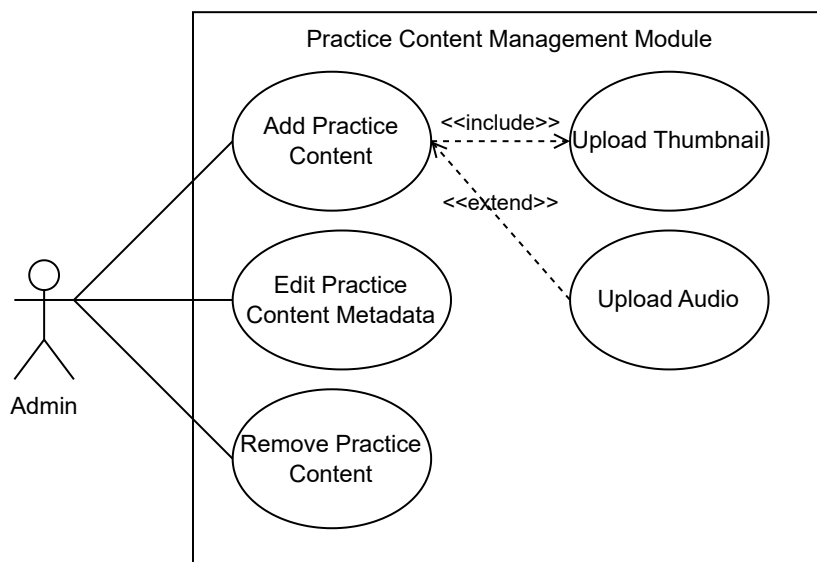


Figure 7: Use Case Diagram of Practice Content Management Module

The Use Case Diagram in Figure 7 for the Practice Content Management Module shows how an Admin manages practice materials in the IELTS Mastermind system. The admin can add, edit, and remove practice content to keep learning materials updated and accurate. When adding new content, the process includes uploading a thumbnail, and may be extended to upload audio when required, such as for Listening or Speaking tasks.

4.2 Use Case Scenarios

This section describes the system's use case scenarios, corresponding to the use cases shown in Section 3.3. Each table defines the actors, preconditions, postconditions, normal flows, alternative flows, and exceptions.



4.2.1 Register

ID and Name	UC-1 Register
Created by	Khang
Date created	04/10/2025
Actors	User
Description	User creates a new account using one of the supported methods: email, or social login (Google/Facebook).
Trigger	User selects “Register” option on the login page.
Preconditions	• System is online. • Registration service is available.
Postconditions	• A new user account is successfully created. • The user is redirected to the profile setup page or the dashboard.
Normal Flow	1. User clicks on “Register” on the login page. 2. System displays registration options (Email, Google, Facebook). 3. User selects a registration method. 3.1 Email registration: User enters personal information (name, email, password) and submits. 3.2 Social login: User chooses Google or Facebook, system redirects to the social provider for authentication. 4. System validates the submitted information or social credentials. 4.1 If valid: System creates a new user account, sends confirmation email. 4.2 For social login: System receives confirmation from the provider and creates the account automatically. 5. User is redirected to the profile setup page or logged in automatically.
Alternative Flows	• At Step 3.1: User provides invalid data → System displays appropriate error messages and prompts re-entry. • At Step 3.2: Social login is denied by the provider → System displays authentication error and returns to registration options. • At Step 4.1: Confirmation email fails to deliver → System still creates the account but notifies user to resend verification later.
Exceptions	• At any step: If the registration service or database is unavailable, the system shows an error message “Something went wrong” and logs the issue. • At Step 4: If the validation service crashes, the system displays a generic system error and stops the registration process.

Table 1: Use Case Table UC-1: Register Account



4.2.2 Login

ID and Name	UC-2 Login
Created by	Khang
Date created	04/10/2025
Actors	User
Description	User logs into the system using registered credentials or social login.
Trigger	User selects “Login” option on the homepage.
Preconditions	• User has a registered account • System is online
Postconditions	• User is authenticated and redirected to dashboard
Normal Flow	1. User selects “Login” option 2. System displays login form and social login buttons 3. User chooses a login method 3.1 User enters credentials (email/phone + password) 3.2 User selects social login (Google/Facebook) 4. System validates the login information 5. User successfully logs in and is redirected to dashboard
Alternative Flows	• At step 3.1: Invalid credentials → system displays error message and allows retry. • At step 3.2: Social login authentication fails → system displays error.
Exceptions	• System or authentication service is down → display maintenance message. • Network connectivity issues → login process fails, user prompted to retry.

Table 2: *Use Case Table UC-2: Login*



4.2.3 Password Reset

ID and Name	UC-3 Password Reset
Created by	Khang
Date created	04/10/2025
Actors	User
Description	User resets their password using email or phone verification.
Trigger	User clicks “Forgot Password”.
Preconditions	• User account exists • System is online
Postconditions	• Password reset is completed successfully, and the user can log in again
Normal Flow	1. User clicks “Forgot Password” 2. System prompts for recovery method (email/phone) 2.1 User enters registered email 2.2 User enters registered phone number 3. System sends OTP or reset link 4. User verifies identity and enters a new password 5. System updates password and confirms reset
Alternative Flows	• At step 2.1/2.2: Invalid or unregistered email/phone → error message displayed. • At step 4: OTP or reset link expired → system offers to resend.
Exceptions	• Email/SMS service unavailable → recovery cannot proceed. • Network failure → user must retry later.

Table 3: *Use Case Table UC-3: Password Recovery*

4.2.4 Browse Practice Content

ID and Name	UC-4 Browse Practice Exercises
Created by	Khang
Date created	05/11/2025
Actors	Learner
Description	Learner views a list of practice exercises associated with a specific skill and may use search or filtering options to find relevant exercises.
Trigger	Learner accesses the <i>Exercise</i> section of a selected skill.
Preconditions	• Learner is authenticated. • Selected skill exists. • Practice exercise service is available.
Postconditions	• The exercise list for the selected skill is displayed.
Normal Flow	1. Learner opens the Exercise section of a specific skill. 2. System loads and displays all available exercises for that skill. 3. Learner scrolls through the list of exercises.
Alternative Flows	• At Step 2 – Search: Learner performs a keyword search → System displays exercises matching the keyword. • At Step 2 – Filter by Topic: Learner selects one or more topics → System displays exercises for those topics. • At Step 2 – Filter by Exercise Type: Learner selects an exercise type (e.g., Multiple Choice, Fill-in-the-blank, Matching) → System displays matching exercises.
Exceptions	• No exercises available for the selected skill → System shows “No exercises available”. • Service timeout → System displays an error message. • Invalid filter criteria → System resets filters and shows default exercise list.

Table 4: Use Case Table UC-4: Browse Practice Exercises



4.2.5 Preview Practice Content

ID and Name	UC-5 Preview Practice Content
Created by	Khang
Date created	05/11/2025
Actors	Learner
Description	Learner reviews metadata and summary of selected practice content before deciding to start.
Trigger	Learner selects a content item from the listing.
Preconditions	• Learner is authenticated. • Selected practice content exists.
Postconditions	• Preview information is displayed.
Normal Flow	1. Learner selects a practice content item. 2. System retrieves metadata and summary details. 3. System displays preview information. 4. Learner decides whether to start content.
Alternative Flows	• At Step 3: Learner selects Start Practice → triggers UC-03 Take Practice Content.
Exceptions	• Content metadata retrieval failure → System displays error. • Content removed/unavailable → System notifies user.

Table 5: Use Case Table UC-5: Preview Practice Content

4.2.6 Bookmark Practice Content

ID and Name	UC-6 Bookmark Practice Content
Created by	Khang
Date created	05/11/2025
Actors	Learner
Description	Learner bookmarks or removes bookmarks on exercises, and views the list of bookmarked exercises.
Trigger	Learner opens the Exercise section of a specific skill.
Preconditions	• Learner is authenticated. • Learner is viewing the Exercise section of a specific skill.
Postconditions	• Exercise is added to or removed from the bookmark list. • Bookmarked exercises list is displayed when requested.
Normal Flow	1. Learner selects an exercise from the exercise list. 2. System displays exercise metadata and a <i>Bookmark</i> button. 3. Learner clicks the <i>Bookmark</i> button. 4. Learner clicks the <i>Bookmark</i> icon located at the top-right corner of a specific exercise. 1.1 If the exercise is not bookmarked → System adds it to the bookmark list. 1.2 If the exercise is already bookmarked → System removes it from the bookmark list. 5. Learner clicks the <i>Bookmarked List</i> menu. 6. System retrieves and displays all bookmarked exercises.
Alternative Flows	• At Step 6: No bookmarked exercises → System displays message “No bookmarked exercises”. • At Step 3.3: Bookmark state unchanged due to error → System notifies failure.
Exceptions	• Bookmark service unavailable → system displays error. • Storage error → system notifies failure.

Table 6: *Use Case Table UC-6: Bookmark Practice Content*

4.2.7 Take Practice Content

ID and Name	UC-7 Take Practice Content
Created by	Khang
Date created	05/11/2025
Actors	Learner
Description	Learner selects a practice test from the exercise list of a specific skill, completes all questions, and submits responses manually or automatically when time expires. The system evaluates the submission and returns band score and feedback.
Trigger	Learner selects a practice test to begin.
Preconditions	• Learner is authenticated. • Learner is viewing the Exercise section of a specific skill.
Postconditions	• Answers are recorded. • Band score is generated. • Feedback and correct answers are presented to the learner.
Normal Flow	1. Learner selects a practice test from the list of exercises. 2. Learner clicks the <i>Enter the Test</i> button. 3. System loads test content and instructions. 4. Learner completes all questions. 5. Learner clicks <i>Submit</i>. 5.1 System automatically submits when time expires. 5.2 Learner submits manually before time ends. 6. System evaluates responses. 6.1 System generates the band score. 6.2 System generates feedback and correct answers. 7. System displays results (score + feedback) to the learner.
Alternative Flows	• At Step 6.1: If the Score Service is unavailable → System displays an error message. • At Step 6.2: If the Feedback Service is unavailable → System displays an error message and skips the feedback retrieval.
Exceptions	• Connection lost → system saves progress for later continuation. • System error → user notified and submission attempt logged.

Table 7: Use Case Table UC-7: Take Practice Content

4.2.8 Take Evaluation Test

ID and Name	UC-8 Take Evaluation Test
Created by	Khang
Date created	05/11/2025
Actors	Learner
Description	Learner completes an evaluation test to determine current proficiency level. System evaluates submitted answers, generates a band score, skill profile, and returns automated feedback.
Trigger	Learner starts the evaluation test.
Preconditions	• Learner is authenticated. • Selected evaluation test is available. • Learner has chosen IELTS skills.
Postconditions	• Score is generated. • Skill profile is created or updated. • Feedback is provided.
Normal Flow	1. Learner chooses IELTS skills. 2. System loads corresponding evaluation test. 3. Learner completes all questions. 4. Learner clicks <i>Submit</i>. 4.1 System auto-submits when time expires. 4.2 Learner submits manually before time ends. 5. System evaluates responses. 6. System generates and displays band score. 7. System generates and displays automated feedback. 8. System generates skill profile. 9. System displays results to learner.
Alternative Flows	• At Step 4.1: System auto-submits due to timeout → continue at Step 5.
Exceptions	• Connection lost → progress saved for resume. • System error → user notified and submission logged.

Table 8: *Use Case Table UC-8: Take Evaluation Test*



4.2.9 Study Plan Generation

ID and Name	UC-9 Study Plan Generation
Created by	Khang
Date created	05/11/2025
Actors	Learner
Description	Learner reviews skill profile, sets target band score, and schedules a test date. System updates study plan accordingly.
Trigger	Learner visits the study plan or skill profile section.
Preconditions	• Learner is authenticated. • Learner has an existing skill profile.
Postconditions	• Study plan updated based on learner inputs.
Normal Flow	1. Learner reviews current skill profile. 2. Learner sets target band score. 3. Learner sets desired test date. 4. System updates and stores the study plan.
Alternative Flows	• At Step 2: Learner skips setting target → system sets default band goal. • At Step 3: Learner skips selecting test date → continues without scheduling.
Exceptions	• System fails to save study plan → user notified.

Table 9: Use Case Table UC-9: Study Plan Setup

4.2.10 Personalized Recommendation

ID and Name	UC-10 Personalized Recommendation
Created by	Khang
Date created	05/11/2025
Actors	Learner, System
Description	System generates learning goals and adaptive recommendations for next practice content and learning materials based on learner profile and targets.
Trigger	System detects new profile data OR learner requests recommendations.
Preconditions	• Learner is authenticated. • Skill profile exists. • Study plan exists.
Postconditions	• Learning goals are generated/updated. • Recommended content and materials available.
Normal Flow	1. System analyzes skill profile + target band. 2. System creates learning goals. 3. System generates adaptive recommendations. 4. System recommends next practice content. 5. System recommends learning materials. 6. Learner views recommendations.
Alternative Flows	• At Step 1: Insufficient data → learner prompted to take evaluation test. • At Step 4–5: No suitable content → fallback suggestions.
Exceptions	• Recommendation service downtime → user notified.

Table 10: *Use Case Table UC-10: Personalized Recommendation*

4.2.11 Automated Feedback

ID and Name	UC-11 Personalized Recommendation
Created by	Nguyễn Đoàn Hải Bằng
Date created	09/12/2025
Actors	System
Description	The system analyzes the user's test responses using AI, constructs prompts, sends them to the AI model, formats the returned feedback, computes band scores, stores all results, and displays them to the user.
Trigger	User submits test responses.
Preconditions	• User submission is valid. • System has internet & AI service connectivity.
Postconditions	• Feedback, structured results, and band scores are saved and shown to the user.
Normal Flow	1. System receives the learner's test responses (Listening/Reading/Writing/Speaking). 2. System loads the appropriate AI prompt template for each required skill. 3. System inserts the user's responses into the corresponding prompt template. 4. System appends additional instruction rules 5. System sends the fully constructed AI prompt to the AI model API. 6. AI model processes the prompt and returns raw feedback + scores. 7. System extracts structured elements (overall score, task response, grammar issues, vocabulary, coherence, etc.). 8. System generates a summary report for the user (strengths, weaknesses, action steps). 9. System stores structured feedback, band scores in the database 10. System displays formatted band scores and feedback in the user interface.
Alternative Flows	• At Step 7: AI response is incomplete → System automatically retries the request with a modified/reinforced prompt • At Step 6: Prompt Too Large (Token Limit Exceeded) → System splits the prompt into multiple smaller segments and processes them sequentially.
Exceptions	• AI endpoint unreachable → System returns an error message and logs the failure.

Table 11: Use Case Table UC-11: Automated FeedBack

4.2.12 Add Practice Content

ID and Name	UC-12 Add Practice Content
Created by	Nguyễn Đoàn Hải Bằng
Date created	09/12/2025
Actors	Admin
Description	Admin creates a new practice content entry, uploads required media (thumbnail and optionally audio), and saves the final practice content.
Trigger	Admin navigates to the Practice Content Management page → clicks “Add New Content”
Preconditions	- Admin is authenticated. - Practice Content Management page is accessible.
Postconditions	- New practice content is saved in the database. - Draft is saved if Admin leaves before completing.
Normal Flow	- System displays a form and asks Admin to select a skill type (Reading, Listening, Speaking, Writing). - Admin enters Instructions & Note Layout in the rich-text editor. - Admin uploads a Thumbnail image. - If the exercise type is Listening, Admin uploads Audio File. - Admin clicks “Add Question” to add questions. - Admin selects a Question Type (MCQ, Short Text, etc.) and enters the Correct Answer. - Admin saves the question → it appears in the question list. - Admin selects the content status: Draft or Published. - Admin clicks Save & Exit. - System validates all required fields. - System saves new practice content to the database - System returns to the Practice Content Management page with updated list.
Alternative Flows	At Step 1 to 8: If Admin leaves the page → System automatically saves a Draft version with current progress. At Step 3 or 4: If thumbnail or audio is not in supported formats → System rejects the file and displays a file-type error.
Exceptions	- Network Timeout → System shows notification. - Database fails to save content → Shows error and keeps content unsaved in browser state.

Table 12: Use Case Table UC-12: Add Practice Content

4.2.13 Edit Practice Content

ID and Name	UC-13 Edit Practice Content
Created by	Nguyễn Đoàn Hải Bằng
Date created	09/12/2025
Actors	Admin
Description	Admin updates existing practice content, including instructions, questions, media files, and metadata. The system loads the existing content, allows edits, and saves revisions as Draft or Published.
Trigger	Admin selects an existing practice item from the Practice Content Management page
Preconditions	• Admin is authenticated. • Practice Content Management page is accessible. • The selected content item exists.
Postconditions	• Updated content is saved (Draft or Published).
Normal Flow	1. Admin clicks Edit (pencil icon) on a specific practice content item. 2. System loads all existing data into the Edit Content form. 3. Admin updates any of the following fields • Instructions in the rich text editor. • Thumbnail (replace or remove). • Audio file (Listening only: upload new or delete existing). • Questions: – Edit an existing question. – Delete a question. – Add a new question. • Exercise Info (title, topic, duration, etc.). 4. Admin selects the content status: Draft or Published. 5. Admin clicks Save & Exit. 6. System validates only the fields that were modified. 7. System updates the practice content in the database. 8. System returns the Admin to the Practice Content Management page with the updated information.
Alternative Flows	• At Step 1 to 4: If Admin leaves the page → System automatically saves a Draft version with current progress. • At Step 3: If thumbnail or audio is not in supported formats → System rejects the file and displays a file-type error.
Exceptions	• Network Timeout → System shows notification. • Database fails to save content → Shows error and keeps content unsaved in browser state.

Table 13: Use Case Table UC-13: Edit Practice Content

4.2.14 Delete Practice Content

ID and Name	UC-14 Delete Practice Content
Created by	Nguyễn Đoàn Hải Bằng
Date created	09/12/2025
Actors	Admin
Description	Admin deletes a practice content item from the system. The system confirms deletion and removes the data permanently.
Trigger	Admin clicks Delete on a practice content item in the management dashboard.
Preconditions	• Admin is authenticated. • Practice Content Management page is accessible. • The selected content item exists.
Postconditions	• Practice Content is deleted
Normal Flow	1. Admin clicks Delete (Trash can icon) on a content item. 2. System displays a confirmation dialog: “Are you sure you want to delete this content?” 3. Admin confirms deletion. 4. System deletes & updates the Practice Content Management list. 5. Admin sees a success message: “Content deleted successfully.”
Alternative Flows	• At Step 2: Admin chooses “Cancel” → System closes dialog and performs no deletion.
Exceptions	• Network Timeout → System shows notification. • Database fails to delete content → Shows error and keeps content unsaved in browser state.

Table 14: Use Case Table UC-14: Remove Practice Content

4.2.15 Create User Account

ID and Name	UC-15 Create User Account
Created by	Quang Duy
Date created	12/10/2025
Actors	Admin, System
Description	Admin creates a new user account by entering user information, assigning one role, and setting the initial account status.
Trigger	Admin navigates to the User Management page and clicks “Add New User”.
Preconditions	• Admin is authenticated. • User Management page is accessible.
Postconditions	• New user account is saved in the database with assigned role and status.
Normal Flow	1. System displays the User Management page with a list of existing users and a Add New User button. 2. Admin clicks Add New User. 3. System displays the Add New User form with fields. 4. Admin enters required user information. 5. Admin selects a role to assign to the user. 6. Initial account status is set to Active by default. 7. Admin clicks Save. 8. System validates that all required fields are filled and that the Email is unique. 9. System saves the new user record in the database with the selected roles and status. 10. System displays a success message and returns to the User Management page with the updated user list.
Alternative Flows	• At Step 4: If Admin leaves the page or clicks Cancel → System discards the entered information and returns to the User Management page. • At Step 9: If required fields are missing or invalid → System highlights invalid fields, displays an error message, and remains on the Add New User form. • At Step 3 or 4: If the Email already exists → System displays an error message and asks Admin to enter a different value.
Exceptions	• Network Timeout → System shows notification. • Database fails to save content → Shows error and keeps content unsaved in browser state.

Table 15: Use Case Table UC-15: Create User Account

4.2.16 Edit User Account

ID and Name	UC-16 Edit User Account
Created by	Quang Duy
Date created	12/10/2025
Actors	Admin, System
Description	Admin updates an existing user account's information such as name, email, role, password, and status using the Edit User form.
Trigger	Admin navigates to the User Management page and clicks the Edit icon for a specific user.
Preconditions	• Admin is authenticated. • User Management page is accessible. • The selected user account already exists in the system.
Postconditions	• User account information is updated in the database with new details, role, and status.
Normal Flow	1. System displays the User Management page with a list of existing users and actions for each user. 2. Admin clicks the Edit icon for a specific user. 3. System displays the Edit User form pre-filled with the current user information. 4. Admin updates one or more fields. 5. If Admin wants to change the password, Admin enters a new password in the Password. 6. Admin clicks Update. 7. System validates that all required fields are filled and that the Email is unique. 8. System updates the user record in the database with the new information. 9. System displays a success message and closes the Edit User form. 10. System refreshes the User Management page and shows the updated user information in the user list.
Alternative Flows	• At Step 3: If Admin clicks Cancel or closes the Edit User form → System discards any changes and returns to the User Management page without updating the user. • At Step 7: If required fields are missing or invalid → System highlights invalid fields, displays an error message. • At Step 7: If the Email already exists → System displays an error message and asks Admin to enter a different value.
Exceptions	• Network Timeout → System shows notification. • Database fails to save content → Shows error and keeps content unsaved in browser state.

Table 16: Use Case Table UC-16: Edit User Account

4.2.17 Delete User Account

ID and Name	UC-17 Delete User Account
Created by	Quang Duy
Date created	12/10/2025
Actors	Admin, System
Description	Admin removes an existing user account from the system using the Delete action on the User Management page.
Trigger	Admin navigates to the User Management page and clicks the Delete button for a specific user.
Preconditions	• Admin is authenticated. • User Management page is accessible. • The selected user account exists in the system.
Postconditions	• Selected user account is removed from the system and no longer appears in the User Management list.
Normal Flow	1. System displays the User Management page with a list of existing users and actions for each user. 2. Admin clicks the Delete button for a specific user. 3. System displays a confirmation dialog asking Admin to confirm the deletion. 4. Admin clicks Confirm. 5. System validates that the user can be deleted. 6. System removes the user account from the database. 7. System displays a success message indicating that the user has been deleted. 8. System refreshes the User Management page and removes the deleted user from the user list.
Alternative Flows	• At Step 3: If Admin clicks Cancel or closes the confirmation dialog → System cancels the delete operation and returns to the User Management page with no changes. • At Step 7: o If business rules prevent deleting the user → System displays an error message.
Exceptions	• Network Timeout → System shows notification. • Database Error on Delete → System shows an error message and keeps the user account in the database.

Table 17: Use Case Table UC-17: Delete User Account

4.3 Activity Diagrams

4.3.1 Password Reset

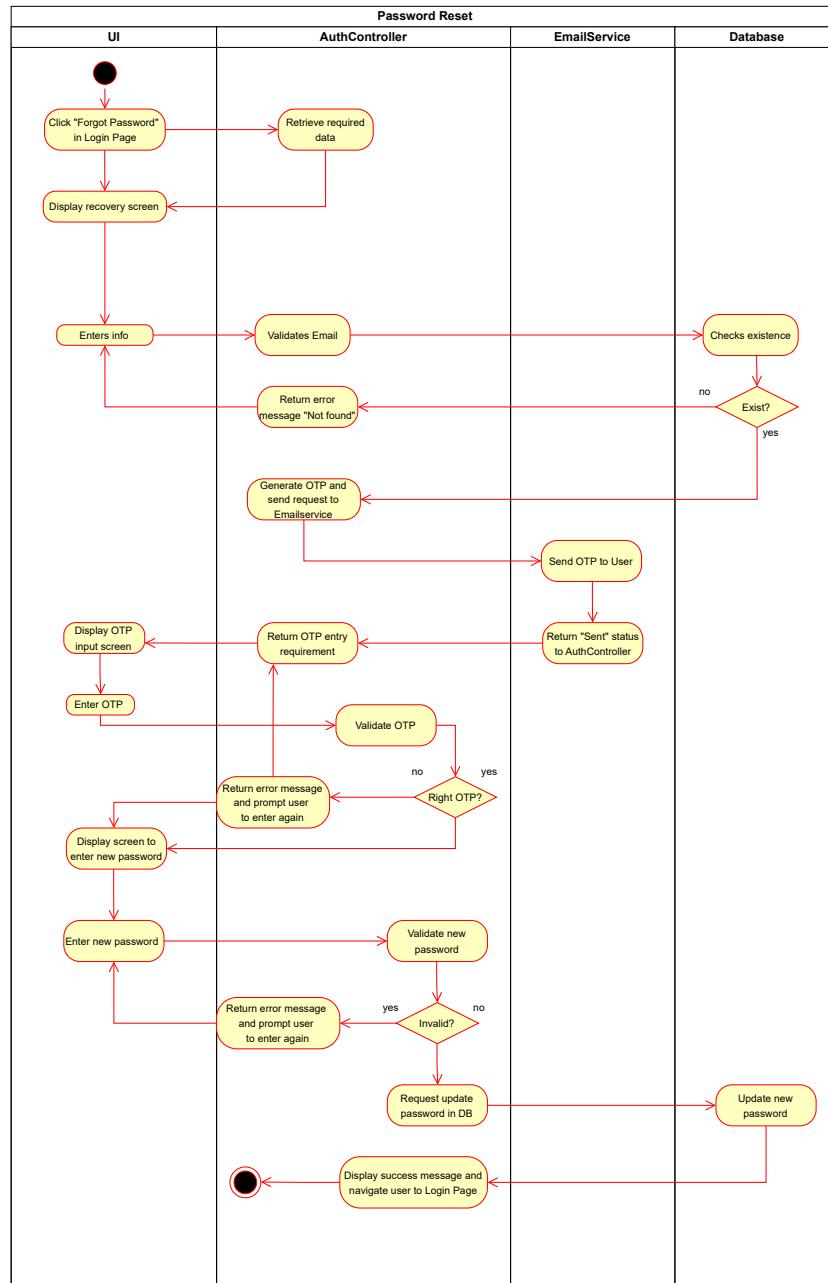


Figure 8: Activity Diagram for Password Reset

This Activity Diagram in Figure 8 shows the full password reset process when a learner forgets their password. The diagram contains the following swimlanes:

- **Learner:** Enters email, receives OTP or reset link, inputs the verification code, sets a new password, and confirms the reset.
- **UI Layer:** Sends reset requests, displays instructions, processes form inputs, and submits OTP or new password to the system.
- **ResetPasswordService:** Validates the email, generates OTP or reset token, sends the reset email, verifies the OTP, and updates the new password in the database.

The diagram highlights two main phases: (1) Email verification and OTP/token generation, and (2) OTP validation and password update. The diagram shows the interaction between the three layers to ensure the learner can complete the entire practice workflow and receive scoring feedback.

4.3.2 Browse & Bookmark Content

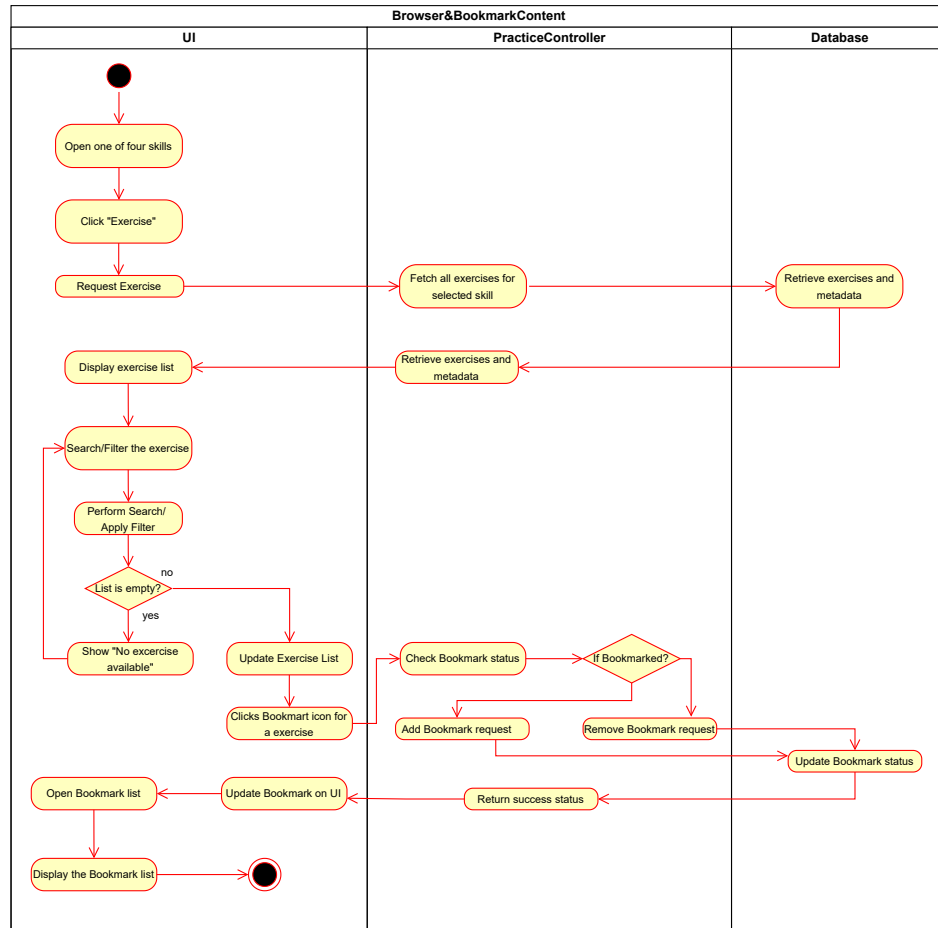


Figure 9: Activity Diagram for Browsing and Bookmarking Content

This Activity Diagram in Figure 9 describes how the learner browses available learning content, applies search or filters, and manages bookmarks. The diagram includes the following swimlanes:

- **Learner:** Scrolls through the content list, enters search keywords, applies filters, opens content details, and adds or removes bookmarks.
- **UI Layer:** Sends search and filter requests, updates the displayed content list, and visually reflects bookmark changes.
- **ContentService:** Retrieves the filtered or searched content from the database, handles bookmark operations, and returns results to the UI.

The diagram clearly illustrates the *search* → *filter* → *display* cycle, including repeated loops when the learner changes search queries or filtering criteria.

4.3.3 Preview & Take Practice

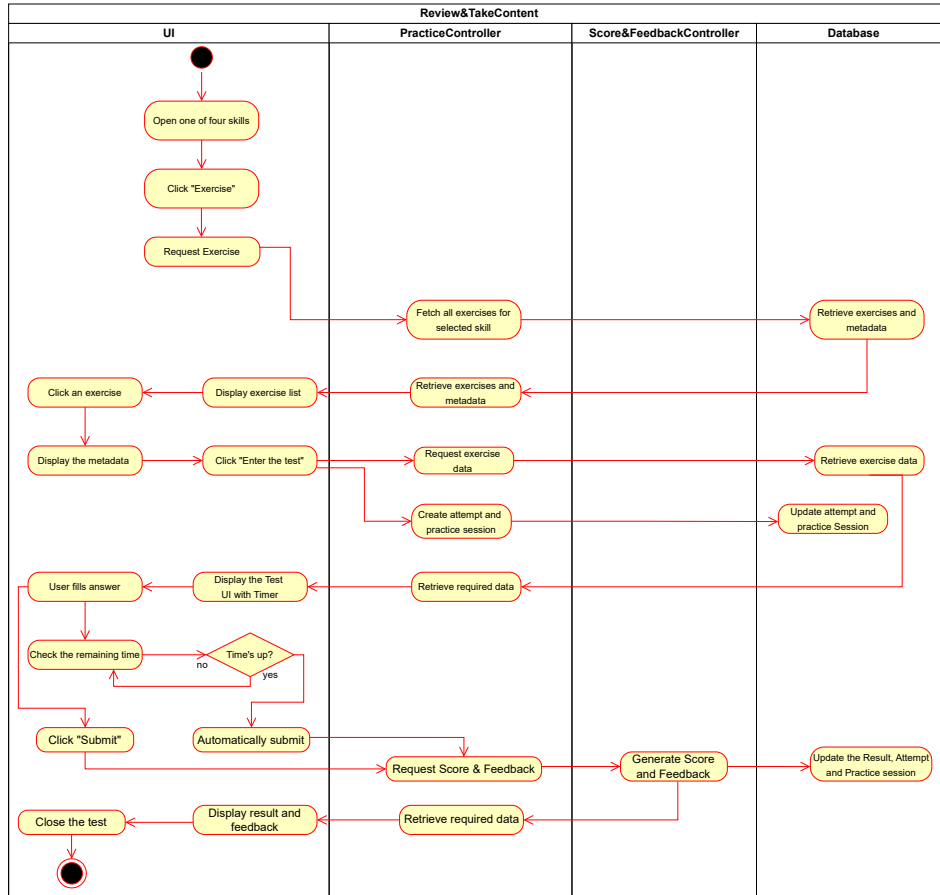


Figure 10: Activity Diagram for Preview and Take Practice

This Activity Diagram in Figure 10 illustrates the complete workflow of how a learner reviews practice content and completes practice exercises. The diagram is divided into the following swimlanes:

- **Learner:** Selects a practice item, starts the exercise, submits answers, and completes the practice session.
- **UI Layer:** Displays the practice content, updates questions step-by-step, sends learner responses to the service, and shows the final score.
- **PracticeService:** Retrieves practice content from the database, validates answers, calculates the score, and stores the result.

4.3.4 Create User Account

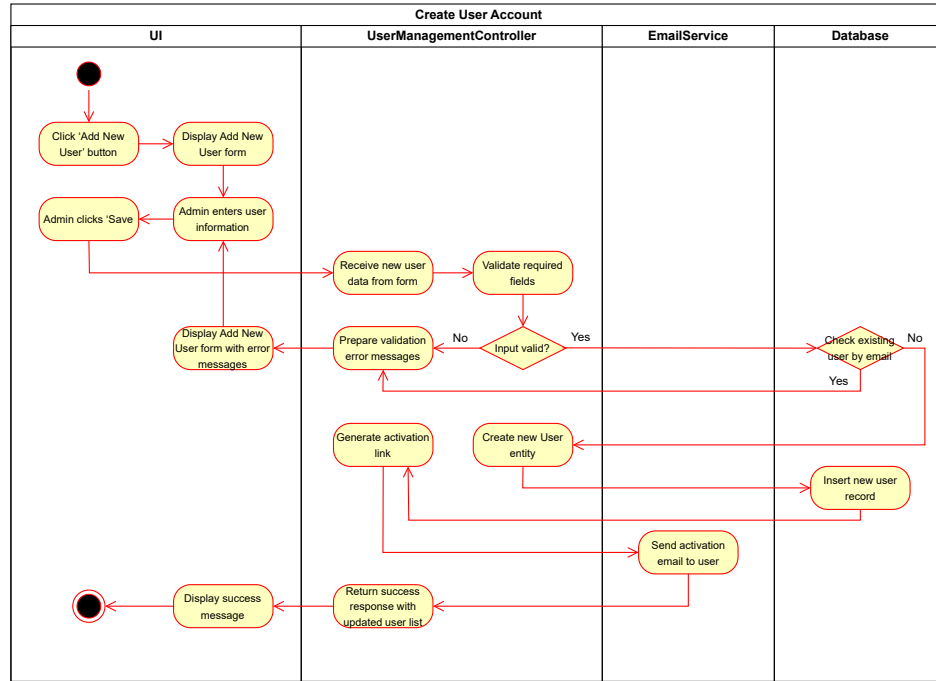


Figure 11: Activity Diagram for Create User Account

This Activity Diagram in Figure 11 illustrates the workflow of how an administrator creates a new user account in the IELTS Mastermind system. The diagram is divided into the following swimlanes:

- **UI Layer:** Displays the User Management page, shows the “Add New User” form, collects the admin’s input, and shows either validation errors or a success message after the operation.
- **UserManagementController:** Receives the submitted user information from the UI, validates the required fields, checks email uniqueness, creates the new user entity, and coordinates saving the account and sending notifications.
- **EmailService:** Sends an activation or welcome email to the newly created user when requested by the controller.
- **Database:** Checks whether the email already exists and stores the new user record with its assigned role and status.

4.3.5 Take Evaluation Test

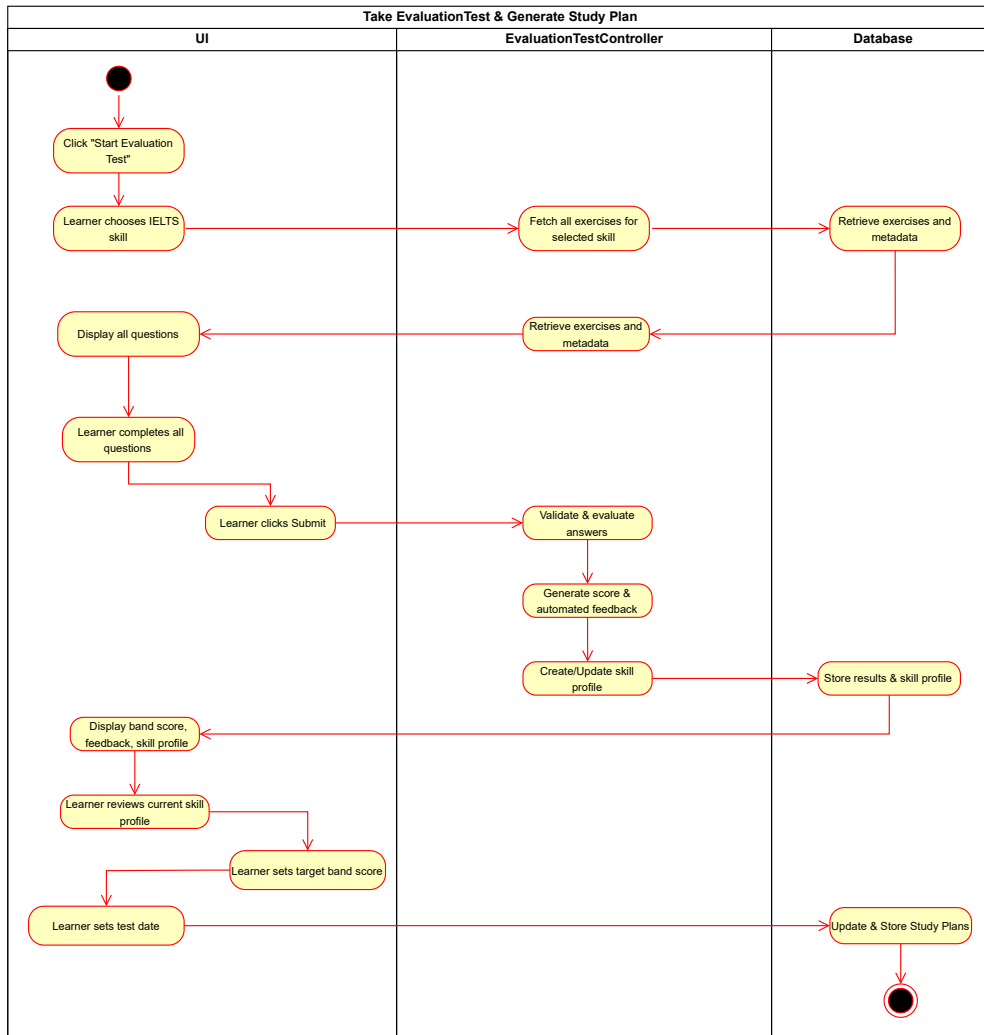


Figure 12: Activity Diagram for Take Evaluation Test & Generate Study Plan

This Activity Diagram in Figure 12 illustrates the workflow of taking an evaluation test and generating a personalized study plan in the IELTS Mastermind system. The diagram shows how a learner completes an evaluation test, receives feedback, and sets learning goals. The process is divided into the following swimlanes:

- **UI Layer:** Enables the learner to start the evaluation test, select an IELTS skill, view and answer all questions, submit responses, review results and feedback, set a target band score, choose a test date, and view the generated study plan.
- **EvaluationTestController:** Retrieves relevant evaluation exercises, validates and evaluates the learner's answers, generates band scores and automated feedback, updates the learner's

skill profile, and creates a personalized study plan based on the learner's goals.

- **Database:** Stores evaluation exercises and metadata, saves test results and updated skill profiles, and persists the generated study plans for future reference.

The workflow starts when the learner initiates the evaluation test and selects an IELTS skill. The system retrieves and displays all related exercises. After the learner completes and submits the test, the system evaluates the responses, generates a band score and feedback, and updates the learner's skill profile. The learner then reviews the results, sets a target band score and test date, and the system generates and stores a personalized study plan accordingly.

4.3.6 Add Practice Content

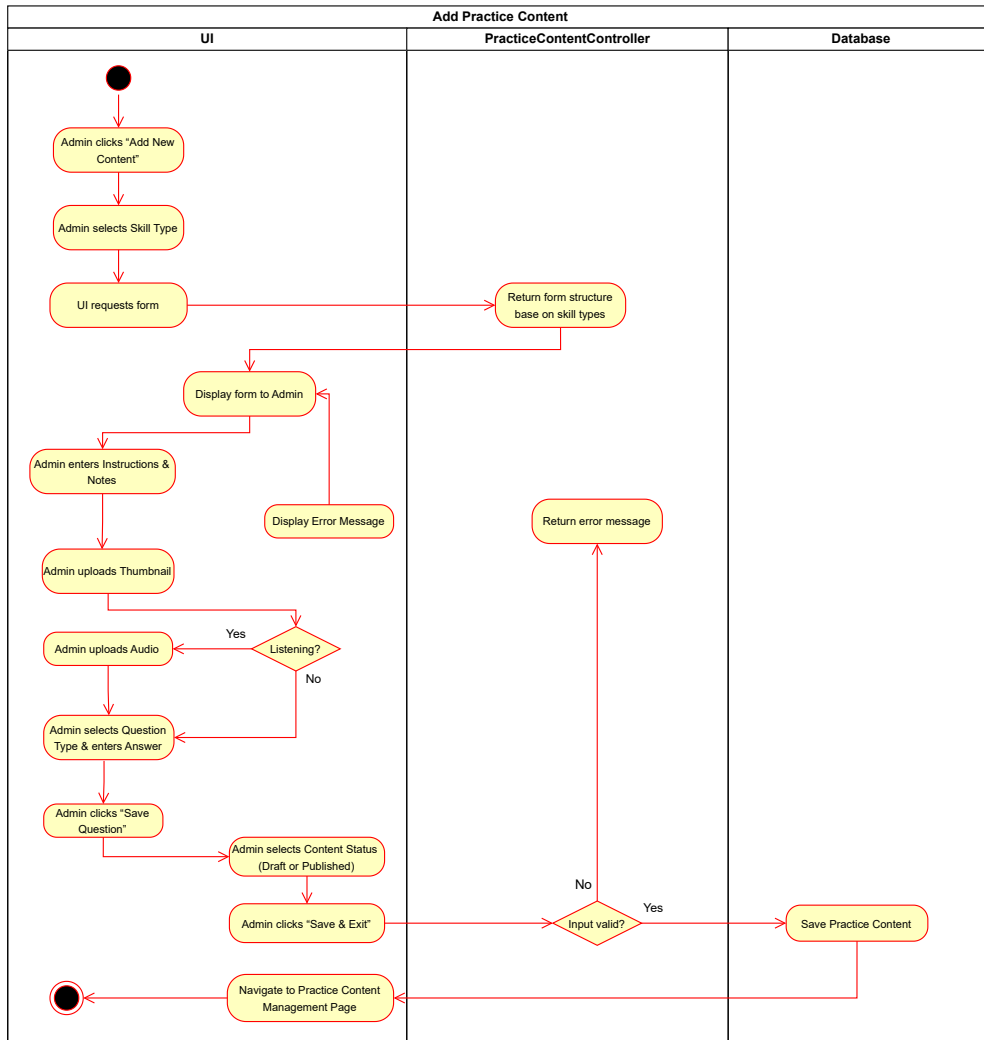


Figure 13: Activity Diagram for Add Practice Content

This Activity Diagram in Figure 13 illustrates the workflow for adding new practice content in the IELTS Mastermind system. The process describes how an administrator creates, configures, and publishes practice questions through the system. The diagram is divided into the following swimlanes:

- **UI Layer:** Allows the administrator to initiate the content creation process, select the skill type, display the dynamic input form, collect content details such as instructions, notes, media files, and questions, and present validation errors or confirmation messages.
- **PracticeContentController:** Handles requests from the UI, returns the appropriate form structure based on the selected skill type, validates the submitted content, manages business rules such as listening-specific requirements, and coordinates saving the practice content.
- **Database:** Stores the practice content, including metadata, questions, media references, and content status (draft or published), after successful validation.

The workflow begins when the administrator selects the option to add new content and chooses a specific skill type. The system then dynamically generates and displays the corresponding input form. The administrator enters instructions, uploads required media (such as audio for listening content), configures questions and answers, and selects the content status. Upon submission, the system validates all inputs. If validation fails, an error message is displayed. If successful, the content is saved to the database and the administrator is redirected to the practice content management page.

4.4 Sequence Diagrams

4.4.1 Password Reset

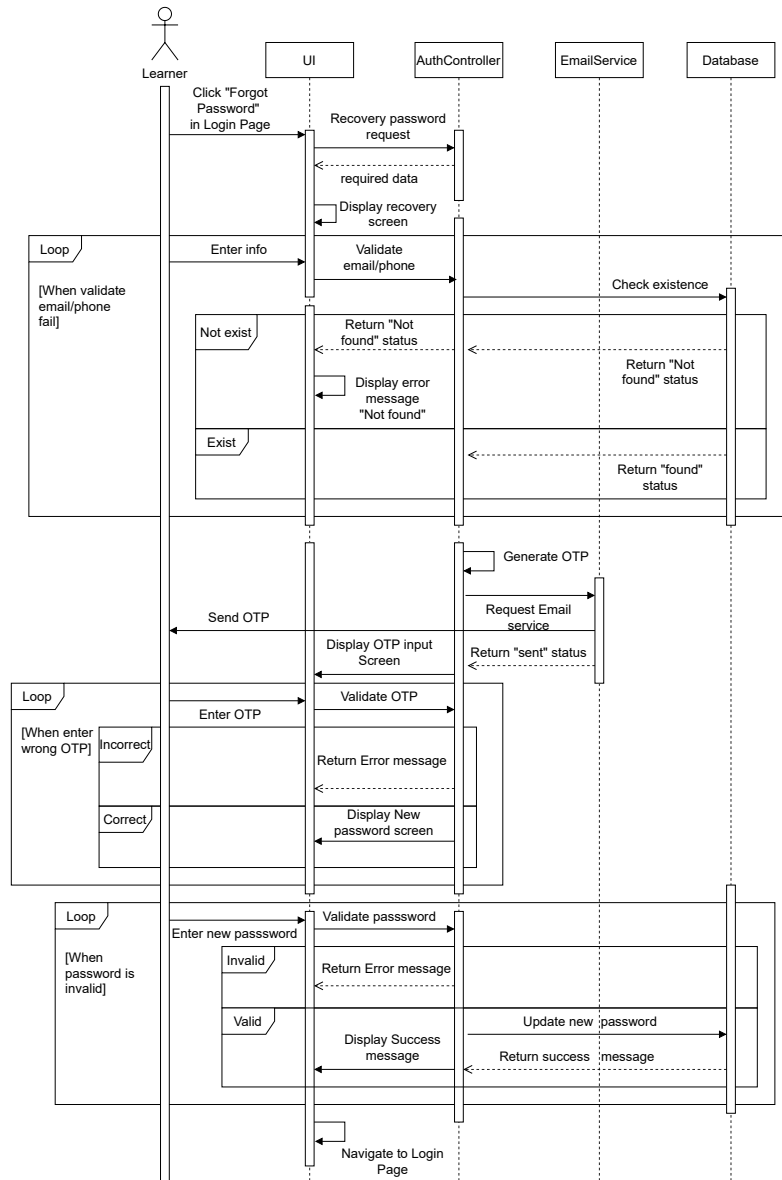


Figure 14: *Sequence Diagram for Password Reset*

This sequence diagram in Figure 14 illustrates the complete password recovery workflow, including user verification, OTP validation, and password update. The interaction is divided into

five lifelines:

- **Learner:** Initiates the password reset flow, enters email/phone, inputs the OTP, and sets a new password.
- **UI:** Displays the appropriate screens during each step and forwards user inputs to the backend.
- **AuthController:** Handles validation, OTP requests, and new password submission.
- **EmailService:** Sends the OTP email to the learner.
- **Database:** Stores user account data and updates the password.

The sequence progresses through the following major stages:

1. **Recovery initiation** The learner selects “Forgot Password” on the Login page. The UI requests recovery information from the *AuthController*, which returns the necessary data to display the recovery screen.
2. **Email/phone validation** The learner enters their email or phone number. The *AuthController* forwards the request to the *EmailService*, which checks the user’s existence in the *Database*.
 - If the account does not exist, an error (“Not found”) is returned and displayed.
 - If the account exists, the flow continues.
3. **OTP generation and delivery** After the account is validated, the *AuthController* generates an OTP and requests the *EmailService* to send it. The UI then displays the OTP input screen.
4. **OTP verification (Loop)** The learner enters the OTP.
 - If the OTP is incorrect, an error message is shown and the learner is prompted to try again.
 - If correct, the UI displays the “New Password” screen.
5. **New password submission (Loop)** The learner enters a new password. The *AuthController* validates the password format:
 - If invalid, an error is returned and the learner must re-enter the password.
 - If valid, the new password is saved into the *Database*.
6. **Completion** Upon successful update, a success message is displayed and the UI navigates the learner back to the Login page.

Overall, the diagram highlights three repeated validation loops: (1) validating email/phone, (2) validating OTP, and (3) validating the new password, ensuring the password reset flow is secure and error-tolerant.

4.4.2 Browse & Bookmark Content

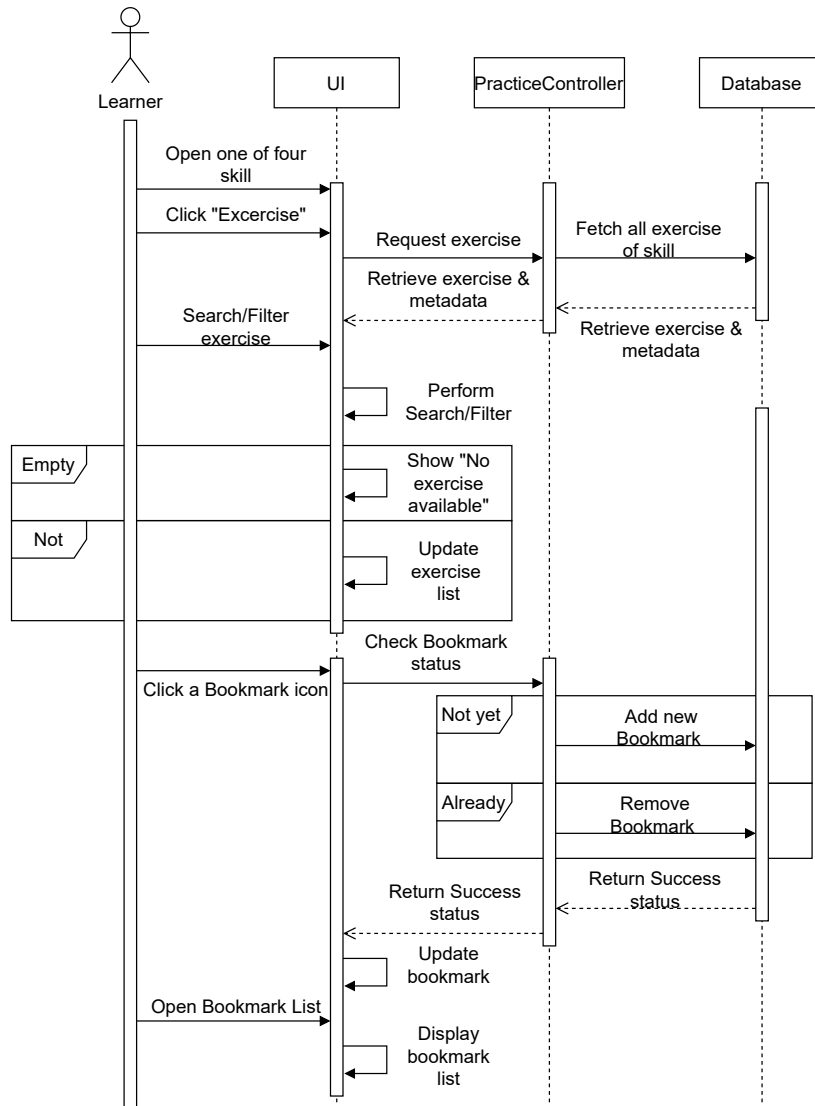


Figure 15: *Sequence Diagram for Browsing and Bookmarking Content*

This sequence diagram in Figure 15 describes the workflow for browsing learning content and bookmarking items for later access. The interaction involves four main lifelines:

- **Learner:** Browses the content list, views item details, and performs bookmarking actions.
- **UI:** Displays content lists and details, and forwards user actions to the backend.

- **ContentController:** Handles content retrieval and bookmark submission requests.
- **Database:** Stores content information and bookmark records for each learner.

The sequence progresses through the following stages:

1. **Content browsing**

The learner navigates to the “Browse Content” screen. The UI requests the content list from the *ContentController*, which retrieves the data from the *Database* and sends it back for display.

2. **Viewing item details**

When the learner selects a specific content item, the UI sends a request for its details. The *ContentController* fetches the required information from the *Database*, allowing the UI to present the full description.

3. **Bookmarking**

When the learner taps the “Bookmark” option, the UI forwards the action to the *ContentController*. The controller updates the bookmark information in the *Database*. If the item was already bookmarked, the system may return an appropriate status message.

4. **Status update**

After the bookmark operation completes, the UI updates the visual state (e.g., highlight bookmark icon) to confirm the action to the learner.

Overall, the diagram highlights the interaction between browsing and personalizing learning content, ensuring that learners can conveniently save important materials for future reference.

4.4.3 Preview & Take Practice

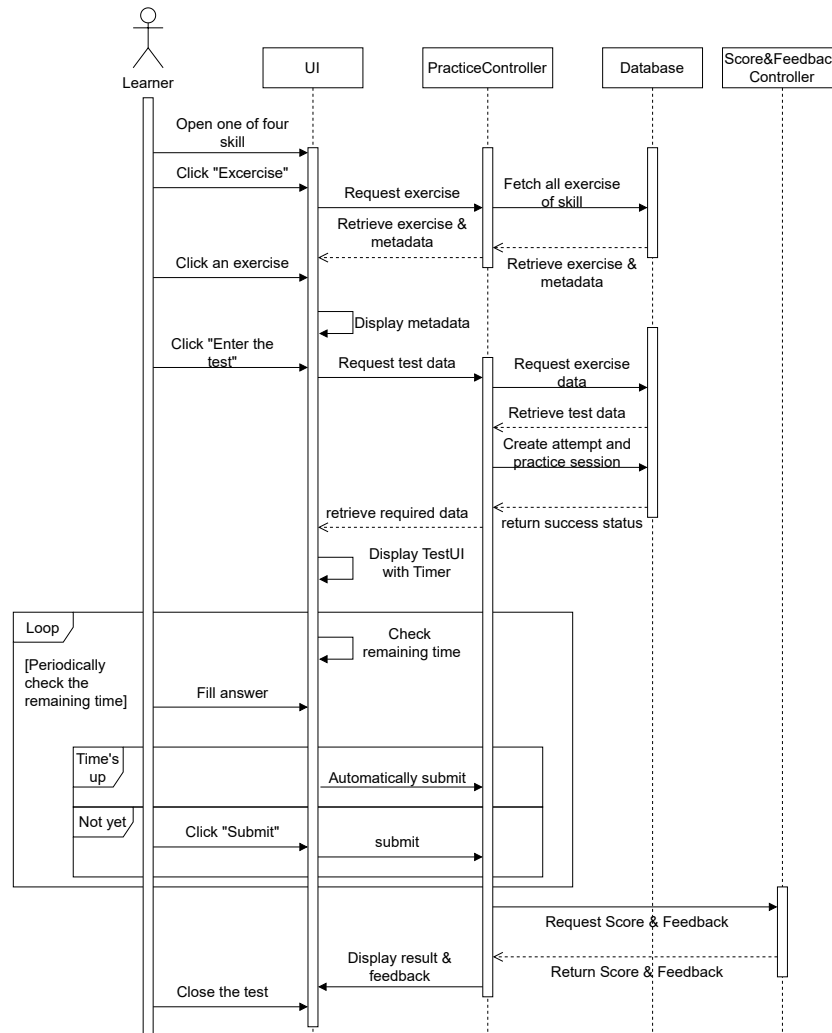


Figure 16: *Sequence Diagram for Preview and Take Practice*

This sequence diagram in Figure 16 illustrates the full workflow for reviewing learning content and taking a practice exercise. The interaction includes five key lifelines:

- **Learner:** Reviews learning materials, answers practice questions, and submits the final response.
- **UI:** Presents content, displays questions, collects answers, and forwards practice submissions.
- **PracticeController:** Retrieves questions, validates answers, and computes the final score.

- **ContentController:** Provides learning material when the learner enters the review stage.
- **Database:** Stores content, questions, correct answers, and user practice results.

The workflow proceeds through several phases:

1. **Reviewing content**

The learner selects a lesson to review. The UI requests the lesson data from the *ContentController*, which queries the *Database* and returns the content for display.

2. **Starting a practice session**

The learner chooses the “Take Practice” option. The UI sends a request to the *PracticeController*, which retrieves the relevant set of questions from the *Database*.

3. **Answering questions (Loop)**

For each question, the learner selects an answer. The UI displays each question sequentially, allowing the learner to proceed through the entire set.

4. **Submitting answers**

After completing all questions, the learner submits the practice. The UI sends the collected answers to the *PracticeController*, which validates them against the correct answers stored in the *Database* and computes the final score.

5. **Displaying results**

The calculated score is returned to the UI, which displays the results to the learner. The system may also store the attempt record in the *Database*.

Overall, the diagram emphasizes the two-part process of reviewing content and completing practice exercises, ensuring a consistent learning flow and accurate assessment of the learner’s understanding.

4.4.4 Create User Account

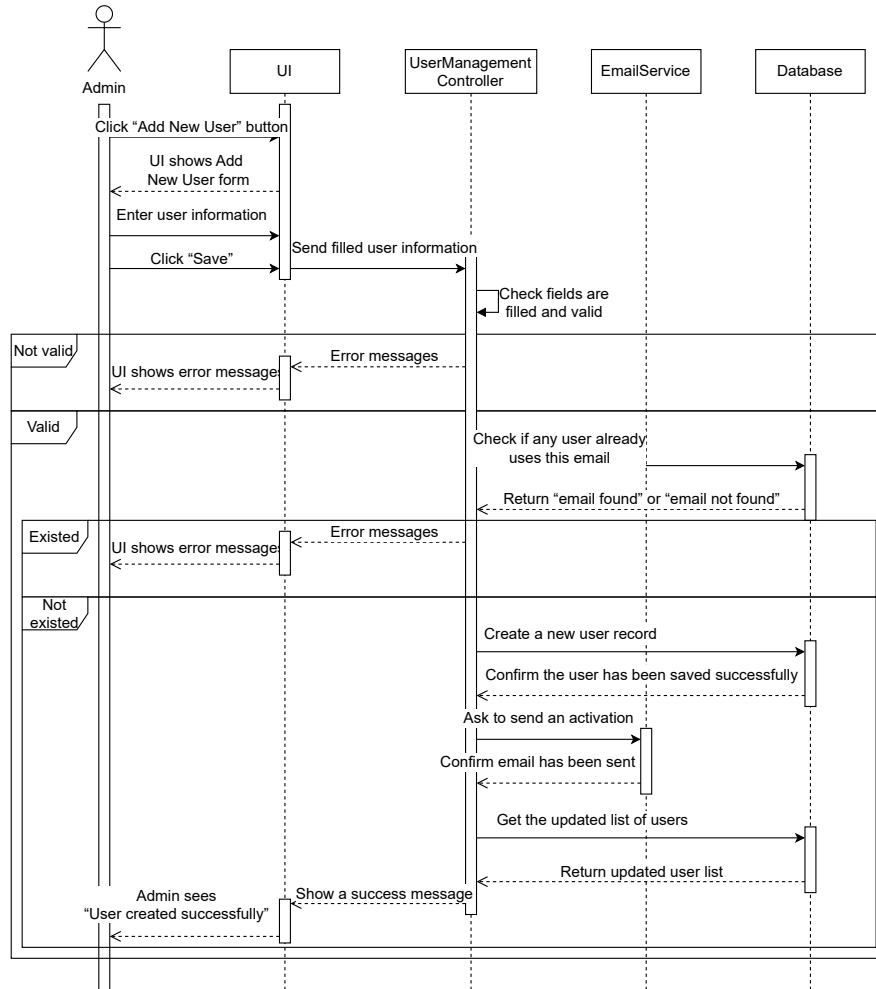


Figure 17: Sequence Diagram for Create User Account

This sequence diagram in Figure 17 describes the workflow for creating a new user account within the system's User Management module. The interaction involves four main lifelines:

- Admin: Initiates the user creation process by accessing the User Management page and submitting new user information.
- UI: Displays the user interface elements (forms, messages) and forwards the admin's actions to the backend.
- UserManagementController: Handles form validation, email uniqueness checks, user creation logic, and optional email notifications.
- Database: Stores user records and performs lookup operations such as email existence checks.

- EmailService: Sends activation or welcome emails to newly created users.

The sequence progresses through the following stages:

1. Accessing the Add User interface

The admin navigates to the User Management page and selects the “Add New User” option. The UI requests the necessary role and default status information from the UserManagementController, which retrieves these values from the Database and then displays the Add New User form to the admin.

2. Submitting user information

After entering the required details (name, email, role, status), the admin submits the form. The UI forwards the completed information to the UserManagementController, which performs field validation. If any required fields are missing or invalid, validation errors are returned to the UI, prompting the admin to correct the input.

3. Email uniqueness verification

For valid input, the controller checks whether the provided email already exists by querying the Database. If the email is found, an error message is returned to the UI, informing the admin that the email is already in use. If the email does not exist, the controller proceeds to create a new user account.

4. Creating the new user record

The UserManagementController requests the Database to store the new user record. After successful insertion, the controller may ask the EmailService to send an activation or welcome email to the newly created user. The EmailService confirms once the email has been sent.

5. Updating the UI and confirming success

The controller retrieves the updated list of users from the Database and sends it to the UI. Finally, the UI displays a success message indicating that the user has been created successfully, along with the refreshed user list.

Overall, the diagram illustrates how the system coordinates validation, data storage, and optional email notifications to ensure the user creation process is handled reliably and interactively for the administrator.

4.4.5 Take Evaluation Test & Generate Study Plan

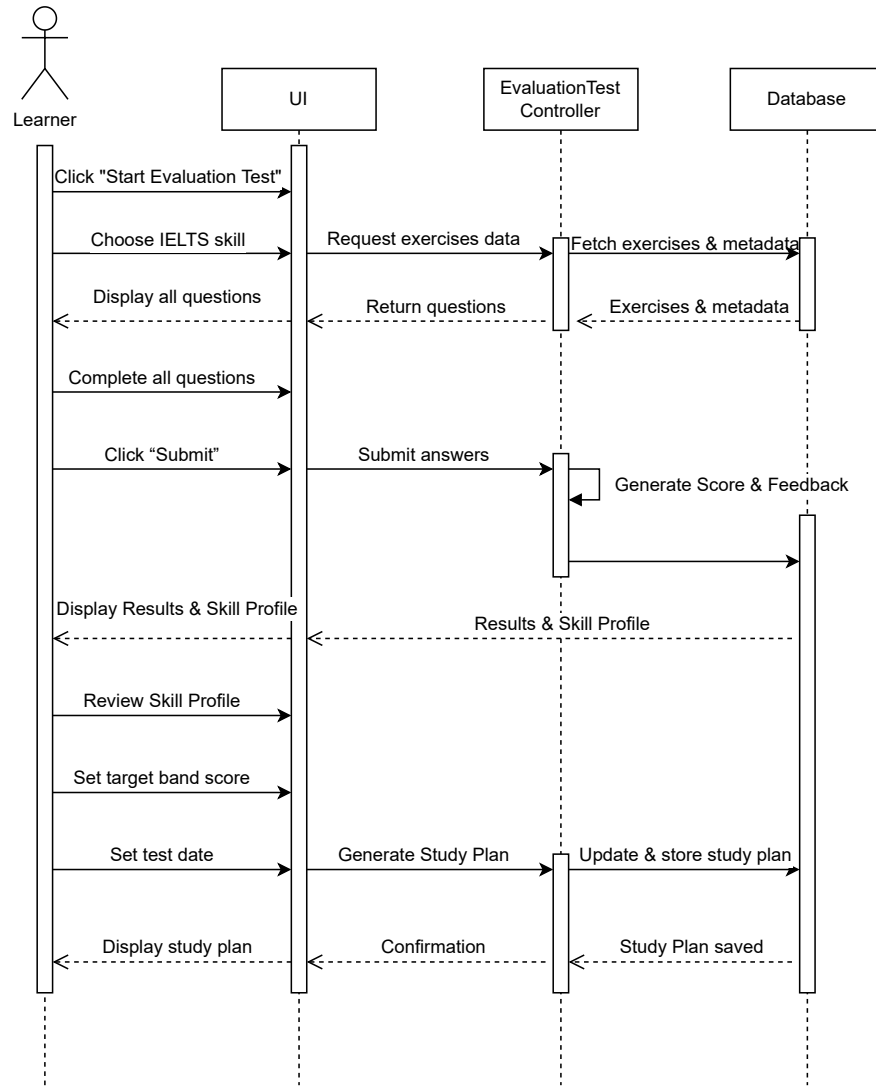


Figure 18: *Sequence Diagram for Take Evaluation Test & Generate Study Plan*

This sequence diagram in Figure 18 describes the workflow for taking an evaluation test and generating a personalized study plan in the IELTS Mastermind system. The interaction involves four primary lifelines:

- **Learner:** Initiates the evaluation test, completes all questions, reviews results, and sets learning goals.
- **UI:** Presents the evaluation test interface, displays questions and results, collects learner responses, and forwards requests to the backend.

- **EvaluationTestController:** Manages test execution, evaluates submitted answers, generates scores and feedback, updates skill profiles, and creates personalized study plans.
- **Database:** Stores evaluation exercises, test results, learner skill profiles, and generated study plans.

The sequence progresses through the following stages:

1. **Starting the evaluation test**

The learner initiates the evaluation test and selects an IELTS skill. The UI sends a request to the EvaluationTestController, which retrieves all relevant exercises and metadata from the Database and returns them to the UI for display.

2. **Completing and submitting the test**

The learner answers all questions and submits the test. The UI forwards the learner's responses to the EvaluationTestController, which validates and evaluates the answers.

3. **Generating results and feedback**

Based on the evaluation, the controller generates a band score and automated feedback. It then updates or creates the learner's skill profile and stores the results in the Database.

4. **Displaying results and reviewing skill profile**

The controller returns the evaluation results to the UI, which displays the band score, feedback, and updated skill profile to the learner for review.

5. **Setting goals and generating a study plan**

The learner sets a target band score and selects a test date. The UI sends this information to the EvaluationTestController, which generates a personalized study plan and stores it in the Database.

6. **Completion and confirmation**

After the study plan has been successfully saved, the controller sends a confirmation response to the UI, and the UI displays the generated study plan to the learner.

In summary, the diagram illustrates how the system integrates evaluation, feedback, and goal-setting mechanisms to provide learners with a data-driven and personalized study plan.

4.4.6 Add Practice Content

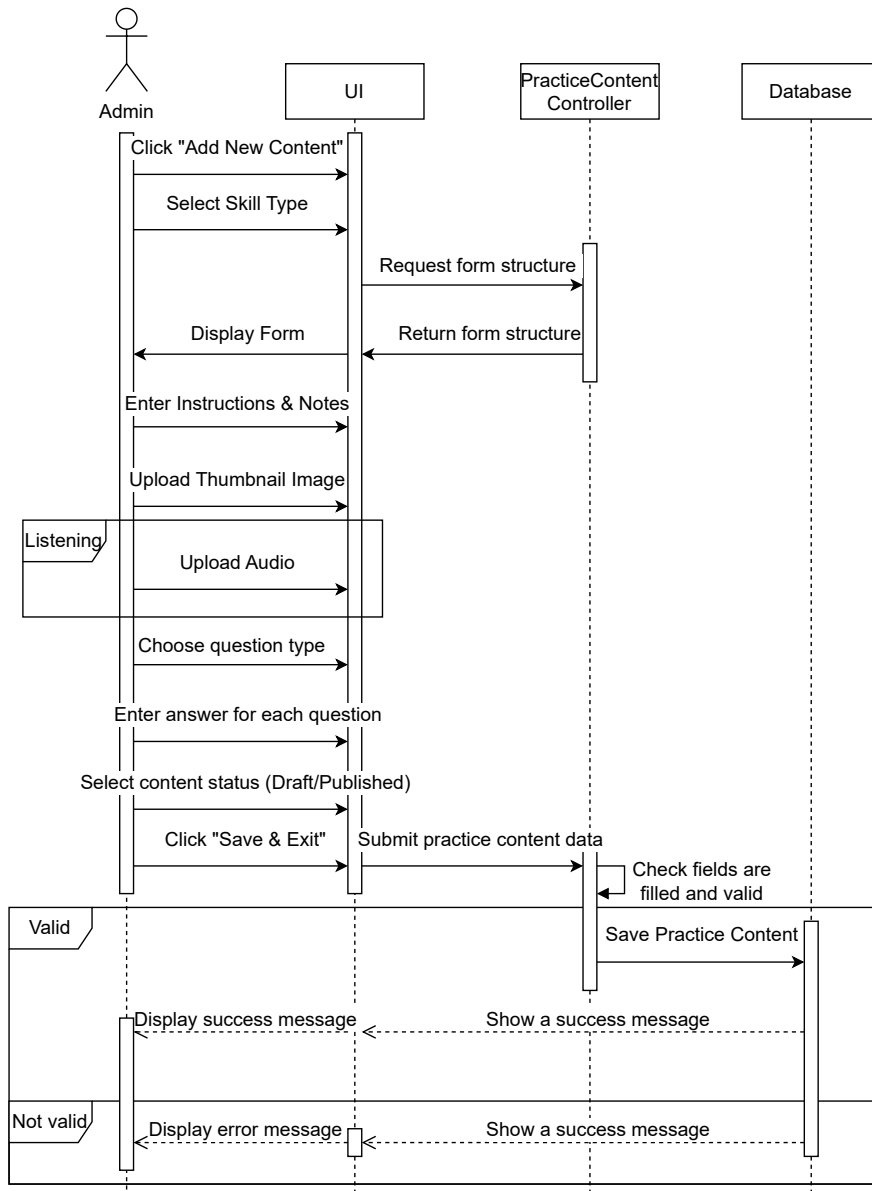


Figure 19: *Sequence Diagram for Add Practice Content*

This sequence diagram in Figure 19 describes the workflow for adding new practice content in the IELTS Mastermind system. The interaction involves four main lifelines:

- **Admin:** Initiates the content creation process by selecting the option to add new practice content and providing all required information.

- **UI:** Displays the content creation interface, dynamically renders the input form based on the selected skill type, collects the administrator's inputs, and shows validation errors or success messages.
- **PracticeContentController:** Handles requests from the UI, returns the appropriate form structure, validates submitted content, applies business rules, and coordinates saving the practice content.
- **Database:** Stores the practice content, including metadata, questions, media references, and publication status.

The sequence progresses through the following stages:

1. Initiating content creation

The administrator selects the "Add New Content" option and chooses a specific skill type. The UI requests the corresponding form structure from the PracticeContentController, which returns the appropriate configuration. The UI then displays the generated form to the administrator.

2. Entering content details

The administrator enters instructions and notes, uploads a thumbnail image, and, for listening content, uploads an audio file. The UI collects these inputs and allows the administrator to configure question types and correct answers.

3. Saving questions and setting content status

After completing the question configuration, the administrator saves the question and selects the content status (draft or published). The administrator then chooses the "Save & Exit" option to submit the full content information.

4. Validation and content persistence

The UI sends the submitted data to the PracticeContentController for validation. If the input is invalid, the controller returns error messages to the UI, which are displayed to the administrator. If the input is valid, the controller requests the Database to save the practice content.

5. Completion and navigation

Once the content has been successfully stored, the controller sends a confirmation response to the UI. The UI then redirects the administrator to the Practice Content Management page, indicating that the operation has completed successfully.

Overall, the diagram demonstrates how the system manages dynamic form generation, validation, and data persistence to ensure that practice content is created accurately and efficiently by the administrator.

4.5 Class Diagram

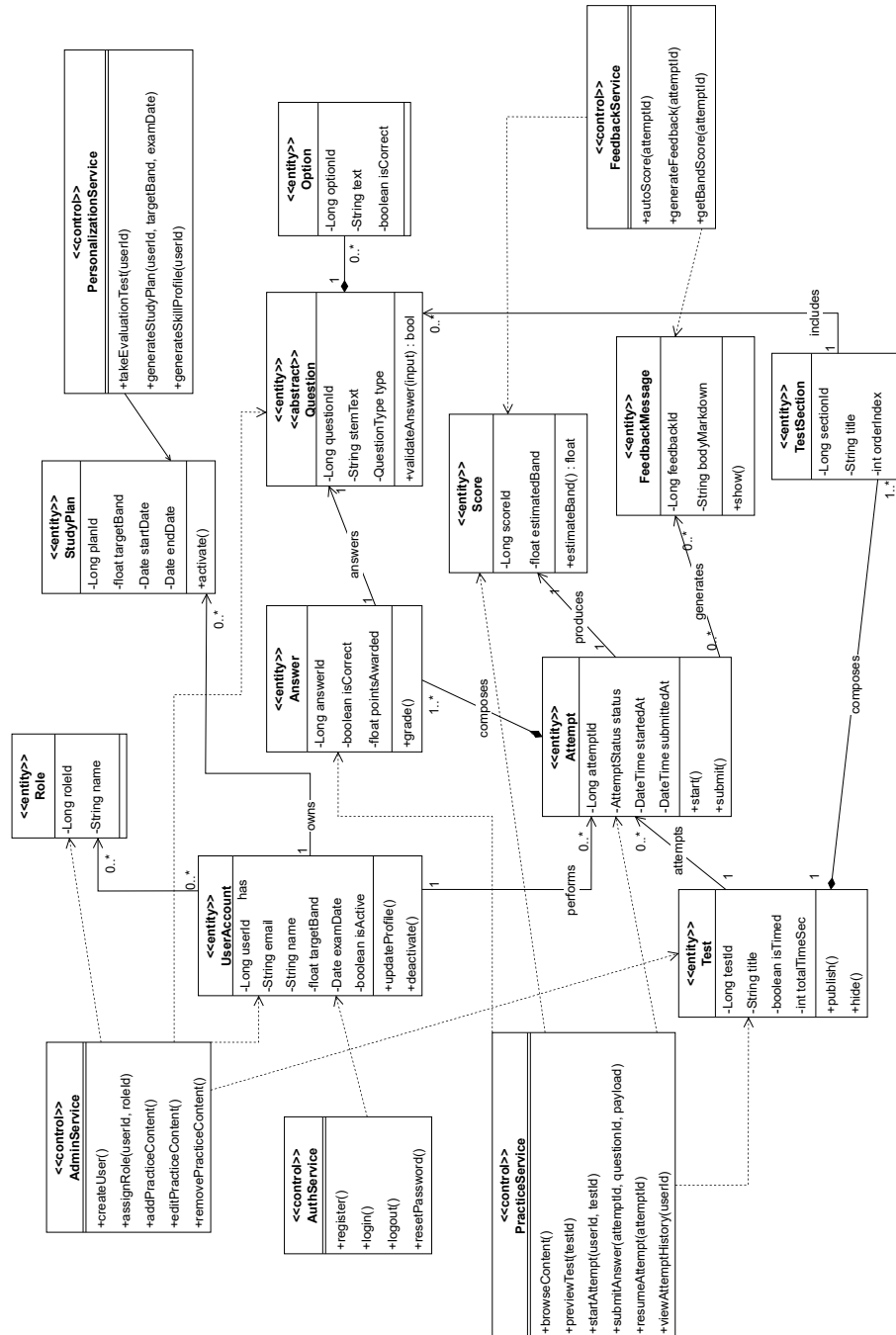


Figure 20: Class Diagram for IELTS Mastermind

The class diagram in Figure 20 illustrates the main entities in the system and the services that operate on them. The entities represent the data that the system needs to store, such as users, practice tests, questions, attempts, scores, study plans, and feedback. The services represent the main features and show how the system uses those entities to deliver functions.

For entities, `UserAccount` stores user information and includes basic actions like updating profile and deactivating the account. Users can have `Role` to support administrator permissions. Practice content is represented by `Test` and `TestSection`, where a `Test` is made up of multiple sections. Each section contains multiple `Question` objects (`Question` is abstract because it can represent different question types), and a question can have multiple `Option` objects for selectable answers.

User learning activity is captured through `Attempt`, `Answer`, and `Score`. A user performs many attempts, each attempt contains multiple answers, and each answer links back to the question being answered. After an attempt is submitted, the system produces a `Score` with an estimated band result. The system also supports planning and guidance with `StudyPlan`, which belongs to the user and stores target band and study dates. Finally, `FeedbackMessage` stores feedback content that can be generated after attempts and shown to the user.

The services in the diagram map directly to the system modules. `AuthService` supports registration, login/logout, and password reset. `PracticeService` covers the practice workflow: browsing content, previewing a test, starting an attempt, submitting answers, resuming an attempt, and viewing attempt history. `PersonalizationService` supports evaluation tests and generates a study plan. `FeedbackService` runs auto-scoring, generates feedback, and returns band score results for an attempt. `AdminService` supports administrator tasks such as creating users, assigning roles, and managing practice content.

5 System Architecture

5.1 Architectural Style Selection

This section presents the rationale behind the architectural style selection for IELTS Mastermind. The architectural decision is made in a structured and incremental manner. First, monolithic and distributed architectural approaches are analyzed and compared based on the system's non-functional requirements. Next, the selected architectural approach is further examined by evaluating different architectural styles to identify the most suitable internal organization for the system.

5.1.1 Monolithic Architecture

A monolithic architecture structures the system as a single deployable unit where all functional modules are developed, deployed, and executed together, typically sharing a single database.

Advantages

- Low communication latency, since components interact through in-process calls.
- Simpler deployment and release process with a single executable artifact.
- Easier testing and debugging due to a single runtime environment.
- Lower infrastructure and operational cost.
- Well suited for small teams and moderate system scale.

Disadvantages

- As the system grows, the codebase can become harder to maintain.
- Limited ability to scale individual components independently.
- Any change requires redeploying the entire application.

5.1.2 Distributed Architecture

A distributed architecture decomposes the system into multiple independent services that communicate through network-based APIs. Each service can be developed, deployed, and scaled independently.

Advantages

- Independent scalability of services.
- Improved fault isolation.
- Flexibility in choosing different technologies per service.

Disadvantages

- Increased latency due to network communication.
- Higher operational complexity, including service orchestration, monitoring, and logging.
- More complex testing, debugging, and deployment pipelines.
- Higher infrastructure and development cost.
- Requires strong expertise in distributed systems.

5.1.3 Architecture Choice

Based on the non-functional requirements defined in Section 3.2, monolithic architecture is selected for IELTS Mastermind.

- **Responsiveness:** A monolithic system minimizes latency by handling requests through fast in-process method calls rather than network communication.
- **Reliability:** Reliability is easier to achieve with a single deployable unit, simplifying deployment, rollback, and recovery processes.
- **Overall cost:** Overall cost is kept low due to reduced infrastructure requirements and operational overhead.

Given the current scale of the system, the expected number of concurrent learners, and the limited development budget, the benefits of a distributed architecture do not outweigh its added complexity. Therefore, a monolithic architecture is the most appropriate choice at this stage.

5.2 Comparison of Monolithic Architectural Styles

After selecting monolithic architecture, different monolithic architectural styles are evaluated based on their suitability for IELTS Mastermind.

5.2.1 Layered Architecture

The layered architecture organizes the system into technical layers such as presentation, business logic, and data access.

- Promotes clear separation of concerns.
- Improves maintainability and readability of the codebase.
- Well suited for CRUD-oriented, web-based information systems.
- Widely adopted and well understood, reducing development risk.

5.2.2 Pipeline Architecture

The pipeline (pipes-and-filters) architecture organizes processing as a sequence of transformation steps, where each stage processes data and passes the result to the next stage.

- Best suited for data processing, batch jobs, and streaming systems.
- Less suitable for interactive, request-response web applications.
- Not aligned with the user-driven workflows of an IELTS learning platform.

5.2.3 Microkernel Architecture

The microkernel architecture consists of a minimal core system with additional functionality implemented as plug-in modules.

- Suitable for product-based systems requiring high extensibility.
- Supports frequent addition or replacement of features as plug-ins.
- Introduces additional architectural complexity.
- Not necessary for IELTS Mastermind, which has a well-defined and stable core feature set.

5.2.4 Comparison

The criteria presented in Table 18 are primarily derived from the system's key non-functional requirements, including responsiveness, reliability, and overall cost. In addition, development complexity is included as a secondary qualitative attribute to reflect the practical impact of each architectural style on implementation effort and long-term maintenance. These criteria are used to evaluate how each monolithic architectural style supports the system's priorities.

Criteria	Layered Architecture	Pipeline Architecture	Microkernel Architecture
Primary focus	Technical separation of concerns	Sequential data processing	Core system with extensible plug-ins
Responsiveness	High – in-process calls and request-response handling	Medium – sequential stage-based processing	Medium – plug-in invocation overhead
Reliability	High – clear layer boundaries simplify fault isolation	Medium – failure in one stage may block pipeline	Medium – depends on plug-in stability
Overall cost	Low – simple structure and low overhead	Medium – coordination and design effort	Medium-High – higher development and maintenance effort
Development complexity	Low – widely used and well understood	Medium – careful pipeline design required	High – complex core-plug-in coordination

Table 18: *Comparison of Monolithic Architectural Styles*



5.3 Chosen Architecture

Based on the evaluation of monolithic architectural styles and the comparison summarized in Table 18, layered architecture is identified as the most suitable monolithic style for IELTS Mastermind. It provides high responsiveness through in-process request handling, strong reliability due to clear separation of concerns, and low overall cost resulting from its simplicity and low operational overhead.

Pipeline architecture is not appropriate because it is designed for data-flow processing rather than interactive web applications. Microkernel architecture introduces unnecessary complexity and cost for a system with a stable and well-defined core feature set.

5.4 Architecture Design

5.4.1 Architecture Diagram

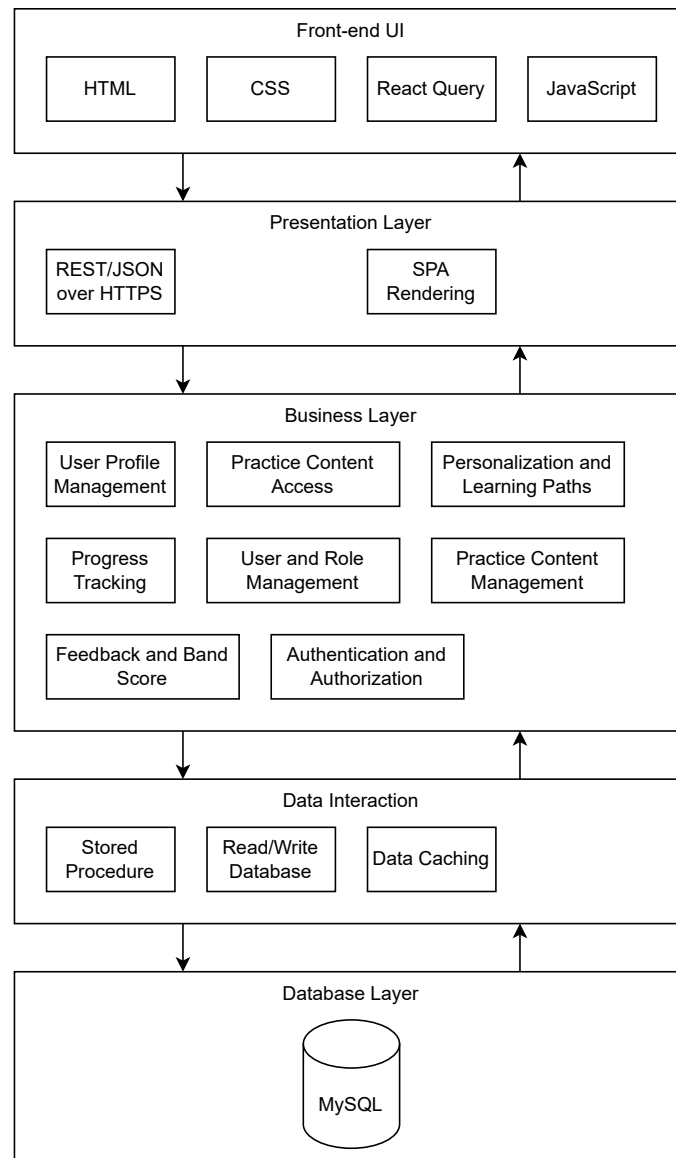


Figure 21: *Architecture Diagram of the System*



The Architecture Diagram in Figure 21 illustrates the overall structure of the IELTS Mastermind System using a layered approach. The system starts with the Front-end UI, built with HTML, CSS, JavaScript, and React Query, which provides an interactive single-page application for users. Requests from the user interface are handled by the presentation layer through REST/JSON APIs over HTTPS, ensuring secure communication. The core system functionalities are managed in the application layer, which includes modules such as authentication and authorization, practice content access, personalization and learning paths, feedback and band score, progress tracking, and user and content management. Data is then processed through the data interaction layer, which handles database operations, caching, and stored procedures. Finally, all data is stored in the database layer using MySQL.

5.4.2 Component Diagram

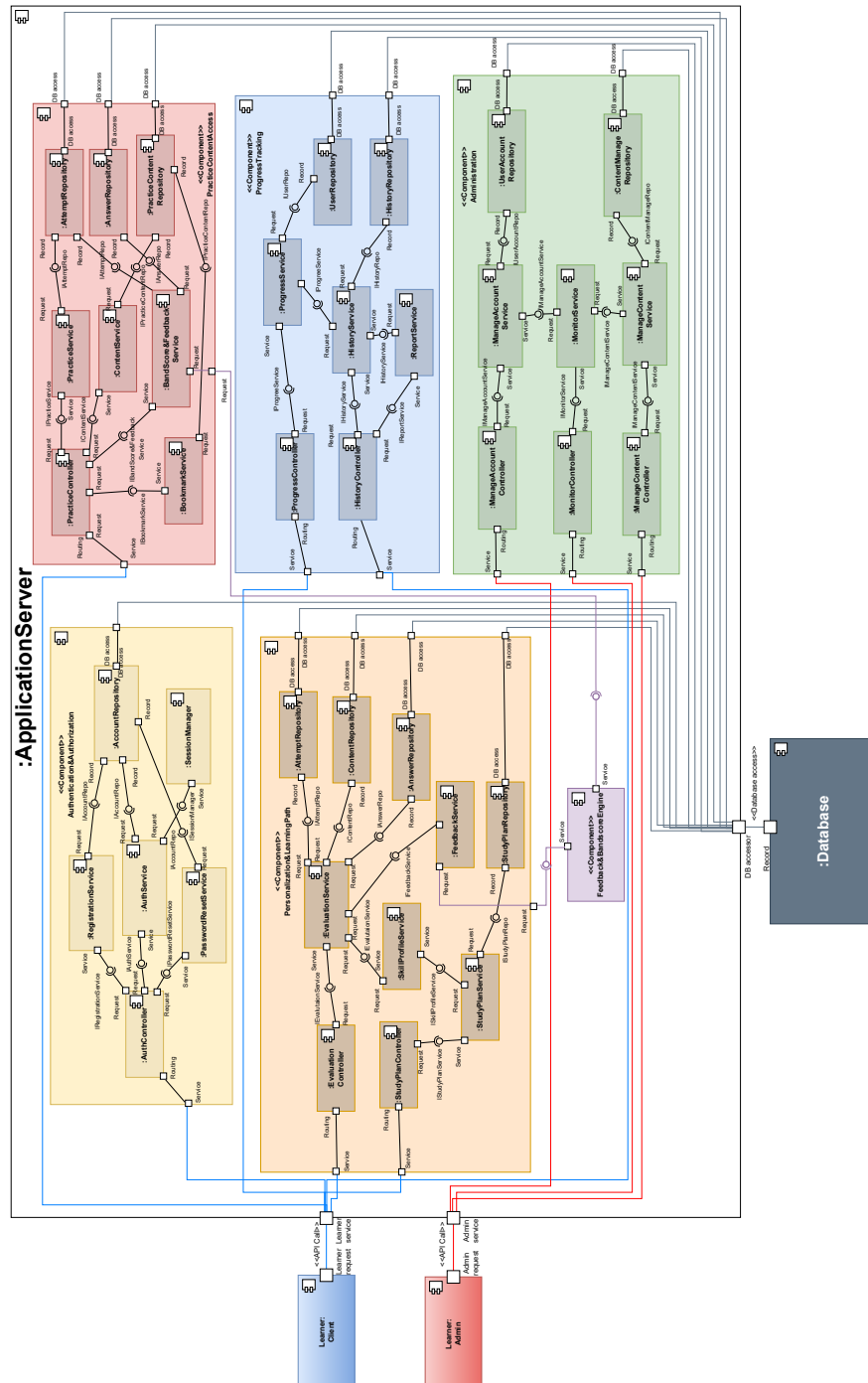


Figure 22: Component Diagram of the System

The Component Diagram in Figure 22 illustrates the overall architecture of the IELTS Mastermind system, which is designed using a modular architecture. In this design, each component represents an independent functional module, encapsulating its own controllers, services, and repositories. The modules communicate with each other through well-defined APIs and interfaces, ensuring loose coupling and high maintainability.

The Authentication & Authorization module is responsible for user identity management, including login, registration, session management, and password reset. It ensures secure access control across the entire system.

The Practice Content Access module handles core learning activities, such as accessing practice materials, submitting answers, bookmarking content, and coordinating with scoring and feedback services to deliver results to learners.

The Feedback & Bandscore Engine module focuses on evaluating user responses and generating band scores and detailed feedback, serving as a central evaluation engine within the system.

The Personalization & Learning Path module supports adaptive learning by evaluating learner performance, building skill profiles, and generating personalized study plans based on historical data and assessment results.

The Progress Tracking module records learning activities, tracks user progress, manages attempt history, and generates reports that help learners monitor their improvement over time.

The Administration module provides system management capabilities for administrators, including user account management, content management, and system monitoring.

Two main user roles, Learner and Admin, interact with the system through client applications by sending requests to the appropriate modules. All modules persist and retrieve data from a shared Database through their respective repositories. This modular design enhances scalability, flexibility, and ease of future extension of the IELTS Mastermind system.

5.5 Data Flow Diagram

5.5.1 Context Diagram

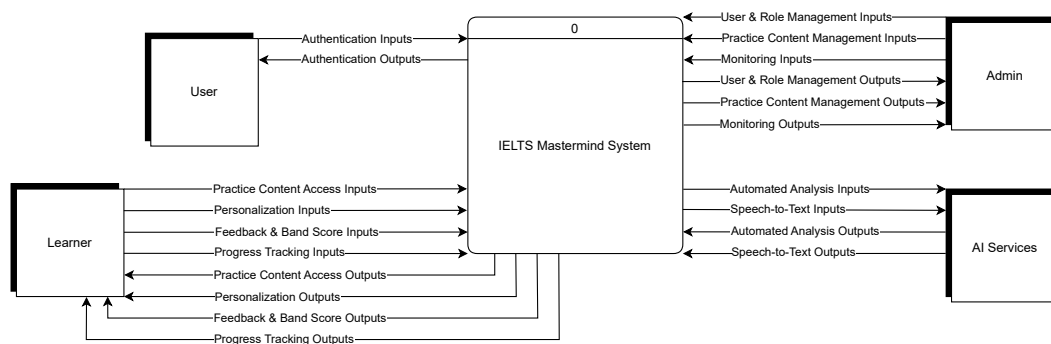


Figure 23: Context Data Flow Diagram of the System



The Context Data Flow Diagram in Figure 23 presents an overall view of how the IELTS Mastermind System interacts with entities. The main system is shown as a single process that exchanges data with four entities: User, Learner, Admin, and AI Services. Users interact with the system through authentication inputs and outputs, while learners provide inputs related to practice content access, personalization, feedback and band scores, and progress tracking, and receive corresponding results from the system. Admins manage the system by sending inputs for user and role management, practice content management, and monitoring, and receive management-related outputs. The system also communicates with AI Services to handle tasks such as automated analysis and speech-to-text processing, receiving analysis results in return.

5.5.2 Diagram 0

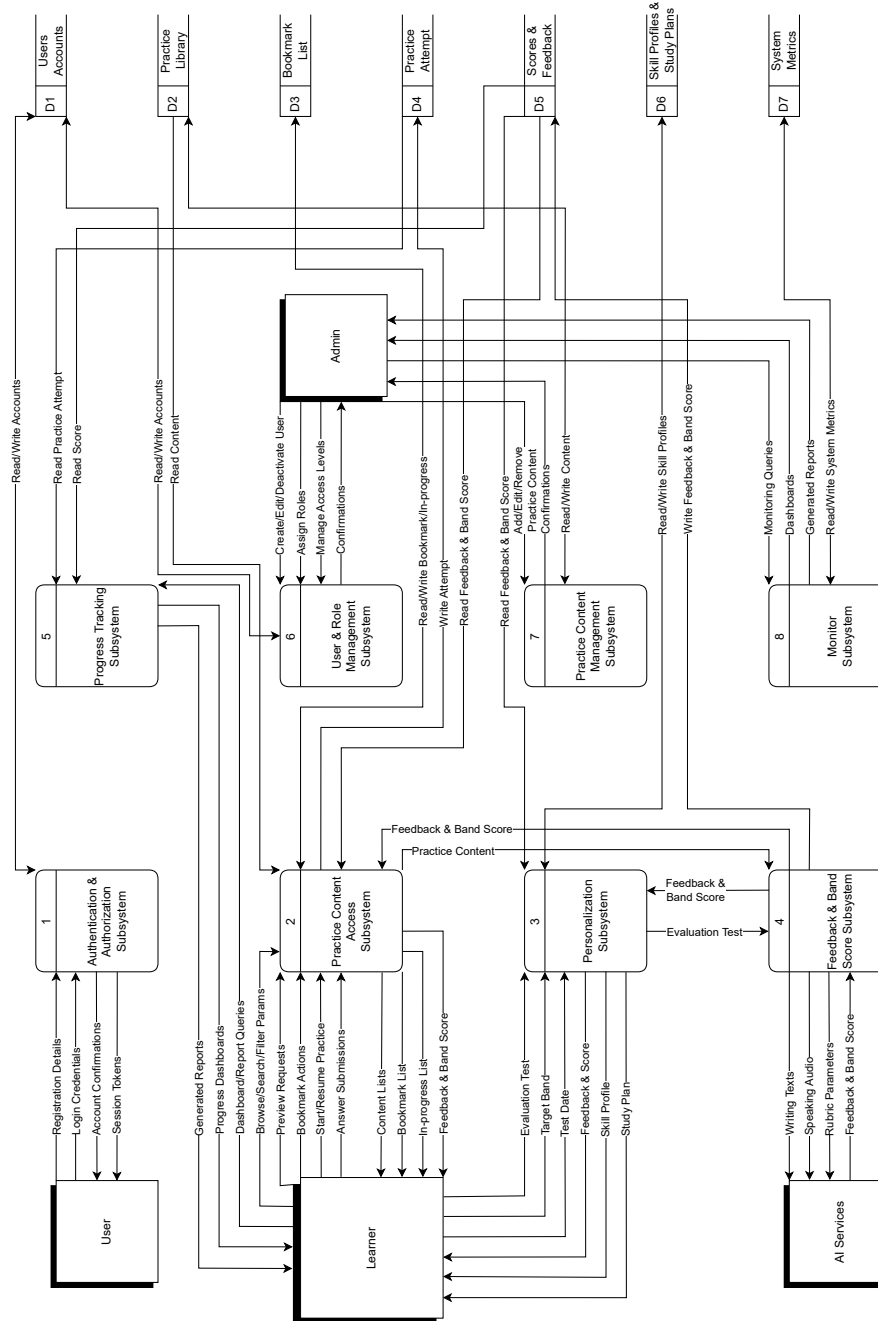


Figure 24: Level-0 Data Flow Diagram of the System

The Level 0 Data Flow Diagram in Figure 24 illustrates the internal structure of the IELTS Mastermind System by breaking it down into its main subsystems and showing how data flows between them. The system includes key subsystems such as Authentication and Authorization, Practice Content Access, Personalization, Feedback and Band Score, Progress Tracking, User and Role Management, Practice Content Management, and Monitoring. Different actors, including User, Learner, Admin, and AI Services, interact with these subsystems by sending inputs and receiving outputs such as authentication results, practice content, feedback, band scores, progress data, and reports.

5.5.3 Authentication and Authorization Module

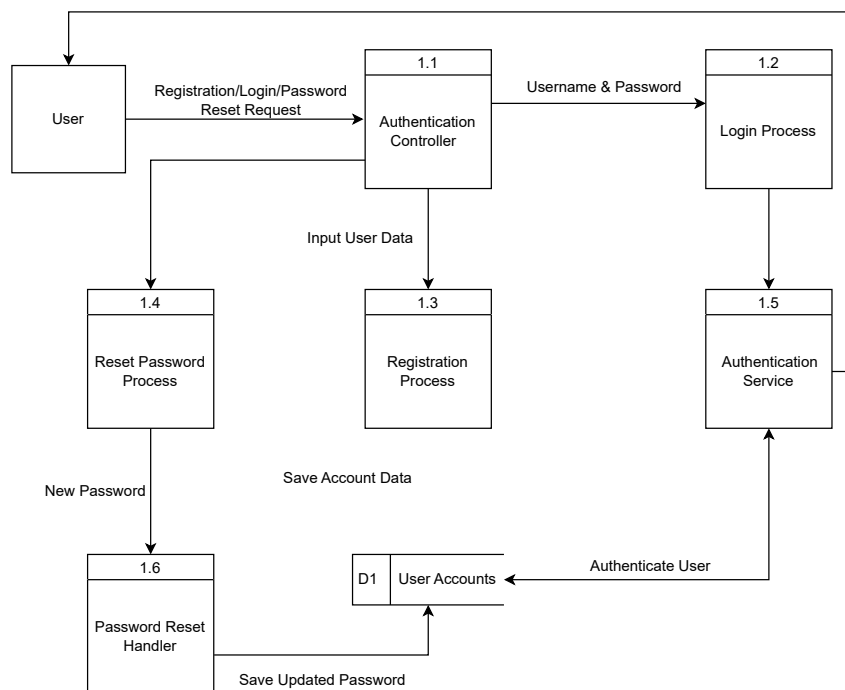


Figure 25: Level-1 Data Flow Diagram of Authentication and Authorization Module

The Level 1 Data Flow Diagram in Figure 25 for the Authentication and Authorization Module shows how user access requests are processed in the IELTS Mastermind system. The User can perform actions such as registration, login, and password reset, which are handled by the Authentication Controller. During registration, user data is processed and stored in the User Accounts data store. For login, the system verifies the username and password through the Login Process and Authentication Service to authenticate the user. In the case of a password reset, the system processes the request, updates the new password, and saves it securely in the user accounts database.

5.5.4 Practice Content Access Module

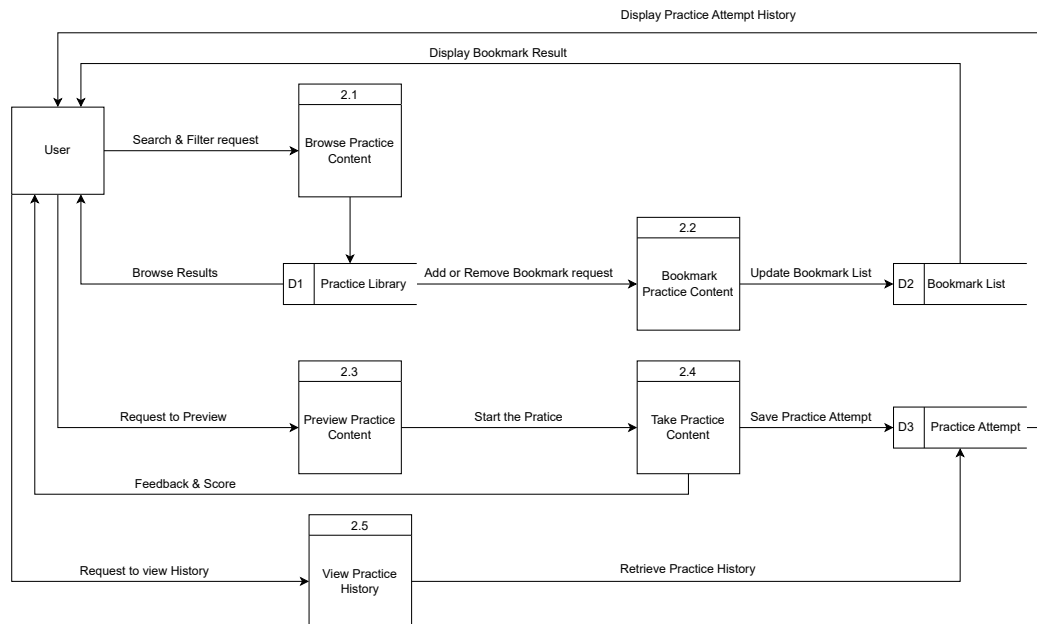


Figure 26: *Level-1 Data Flow Diagram of Practice Content Access Module*

The Level 1 Data Flow Diagram in Figure 26 for the Practice Content Access Module explains how users interact with practice materials in the IELTS Mastermind system. Users can search and filter practice content, which is processed by the Browse Practice Content function using data from the Practice Library. Users may preview practice content before starting and can choose to bookmark items, with bookmark data stored in the Bookmark List. When users take a practice session, their attempts are saved in the Practice Attempt data store, and they receive feedback and scores after completion. The diagram also shows how users can view their practice history, allowing them to review past attempts.

5.5.5 Personalization and Learning Paths Module

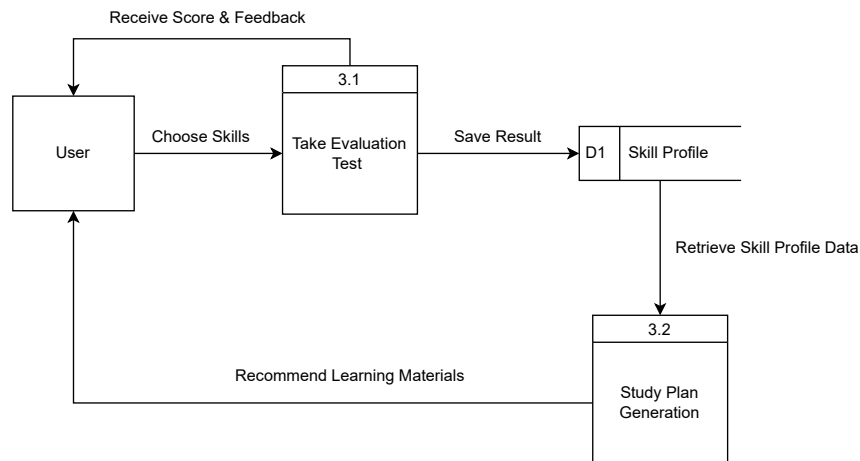


Figure 27: *Level-1 Data Flow Diagram of Personalization and Learning Paths Module*

The Level 1 Data Flow Diagram in Figure 27 for the Personalization and Learning Paths Module shows how the system creates a personalized study experience for users. The User begins by choosing skills and taking an evaluation test, after which the system processes the results and stores them in the Skill Profile data store. Based on the saved skill profile, the system generates a study plan by analyzing the user's strengths and weaknesses. The user then receives scores, feedback, and recommended learning materials that match their learning needs.

5.5.6 Feedback and Band Score Module

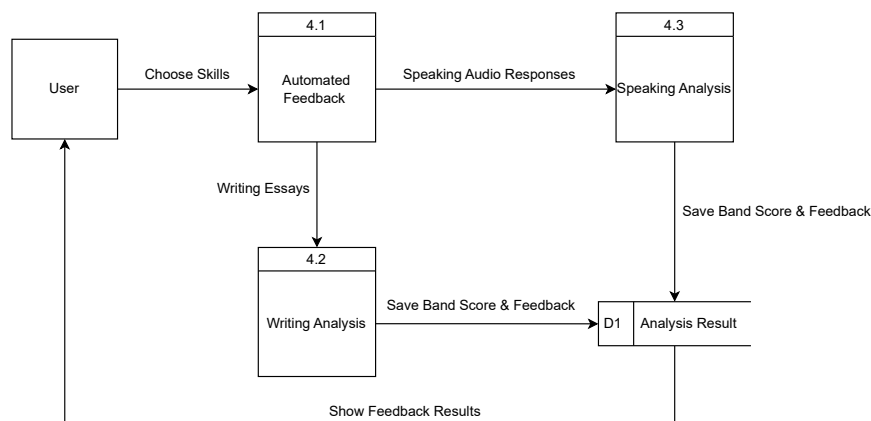


Figure 28: *Level-1 Data Flow Diagram of Feedback and Band Score Module*

The Level 1 Data Flow Diagram in Figure 28 for the Feedback and Band Score Module shows how the system evaluates user responses and provides feedback. The User selects the skills to be assessed and submits responses, such as writing essays or speaking audio. These inputs are processed through Writing Analysis and Speaking Analysis, where the system evaluates performance and generates band scores and feedback. The analysis results are then stored in the Analysis Result data store. Finally, the system returns the feedback results to the user.

5.5.7 User and Role Management Module

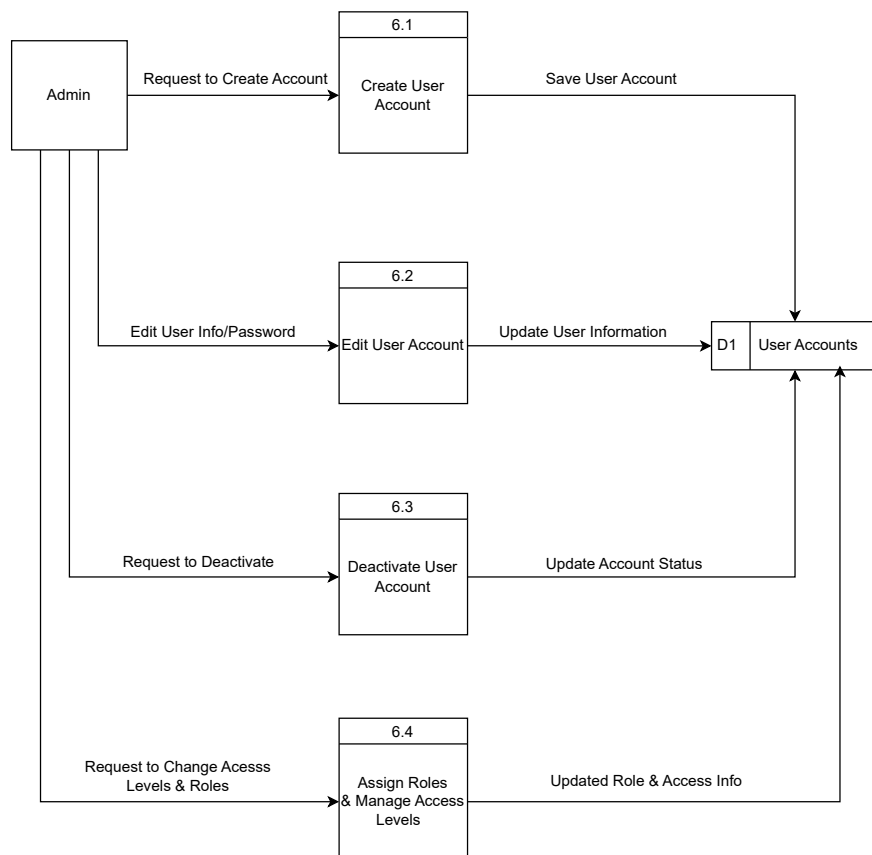


Figure 29: *Level-1 Data Flow Diagram of User and Role Management Module*

The Level 1 Data Flow Diagram in Figure 29 for the User and Role Management Module explains how an Admin manages user accounts in the IELTS Mastermind system. The admin can create new user accounts, edit user information or passwords, deactivate accounts, and assign roles and access levels. Each request is processed by the corresponding function and the updated information is stored in the User Accounts data store.

5.6 Deployment Diagram

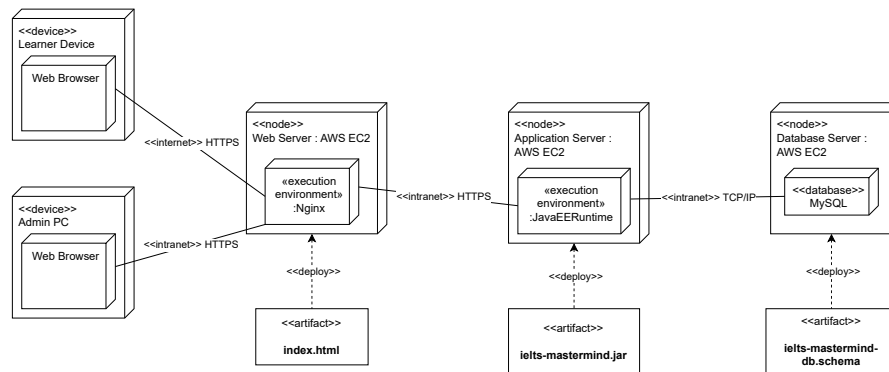


Figure 30: *Deployment Diagram of the System*

The deployment diagram in Figure 30 shows IELTS Mastermind as a simple web system deployed on AWS. Users access the system from two types of client devices (Learner Device and Admin PC) using a web browser. These clients connect to the system through the Internet using HTTPS.

Inside the cloud, the system is separated into three servers. The Web Server (AWS EC2) runs Nginx and hosts the frontend artifact. This layer handles incoming web requests and serves the website interface to users.

The Application Server (AWS EC2) runs a Java EE Runtime and deploys the backend artifact. This server processes business logic such as authentication, practice workflow, scoring, and managing user progress. The diagram also indicates communication between internal servers via the intranet.

Finally, the Database Server (AWS EC2) hosts MySQL, where the system stores core data such as users, practice content, attempts, and results. The database schema artifact is shown as deployed to the database server, meaning the database structure is installed there to support the application.

5.7 Technologies Used

5.7.1 Front-end

We chose React because it makes building interactive web interfaces straightforward. Its component model encourages reusable, testable UI pieces. Its large ecosystem speeds up development, and its virtual DOM helps keep interfaces fast. Its strong community support and documentation make troubleshooting easier.

5.7.2 Back-end

We chose Spring Boot because it is a mature framework for building web APIs and backend services quickly. It provides ready-to-use configuration, production-ready features, and easy



integration with databases. Java's performance, strong typing, and broad library support make it a safe choice for IELTS Mastermind.

5.7.3 Database

We chose MySQL because it is a widely used relational database known for stability, and good performance for common workloads. It fits our data and keeps it safe and consistent. It is easy to back up, and it connects smoothly to Spring Boot through standard libraries.

5.7.4 AI Model Comparison

AI Model	Strengths	Limitations
GPT-5 / GPT-5.2 (OpenAI)	• Accurate Listening and Reading answer analysis • Advanced Writing evaluation (grammar, vocabulary, coherence, cohesion, idea development) • High-quality Speaking feedback with speech-to-text integration • Detailed, structured feedback aligned with IELTS band descriptors • Consistent band score estimation across all skills	• High computational and deployment cost • Band scores are estimated, not official IELTS results • Requires strict prompt control to follow IELTS rubrics
Claude 4.x (Anthropic)	• Strong long-form Writing analysis • Excellent evaluation of coherence, cohesion, and idea development • Clear and learner-friendly feedback explanations	• Less precise grammar error detection • Weaker alignment with IELTS band descriptors • Not suitable for pronunciation analysis

AI Model	Strengths	Limitations
Gemini 2.5 Pro (Google)	• Strong contextual understanding for Reading and Writing • Effective grammar correction and paraphrasing • Useful for question-type performance analysis	• Feedback may be generic • Inconsistent band score estimation • Limited Speaking assessment capability
Speech-to-Text and GPT-5 (Hybrid)	• Automatic transcription for Speaking responses • Analysis of fluency, grammar, and vocabulary • Scalable Speaking feedback generation	• Approximate pronunciation assessment • Accent variation affects transcription accuracy • Higher system complexity
Rule-based and AI (Hybrid System)	• Highly accurate Listening and Reading answer checking • Stable question-type performance analysis • Transparent and consistent scoring logic	• Limited flexibility for creative Writing • Less natural feedback language • High initial development effort

Table 19: *Comparison of Newest AI Models for IELTS Automated Feedback Module*

Conclusion

Based on the comparison above, **GPT-5 / GPT-5.2 (OpenAI)** is selected as the primary model for the automated IELTS feedback system. This model provides the most comprehensive support across all IELTS components, including Listening and Reading answer analysis, advanced Writing evaluation, and high-quality feedback generation aligned with IELTS band descriptors.

For the Speaking module, a **Speech-to-Text + GPT-5 hybrid approach** is adopted to enable speech transcription and analysis of fluency, grammar, and vocabulary. This combination ensures accurate, scalable, and consistent feedback generation, making it the most suitable solution for the proposed IELTS automated feedback module.

6 System Design

6.1 Module Design

The system is organized into six main modules. Each module encapsulates a specific set of functionalities to ensure a clean separation of concerns, scalability, and maintainability. These modules collaborate to provide end-to-end support for user learning, personalized training, and performance analytics. The six modules are:

- Authentication & Authorization
- Practice & Content Delivery
- Personalization & Learning Path
- Feedback & Band Scoring
- Progress Tracking
- Administration & Content Management

6.1.1 Authentication & Authorization

This module manages user identity and access control across the system. It supports account registration via email, phone, or third-party providers (Google, Facebook), as well as login, logout, and password recovery.

Main functions:

- User registration
- Login / Logout
- Password reset

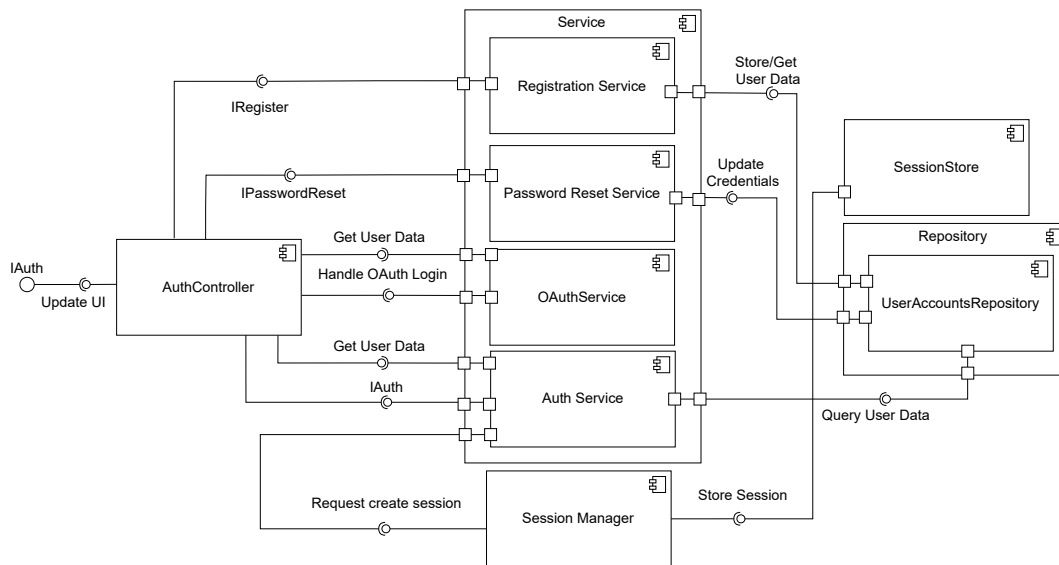


Figure 31: *Component Diagram of Authentication & Authorization Module*

• AuthController

- Acts as the central controller that coordinates all authentication and authorization workflows.
- Interacts with the user interface via **IAuth** and **UpdateUI**.
- Delegates user requests to specialized services:
 - * **Registration Service** – handles user sign-up processes.
 - * **Password Reset Service** – manages password recovery.
 - * **OAuth Service** – supports external logins (Google, Facebook, etc.).
 - * **Auth Service** – processes login, logout, and retrieves user data.
- Receives results from these services and updates the UI accordingly.

• Service

- **Registration Service**
 - * Handles registration requests from **AuthController** via **IRegister**.
 - * Validates user input such as email format, password strength, or phone number.
 - * Stores valid user information into **UserAccountsDB**.
 - * Reports registration results back to the **AuthController**.
- **Password Reset Service**
 - * Handles password reset operations through **IPasswordReset**.
 - * Sends reset links or OTP codes to the user's registered email or phone.
 - * After verification, updates user credentials in **UserAccountsDB**.
 - * Returns process status to the **AuthController**.
- **OAuth Service**

- * Manages third-party login requests (**Handle OAuth Login**).
- * Validates OAuth tokens and retrieves user data from external providers.
- * Returns authenticated user data to the **AuthController**.
- **Auth Service**
 - * Handles authentication tasks such as login and logout (**IAuth**).
 - * Retrieves and verifies user credentials from **UserAccountsDB**.
 - * On successful login, sends a **Request create session** to the **Session Manager**.
 - * Queries user data when requested by the **AuthController** (**Get User Data**).
 - * Collaborates with the **Session Manager** to manage session tokens and maintain authentication persistence.
- **Session Manager**
 - Manages all user session-related operations.
 - Creates and stores user sessions upon request from the **Auth Service**.
 - Saves session tokens and expiration times in **SessionDB** (**Store Session**).
 - Provides functionality to query or validate existing sessions (**Query User Data**).
- **Repository**
 - **UserAccountRepository**
 - * Encapsulates all persistence operations related to user accounts.
 - * Stores and retrieves user credentials, personal information, and OAuth bindings.
 - * Is accessed by **Registration Service**, **Password Reset Service**, **OAuth Service**, and **Auth Service**.
 - * Provides a consistent data access abstraction, isolating services from direct database interactions.
- **In-Memory Session Store**
 - Maintains active user sessions in server memory.
 - Stores session tokens, authentication states, and expiration timestamps.
 - Is managed exclusively by the **Session Manager**.
 - Enables fast session lookup and validation without persistent database access.
 - Session data is temporary and may be cleared when the server restarts.

6.1.2 Practice & Content Delivery

This module provides IELTS practice materials (Reading, Listening, Writing, and Speaking) and enables learners to interact with them. Learners can browse, search, and filter content, view item details, bookmark content, and start practice sessions. Submitted answers are stored, and basic scoring and feedback is generated. Users can also review attempt history.

Main functions:

- Browse / search / filter content

- Content preview & metadata display
- Bookmark management
- Start / resume practice
- Answer submission
- Store attempts
- Band score and automated feedback
- Attempt history

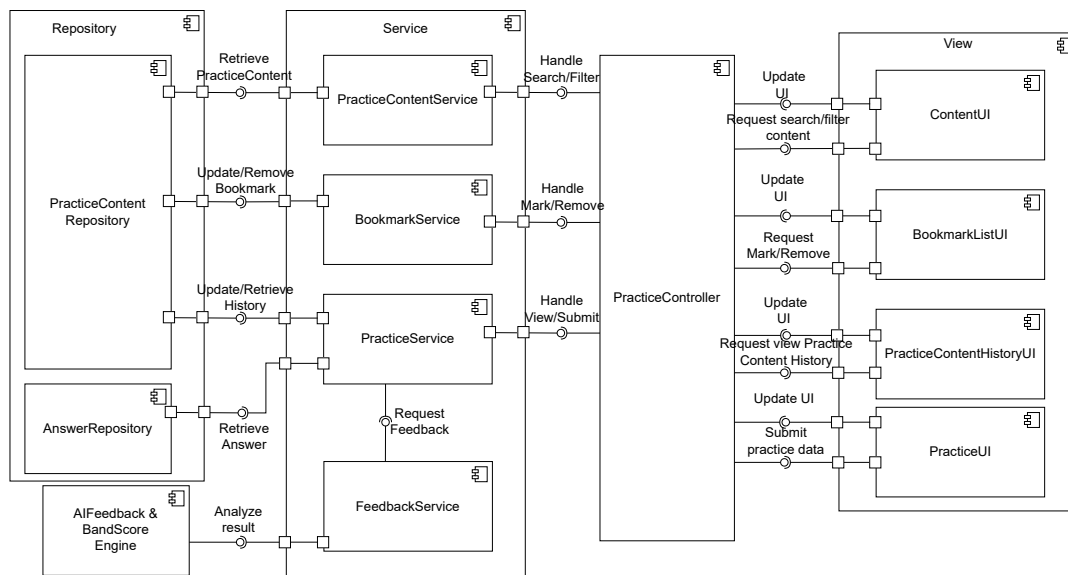


Figure 32: Component Diagram of Practice & Content Delivery Module

• Controller

- **PracticeController:** acts as the central coordinator between the services and the user interface. It handles user requests such as:
 - * Requesting filtered or searched content (via *PracticeContentService*)
 - * Bookmarking or removing content (via *BookmarkService*)
 - * Starting or submitting a practice (via *PracticeService*)
 - * Viewing attempt history (via *PracticeService*)

The controller updates UI components after receiving data or results from services.

• Service

- **PracticeContentService:** retrieves and filters practice content based on learner input. Communicates with *PracticeContentDB* to get relevant exercises.

- **BookmarkService**: manages user bookmarks, allowing adding or removing saved practice content. Communicates with both the *User Profile Database* (not shown) and *PracticeController*.
- **PracticeService**: handles practice session logic—loading questions, submitting answers, and managing session history. Interacts with *AnswerDB* and triggers requests for feedback through *FeedbackService*.
- **FeedbackService**: processes scoring and feedback by sending user answers to the *AIFeedbackEngine* for analysis, then returning results to the *PracticeController*.
- **View**
 - **ContentUI**: displays practice materials, search, and filter options. Sends user requests to *PracticeController* and updates when new content is loaded.
 - **BookmarkListUI**: shows all bookmarked practice content. Sends bookmark/unbookmark requests and updates dynamically.
 - **PracticeContentHistoryUI**: lists past attempts and retrieved results. Requests data from *PracticeController* and refreshes when new history is available.
 - **PracticeUI**: provides the main practice interface, where learners answer questions and submit their responses. Sends submission requests to *PracticeController* and receives real-time feedback.
- **Repository**
 - **PracticeContentRepository**
 - * Manages persistence and retrieval of IELTS practice questions, metadata, and related media.
 - * Supports search, filtering, and content lookup operations.
 - * Is accessed by *PracticeContentService* to provide data for browsing and previewing practice materials.
 - **AnswerRepository**
 - * Stores user-submitted answers and official reference answers.
 - * Provides access to answer data for scoring and result comparison.
 - * Is used by *PracticeService* during answer submission and evaluation.
 - **BookmarkRepository**
 - * Handles persistence of bookmarked practice content for each learner.
 - * Supports adding, removing, and retrieving bookmarked items.
 - * Serves as the data access layer for *BookmarkService*.
 - **PracticeAttemptRepository**
 - * Stores practice session attempts and related performance results.
 - * Supports retrieval of attempt history for review purposes.
 - * Is accessed by *PracticeService* to manage practice history.
- **AIFeedback&BandScoreEngine**
 - Performs automated analysis of evaluation and practice submissions.
 - Generates detailed qualitative feedback on user answers across different skills.

- Estimates IELTS band scores based on scoring rubrics and performance patterns.
- Produces skill-level insights that support skill profile updates and learning path adaptation.
- Returns feedback results and band score estimations to **FeedbackService** for storage and UI updates.

6.1.3 Personalization & Learning Path

This module personalizes the learning journey based on the learner's proficiency and goals. Skill profiles are generated from evaluation tests and ongoing performance data. Using these profiles, the platform produces personalized study plans and recommendations aligned with users' target band scores and exam timelines. The skill profile is continuously updated as new results are recorded.

Main functions:

- Evaluation test → skill profile generation
- Personalized study plan
- Target band & exam date setup
- Adaptive learning recommendations
- Continuous skill profile updates

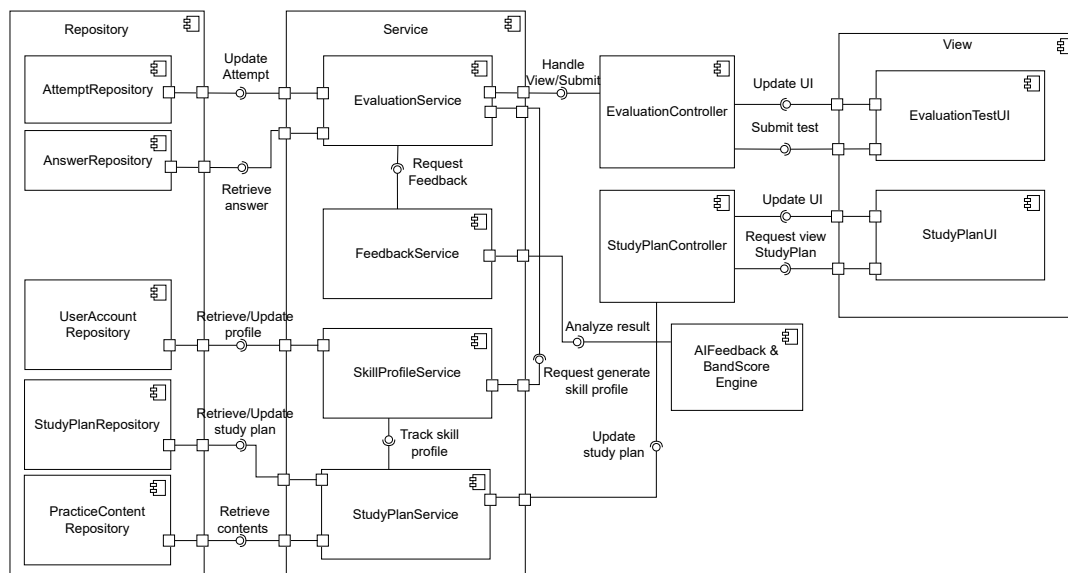


Figure 33: *Component Diagram of Personalization & Learning Path Module*

• Controller

– **EvaluationController**

- * Handles user interactions related to the Evaluation Test feature.
- * Communicates with **EvaluationTestUI** via **Handle View/Submit** and **Update UI** events.
- * Submits evaluation test responses to the **EvaluationService**.
- * Requests feedback generation from **FeedbackService** and skill profile creation from **SkillProfileService**.
- * Updates the UI with test results and skill profile insights.

– **StudyPlanController**

- * Manages all interactions related to study plan creation and viewing.
- * Communicates with **StudyPlanUI** to handle **Request View Study Plan** and **Update UI**.
- * Requests skill profile analysis from **SkillProfileService** to generate personalized study plans.
- * Sends requests to **StudyPlanService** to create or update the learner's study plan.
- * Receives updated study plans and reflects them to the UI.

• **Service**

– **EvaluationService**

- * Processes user evaluation test submissions.
- * Stores and retrieves test attempts from **AttemptDB** and **AnswerDB**.
- * Requests feedback generation from **FeedbackService** for completed evaluations.
- * Returns evaluation results to **EvaluationController**.

– **FeedbackService**

- * Handles all feedback-related operations for evaluation tests.
- * Sends requests to the **AIFeedbackEngine** for automated feedback and band estimation.
- * Stores generated feedback and band scores.
- * Returns processed feedback results to both **EvaluationService** and **SkillProfileService**.

– **SkillProfileService**

- * Generates and maintains the learner's skill profile.
- * Analyzes feedback results received from **FeedbackService**.
- * Retrieves and updates skill profile data stored in **UserAccountDB**.
- * Provides skill insights to the **StudyPlanService** and **StudyPlanController**.

– **StudyPlanService**

- * Creates, updates, and retrieves personalized study plans.
- * Uses learner skill profiles and content information to design tailored learning paths.
- * Retrieves relevant practice content from **PracticeContentDB**.
- * Saves and updates study plan data in **StudyPlanDB**.
- * Works closely with **SkillProfileService** to adapt the study plan as the learner progresses.

• **View**

- **EvaluationTestUI**
 - * Presents evaluation test questions and time controls to the learner.
 - * Sends user submissions (answers, audio uploads) to **EvaluationController** via **Submit test**.
 - * Receives updates and feedback display requests via **Update UI** from **EvaluationController**.
- **StudyPlanUI**
 - * Displays personalized study plans, recommendations, and progress indicators.
 - * Allows users to request viewing or refreshing their study plan via **Request View Study Plan**.
 - * Receives updated study plans and visualizations via **Update UI** from **StudyPlanController**.
- **Repository**
 - **AttemptRepository**
 - * Manages persistence of evaluation test attempts and progress status.
 - * Provides read and write operations for evaluation-related attempts.
 - * Is accessed by **EvaluationService** during evaluation processing.
 - **AnswerRepository**
 - * Stores and retrieves detailed answer submissions for evaluation tests.
 - * Supports access to user responses for feedback generation and skill analysis.
 - * Serves as a data source for **EvaluationService** and **FeedbackService**.
 - **UserAccountRepository**
 - * Encapsulates persistence logic for learner profiles and skill-related data.
 - * Stores and updates skill profiles, learning history, and target information.
 - * Is used by **SkillProfileService** to maintain learner proficiency data.
 - **StudyPlanRepository**
 - * Manages storage of personalized study plans generated for learners.
 - * Supports creation, update, and retrieval of study plan data.
 - * Acts as the data access layer for **StudyPlanService**.
 - **PracticeContentRepository**
 - * Provides access to official and recommended practice content.
 - * Supplies content metadata used in personalized study plan generation.
 - * Is accessed by **StudyPlanService** when constructing learning paths.
- **AIFeedback&BandScoreEngine**
 - Performs automated analysis of evaluation and practice submissions.
 - Generates detailed qualitative feedback on user answers across different skills.
 - Estimates IELTS band scores based on scoring rubrics and performance patterns.
 - Produces skill-level insights that support skill profile updates and learning path adaptation.
 - Returns feedback results and band score estimations to **FeedbackService** for storage and UI updates.

6.1.4 Feedback & Band Scoring

This module automatically evaluates Writing and Speaking tasks using AI. For Writing, it analyzes grammar, vocabulary, structure, coherence, and task relevance. For Speaking, submitted audio is transcribed and assessed for pronunciation, fluency, grammar, and vocabulary. The system generates estimated band scores and detailed feedback to guide improvement. Results are stored for learner review and performance tracking.

Main functions:

- Writing analysis
- Speaking analysis (speech → transcription → scoring)
- Automated band scoring
- Feedback generation
- Store scoring results and feedback

6.1.5 Progress Tracking

This module tracks and analyzes learning progress. It aggregates performance data, practice history, and study time, presenting them through dashboards and reports. Learners can monitor skill improvement, review detailed attempt history, and compare performance over time. Data from practice attempts and scoring results feed into this module to support analytics.

Main functions:

- Skill progress tracking
- Dashboard visualization
- Study time tracking
- Attempt history
- Reporting & analytics

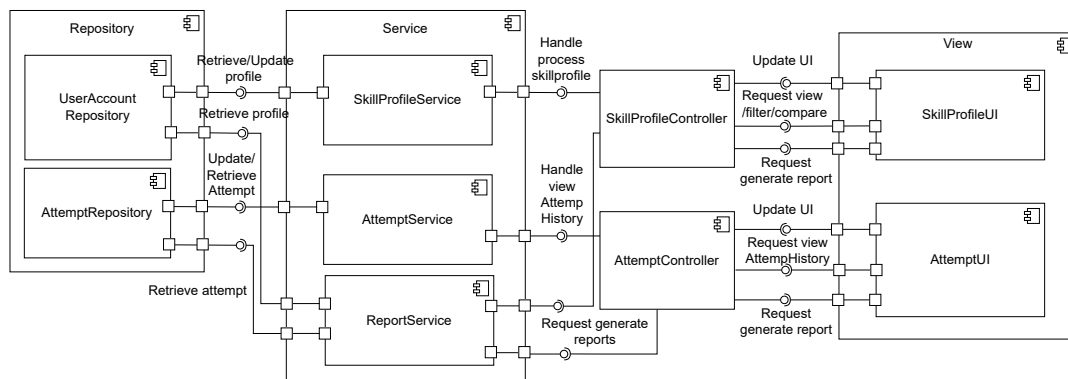


Figure 34: *Component Diagram of Progress Tracking Module*

- **Controller**

- **SkillProfileController**

- * Handles all user interactions related to viewing and analyzing skill progress.
 - * Communicates with **SkillProfileUI** through **Request View/Filter/Compare Skill Profile** and **Update UI**.
 - * Requests skill profile processing and retrieval from the **SkillProfileService**.
 - * Sends requests to **ReportService** to generate detailed performance reports.
 - * Updates the UI with skill analytics, comparison insights, and generated reports.

- **AttemptController**

- * Manages user requests related to viewing practice attempt history.
 - * Communicates with **AttemptUI** via **Request View Attempt History** and **Update UI**.
 - * Retrieves the learner's attempt history by calling the **AttemptService**.
 - * Sends report generation requests to **ReportService** for detailed attempt analytics.
 - * Returns processed attempt history and reports back to the UI for visualization.

- **Service**

- **SkillProfileService**

- * Manages the processing and retrieval of the learner's skill profile.
 - * Retrieves existing skill profile data from **UserAccountDB**.
 - * Updates skill progress based on new attempts or performance improvements.
 - * Returns analyzed skill profile data to **SkillProfileController**.

- **AttemptService**

- * Handles retrieval and updates related to user practice attempts.
 - * Fetches attempt data from **AttemptDB**.
 - * Provides attempt history details to **AttemptController** for UI display.
 - * Updates attempt records when new results or re-evaluations occur.

- **ReportService**

- * Generates reports for both skill profile progress and attempt history.
 - * Collects necessary data from **SkillProfileService** and **AttemptService**.
 - * Compiles insights into structured reports for visualization in the UI.
 - * Returns completed reports to both **SkillProfileController** and **AttemptController**.

- **View**

- **SkillProfileUI**

- * Displays the learner's skill profile, progress charts, and comparative performance analytics.
 - * Allows users to request viewing, filtering, or comparing skill profiles via UI actions.
 - * Receives processed skill data, insights, and reports from **SkillProfileController** for visualization.

- **AttemptUI**

- * Displays the learner's practice attempt history and detailed performance breakdown.

- * Allows users to request viewing of past attempts or to generate attempt performance reports.
- * Receives updated attempt history and generated reports via `Update UI` from `AttemptController`.

• Repository

– UserAccountRepository

- * Provides data access operations for user profile information related to learning progress.
- * Encapsulates persistence logic for skill profiles, performance metrics, and user-related progress data.
- * Is used by `SkillProfileService` to retrieve and update skill progress information.

– AttemptRepository

- * Manages persistent storage of user practice attempt data.
- * Provides retrieval and update operations for attempt history and performance records.
- * Serves as the data source for `AttemptService` and reporting-related analytics.

6.1.6 Administration & Content Management

This module enables system administrators to manage users, roles, permissions, and practice content. Administrators can upload, update, categorize, or remove practice materials, as well as edit metadata. User and role management ensures proper access control and content delivery. This module guarantees that learning resources remain accurate and up-to-date.

Main functions:

- User & role management
- Permission management
- Practice content CRUD
- Metadata & category management

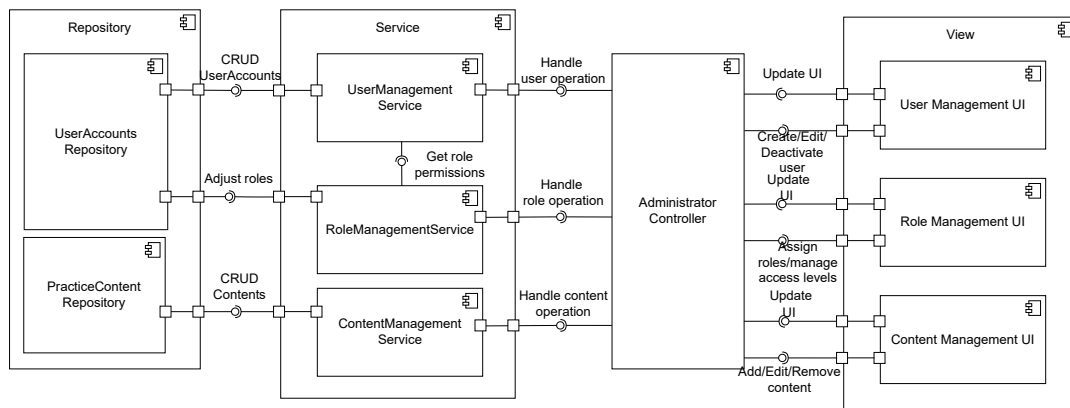


Figure 35: *Component Diagram of Administration & Content Management Module*

- **Controller**

- **AdministratorController**

- * Handles all admin interactions related to managing users, roles, and practice content.
 - * Communicates with the respective UI components via:
 - **Update UI** for displaying admin actions.
 - **Create/Edit/Deactivate User** operations for user management.
 - **Assign roles/manage access levels** for role and permission configuration.
 - **Add/Edit/Remove content** for practice content management.
 - * Delegates requests to:
 - **UserManagementService** for handling user account operations.
 - **RoleManagementService** for role and permission assignments.
 - **ContentManagementService** for managing practice content data.
 - * Receives processed data from services and updates the respective UIs.

- **Service**

- **UserManagementService**

- * Handles CRUD operations for user accounts.
 - * Communicates with **UserAccountsDB** to create, edit, deactivate, or retrieve user information.
 - * Returns updated user list and account details back to the controller.

- **RoleManagementService**

- * Manages roles, permissions, and access-level rules for different user types.
 - * Retrieves and updates role permissions in **UserAccountsDB**.
 - * Returns role configuration and permission data to **AdministratorController**.

- **ContentManagementService**

- * Handles CRUD operations for practice content.
 - * Communicates with **PracticeContentDB** to add, edit, or remove content.
 - * Provides updated content lists to the controller for UI rendering.

- **View**

- **User Management UI**

- * Provides interfaces for admins to create, edit, view, or deactivate user accounts.
 - * Sends admin actions (create/edit/deactivate) to **AdministratorController**.
 - * Displays user lists, user details, and operation results returned from the controller.

- **Role Management UI**

- * Allows admins to assign roles and manage access levels for different users.
 - * Sends role assignment and permission-change requests to **AdministratorController**.

- **Repository**

- **UserAccountRepository**

- * Provides data access operations for user accounts, roles, and permissions.



- * Encapsulates all persistence logic related to user profiles, account status, and role assignments.
 - * Is used by `UserManagementService` and `RoleManagementService` to perform CRUD operations and access control updates.
- **PracticeContentRepository**
- * Manages data persistence for practice content and related metadata.
 - * Supports CRUD operations for learning materials, categories, and visibility settings.
 - * Serves as the data access layer for `ContentManagementService`.

6.2 Database Design

6.2.1 Entity Relationship Diagram

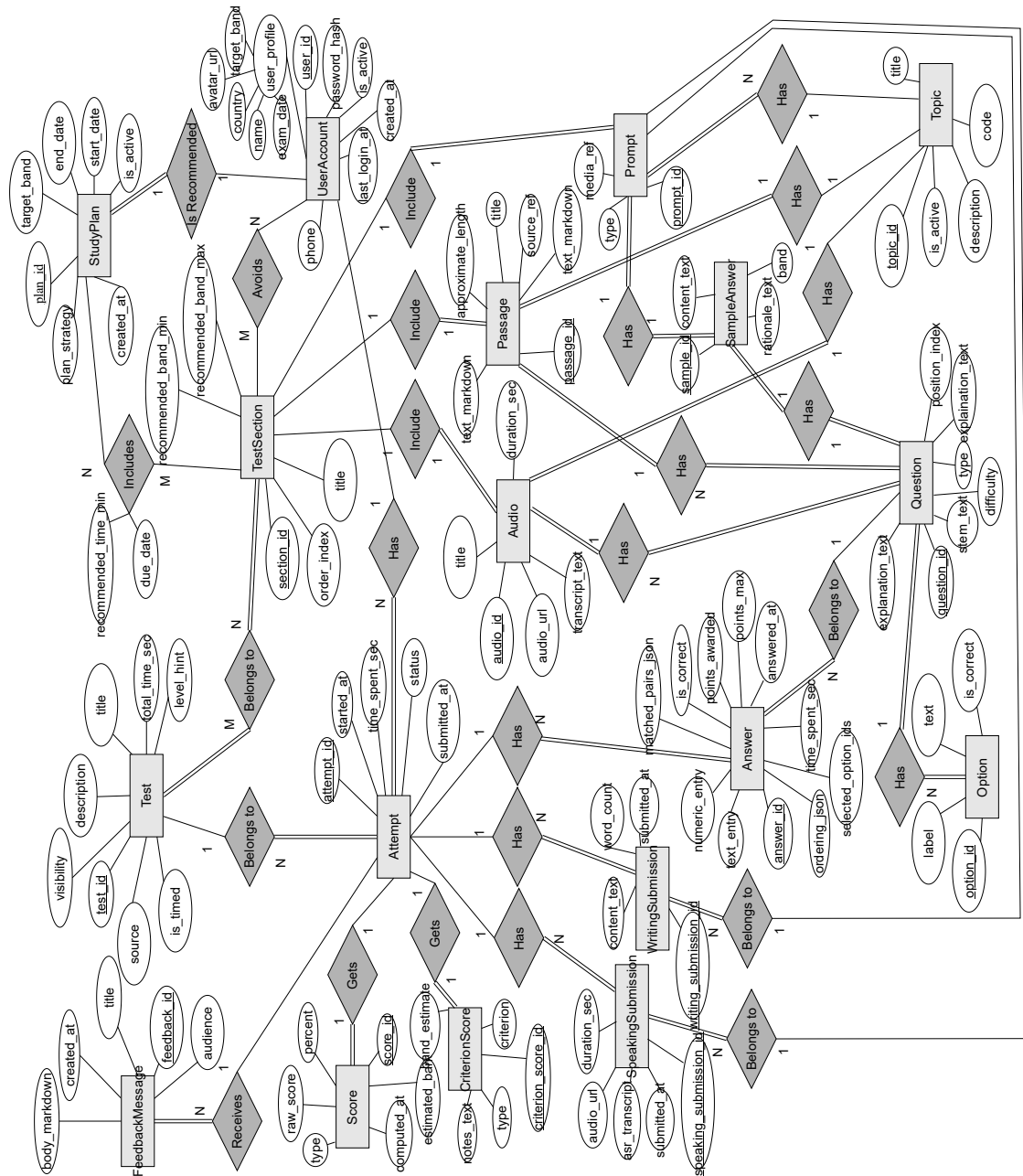


Figure 36: Entity Relationship Diagram of the System

The Entity–Relationship Diagram (ERD) in Figure 36 describes the data structure of the IELTS Mastermind system by defining its entities, attributes, and relationships.

UserAccount represents system users, including learners and administrators. *Attributes:* user_id, name, phone, password_hash, avatar_url, user_profile, target_band, exam_date, country, is_active, created_at, last_login_at. *Relationships:* A UserAccount can create multiple Attempts (1–N), receive multiple FeedbackMessages (1–N), and be assigned multiple StudyPlans (1–N).

Topic represents a high-level classification of learning content. *Attributes:* topic_id, code, title, description, is_active. *Relationships:* A Topic can include multiple Tests (1–N).

Test represents a practice test or assessment. *Attributes:* test_id, title, description, source, is_timed, total_time_sec, level_hint, visibility. *Relationships:* A Test belongs to one Topic (N–1) and consists of multiple TestSections (1–N).

TestSection represents a logical section within a test, such as Reading, Listening, Writing, or Speaking. *Attributes:* section_id, title, duration_sec, order_index. *Relationships:* A TestSection belongs to one Test (N–1) and may include multiple Passages, Audios, and Questions (1–N).

Passage represents reading material used in reading sections. *Attributes:* passage_id, title, text_markdown, approximate_length, source_ref. *Relationships:* A Passage belongs to one TestSection (N–1) and can be associated with multiple Questions (1–N).

Audio represents audio resources for listening or speaking tasks. *Attributes:* audio_id, title, audio_url, transcript_text. *Relationships:* An Audio item belongs to one TestSection (N–1) and may be linked to multiple Questions (1–N).

Question represents an assessment question. *Attributes:* question_id, stem_text, type, difficulty, position_index, recommended_band_min, recommended_band_max, explanation_text. *Relationships:* A Question belongs to one TestSection (N–1), may reference a Passage or Audio (N–1), and can have multiple Options and SampleAnswers (1–N).

Option represents a selectable answer choice for multiple-choice questions. *Attributes:* option_id, text_label, is_correct. *Relationships:* An Option belongs to one Question (N–1).

Prompt represents task instructions for writing or speaking questions. *Attributes:* prompt_id, type, text_markdown, media_ref. *Relationships:* A Prompt is associated with one Question (1–1).

SampleAnswer represents reference answers used for evaluation or guidance. *Attributes:* sample_id, content_text, rationale_text. *Relationships:* A SampleAnswer belongs to one Question (N–1).

Attempt represents a user’s attempt at completing a test or practice activity. *Attributes:* attempt_id, started_at, submitted_at, time_spent_sec, status. *Relationships:* An Attempt belongs to one UserAccount (N–1), includes multiple Answers (1–N), and produces one Score (1–1).

Answer stores a user’s response to a question. *Attributes:* answer_id, selected_option_ids, numeric_entry, text_entry, ordering_json, matched_pairs_json, points_awarded, is_correct, submitted_at. *Relationships:* An Answer belongs to one Attempt (N–1) and one Question (N–1), and may reference a WritingSubmission or SpeakingSubmission (1–1).

WritingSubmission stores written responses for writing tasks. *Attributes:* writing_submission_id, content_text, word_count, submitted_at. *Relationships:* A WritingSubmission is associated with one Answer (1–1).



SpeakingSubmission stores speaking responses. *Attributes:* `speaking_submission_id`, `audio_url`, `asr_transcript`, `duration_sec`. *Relationships:* A **SpeakingSubmission** is associated with one **Answer** (1–1).

Score represents the evaluation result of an attempt. *Attributes:* `score_id`, `raw_score`, `percent`, `estimated_band`, `computed_at`. *Relationships:* A **Score** belongs to one **Attempt** (1–1) and can have multiple **CriterionScores** (1–N).

CriterionScore represents detailed scoring for individual evaluation criteria. *Attributes:* `criterion_score_id`, `criterion`, `band_estimate`, `notes_text`. *Relationships:* A **CriterionScore** belongs to one **Score** (N–1).

FeedbackMessage represents feedback provided to users. *Attributes:* `feedback_id`, `title`, `body_markdown`, `audience`, `created_at`. *Relationships:* A **FeedbackMessage** is received by one **UserAccount** (N–1) and may relate to an **Attempt** or **Score**.

StudyPlan represents a personalized learning plan. *Attributes:* `plan_id`, `target_band`, `start_date`, `end_date`, `plan_strategy`, `recommended_time_min`, `due_date`, `created_at`, `is_active`. *Relationships:* A **StudyPlan** is assigned to one **UserAccount** (N–1).

6.2.2 Relational Schema

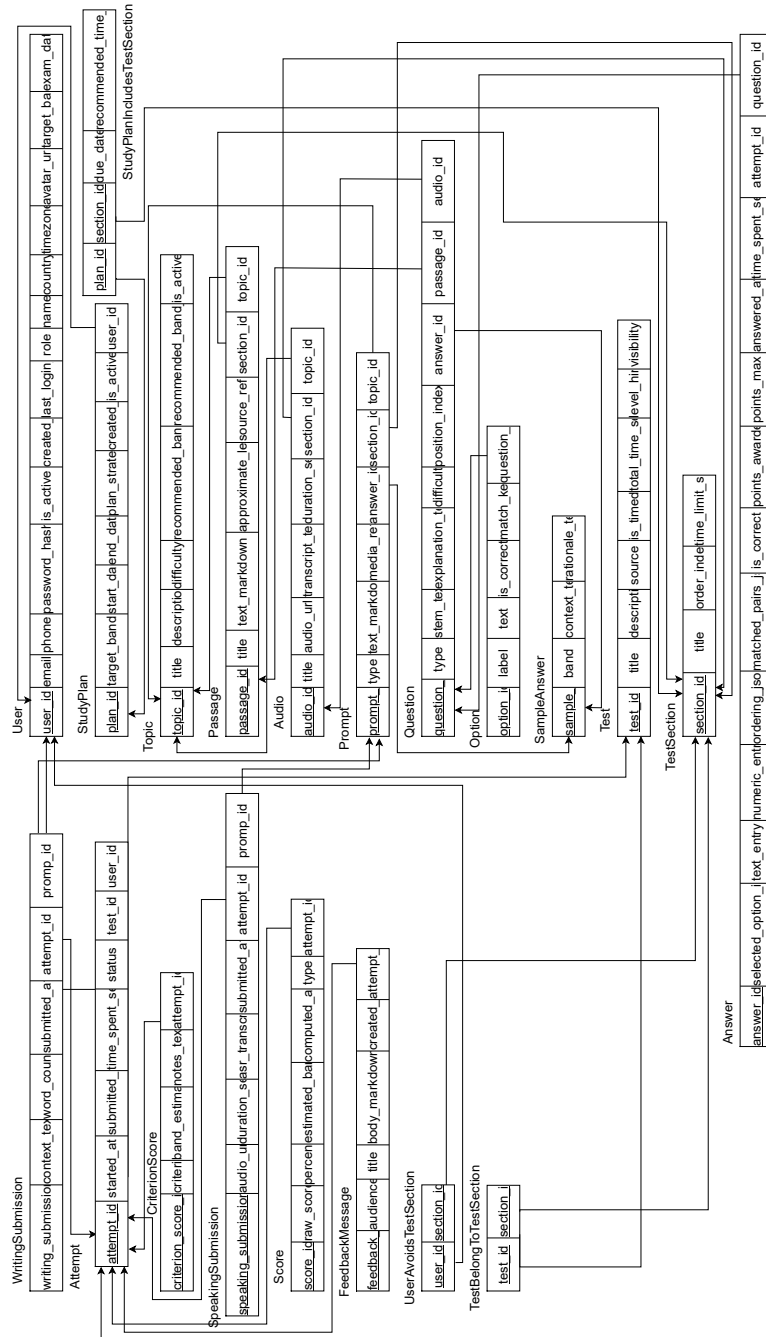


Figure 37: Relational Schema of the System

6.3 API Design

6.3.1 Authentication and Authorization

Register Using Email and Password

- Function: POST /auth/register/email
- Parameters: { email, password }
- Returns:
 - 201 Created: { "userId": "12345", "email": "test@gmail.com" }
- Errors:
 - 400 Bad Request: Email already used
 - 401 Unauthorized: Missing or invalid email

Register/Login via Google Email

- Function: POST /auth/register/google
- Parameters: { token }
- Returns:
 - 200 OK: { "userId": "12345", "email": "abc@gmail.com" }
- Errors:
 - 401 Unauthorized: Missing or invalid token

Register/Login via Facebook

- Function: POST /auth/register/facebook
- Parameters: { token }
- Returns:
 - 200 OK: { "userId": "12345", "email": "user@facebook.com" }
- Errors:
 - 401 Unauthorized: Missing or invalid token

Login

- Function: POST /auth/login
- Parameters: { email, password }
- Returns:

- 200 OK: { "accessToken": "...", "refreshToken": "..." }

- Errors:

- 401 Unauthorized

Logout

- Function: POST /auth/logout

- Parameters: { refreshToken }

- Returns:

- 200 OK

- Errors:

- 400 Bad Request: Missing or invalid refresh token

- 401 Unauthorized: Token expired or unauthorized

Send Password Reset Link

- Function: POST /auth/password-reset/request

- Parameters: { emailOrPhone }

- Returns:

- 200 OK: { "message": "Reset code sent" }

- Errors:

- 400 Bad Request: Missing or invalid email

- 404 Not Found: Account not found for the provided email

- 429 Too Many Requests: Too many reset attempts, please try again later

Confirm Reset Token and Set New Password

- Function: POST /auth/password-reset/confirm

- Parameters: { token, newPassword }

- Returns:

- 200 OK: { "message": "Password updated" }

- Errors:

- 400 Bad Request: Missing or invalid token, or password does not meet policy

- 404 Not Found: Token or user not found

- 410 Gone: Token expired

- 429 Too Many Requests: Too many attempts, please try again later

6.3.2 User Profiles

Get User Profile

- Function: GET /user/profile
- Parameters: None (requires authentication)
- Returns:
 - 200 OK: { "name": "abc", "email": "abc@gmail.com", "country": "VN", "dob": "2000-01-01", "avatar": "url" }
- Errors:
 - 401 Unauthorized: Missing or invalid access token

Update User Profile

- Function: PUT /user/profile
- Parameters: { name, country, dob, avatar }
- Returns:
 - 200 OK: { "message": "Profile updated" }
- Errors:
 - 400 Bad Request: Invalid data or unsupported field type
 - 401 Unauthorized: Missing or invalid access token

Get Study Plan

- Function: GET /user/study-plan
- Parameters: None
- Returns:
 - 200 OK: { "targetBand": 7.5, "examDate": "2025-03-20", "goals": [...] } }
- Errors:
 - 401 Unauthorized

Update Study Plan

- Function: PUT /user/study-plan
- Parameters: { targetBand, examDate, goals }
- Returns:

- 200 OK: { "message": "Study plan updated" }

- Errors:

- 400 Bad Request: Invalid or incomplete study plan data
- 401 Unauthorized

Get Skill Profile

- Function: GET /user/skill-profile

- Parameters: None

- Returns:

- 200 OK: { "listening": 6.0, "reading": 7.0, "writing": 6.5, "speaking": 6.0 }

- Errors:

- 401 Unauthorized

Get Attempt History

- Function: GET /user/attempts

- Parameters: { type, dateFrom, dateTo } (optional filters)

- Returns:

- 200 OK: List of attempts across all modules

- Errors:

- 400 Bad Request: Invalid query format
- 401 Unauthorized

Get Specific Attempt Details

- Function: GET /user/attempts/{attemptId}

- Parameters: None

- Returns:

- 200 OK: Detailed attempt information

- Errors:

- 401 Unauthorized
- 404 Not Found: Attempt ID not found

6.3.3 Practice Content Access

Browse Practice Content

- Function: GET /practice/content
- Parameters: { search, filter, skill, level, tags } (optional)
- Returns:
 - 200 OK: { "contentId": "123", "title": "Reading Test 1", "skill": "reading", "level": "intermediate", "tags": ["true-false-not-given"] }
- Errors:
 - 400 Bad Request: Invalid filter or query parameter

Get Practice Content Preview

- Function: GET /practice/content/{contentId}/preview
- Parameters: None
- Returns:
 - 200 OK: { "title": "Listening Test 2", "description": "Practice the IELTS Listening Test.", "questionCount": 40 }
- Errors:
 - 404 Not Found: Content ID not found

Get Practice Content Metadata

- Function: GET /practice/content/{contentId}/metadata
- Parameters: None
- Returns:
 - 200 OK: Metadata including difficulty, skill type, time limit
- Errors:
 - 404 Not Found: Content ID not found

Start New Practice Session

- Function: POST /practice/content/{contentId}/start
- Parameters: None
- Returns:



- 201 Created: { "sessionId": "abc123" }

- Errors:

- 401 Unauthorized
- 404 Not Found: Content ID not found

Resume Practice Session

- Function: POST /practice/session/{sessionId}/resume
- Parameters: None
- Returns:
 - 200 OK: { "sessionId": "abc123", "progress": 65, "answers": ... }
- Errors:
 - 401 Unauthorized
 - 404 Not Found: Session not found or expired

Submit Practice Answers

- Function: POST /practice/session/{sessionId}/submit
- Parameters: { answers }
- Returns:
 - 200 OK: { "message": "Submitted successfully" }
- Errors:
 - 400 Bad Request: Invalid answer format
 - 401 Unauthorized
 - 404 Not Found: Session not found

Auto-Submit on Timer Expiry

- Function: POST /practice/session/{sessionId}/submit-auto
- Parameters: None
- Returns:
 - 200 OK
- Errors:
 - 404 Not Found: Session not found or expired

Get Band Score

- Function: GET /practice/session/{sessionId}/band-score
- Parameters: None
- Returns:
 - 200 OK: { "bandScore": 6.5 }
- Errors:
 - 404 Not Found: Session not found

Get Detailed Feedback

- Function: GET /practice/session/{sessionId}/feedback
- Parameters: None
- Returns:
 - 200 OK: { "sessionId": "abc123", "bandScore": 6.5, "strengths": [...], "weaknesses": [...], "questionFeedback": [...] }
- Errors:
 - 404 Not Found: Session not found

6.3.4 Personalization and Learning Paths

Start Evaluation Test

- Function: POST /evaluation/start
- Parameters: None
- Returns:
 - 201 Created: { "sessionId": "eval123" }
- Errors:
 - 401 Unauthorized

Submit Evaluation Test Answers

- Function: POST /evaluation/{sessionId}/submit
- Parameters: { answers }
- Returns:
 - 200 OK: { "message": "Submitted successfully" }
- Errors:

- 400 Bad Request: Invalid answer schema
- 401 Unauthorized
- 404 Not Found: Session not found

Get Evaluation Band Score

- Function: GET /evaluation/{sessionId}/band-score
- Parameters: None
- Returns:
 - 200 OK: { "bandScore": 6.0 }
- Errors:
 - 401 Unauthorized
 - 404 Not Found: Session not found

Get Evaluation Test Feedback

- Function: GET /evaluation/{sessionId}/feedback
- Parameters: None
- Returns:
 - 200 OK: Detailed strengths, weaknesses, and item-level analysis
- Errors:
 - 401 Unauthorized
 - 404 Not Found: Session not found

Generate Skill Profile

- Function: POST /skill-profile/generate
- Parameters: None
- Returns:
 - 200 OK: { "status": "Skill profile generated" }
- Errors:
 - 401 Unauthorized
 - 409 Conflict: Skill profile generation already in progress

Get Full Skill Profile

- Function: GET /skill-profile
- Parameters: None
- Returns:
 - 200 OK: { "listening": 6.0, "reading": 7.0, "writing": 6.5, "speaking": 6.0 }
- Errors:
 - 401 Unauthorized

Generate Personalized Study Plan

- Function: POST /study-plan/generate
- Parameters: { targetBand, examDate }
- Returns:
 - 200 OK: { "studyPlan": "weeks": 10, "recommendedActivities": [...] }
- Errors:
 - 400 Bad Request: Invalid or missing target band/exam date
 - 401 Unauthorized
 - 409 Conflict: Existing study plan already generated

6.3.5 User & Role Management

Create User

- Function: POST /api/user
- Parameters: { email, phone, password, role, firstname, lastname, country, timezone, avatarUrl, targetBand, examDate }
- Returns:
 - 201 Created: { "userId": "uuid", "email": "jane.doe@example.com" }
- Errors:
 - 400 Bad Request: Invalid user data
 - 409 Conflict: Email already exists

Get All Users

- Function: GET /api/user
- Parameters: None
- Returns:

- 200 OK: List of users

- Errors:

- 401 Unauthorized: Unauthorized access

Get User By ID

- Function: GET /api/user/{id}

- Parameters: { id }

- Returns:

- 200 OK: User detail

- Errors:

- 404 Not Found: User not found

Update User By ID

- Function: PUT /api/user/{id}

- Parameters: { id, updated user information }

- Returns:

- 200 OK: User updated successfully

- Errors:

- 400 Bad Request: Invalid user data
- 404 Not Found: User not found

Delete User By ID

- Function: DELETE /api/user/{id}

- Parameters: { id }

- Returns:

- 204 No Content

- Errors:

- 404 Not Found: User not found
- 401 Unauthorized: Unauthorized access

6.3.6 Practice Content Management

Create Practice Content

- Function: POST /api/practice-content
- Parameters: Practice content object (skill, title, instructions, task, status, tags, media URLs, questions)
- Returns:
 - 201 Created: { "id": "uuid", "status": "DRAFT" }
- Errors:
 - 400 Bad Request: Invalid request body
 - 401 Unauthorized: Unauthorized access

Get All Practice Content

- Function: GET /api/practice-content
- Parameters: None
- Returns:
 - 200 OK: List of practice content
- Errors:
 - 401 Unauthorized: Unauthorized access

Get Practice Content By ID

- Function: GET /api/practice-content/{id}
- Parameters: { id }
- Returns:
 - 200 OK: Practice content detail
- Errors:
 - 404 Not Found: Practice content not found
 - 401 Unauthorized: Unauthorized access

Update Practice Content By ID

- Function: PUT /api/practice-content/{id}
- Parameters: { id, updated practice content object }

- Returns:
 - 200 OK: Practice content updated successfully
- Errors:
 - 400 Bad Request: Invalid request body
 - 404 Not Found: Practice content not found
 - 401 Unauthorized: Unauthorized access

Delete Practice Content By ID

- Function: DELETE /api/practice-content/{id}
- Parameters: { id }
- Returns:
 - 204 No Content
- Errors:
 - 404 Not Found: Practice content not found
 - 401 Unauthorized: Unauthorized access

Upload Thumbnail

- Function: POST /api/files/thumbnail
- Parameters: { file (image) }
- Returns:
 - 201 Created: { "url": "/files/thumbnails/{filename}" }
- Errors:
 - 400 Bad Request: Invalid file
 - 413 Payload Too Large: File size exceeded

Upload Audio

- Function: POST /api/files/audio
- Parameters: { file (audio) }
- Returns:
 - 201 Created: { "url": "/files/audio/{filename}" }
- Errors:
 - 400 Bad Request: Invalid file

- 413 Payload Too Large: File size exceeded

Get Thumbnail

- Function: GET /files/thumbnails/{filename}
- Parameters: { filename }
- Returns:
 - 200 OK: Image file
- Errors:
 - 404 Not Found: File not found

Get Audio

- Function: GET /files/audio/{filename}
- Parameters: { filename }
- Returns:
 - 200 OK: Audio file
- Errors:
 - 404 Not Found: File not found

6.4 UI/UX Design

6.4.1 Full System UI Workflow Diagram

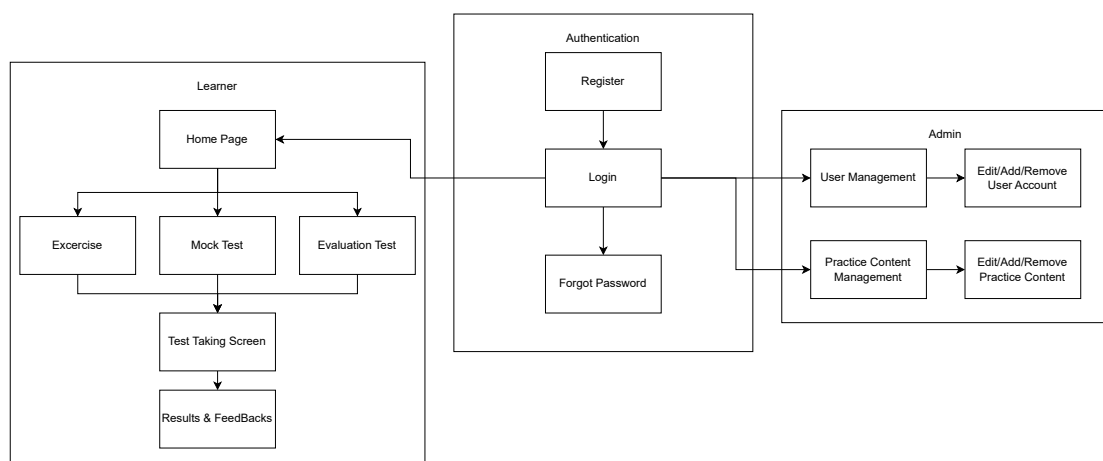


Figure 38: Full System UI Workflow Diagram

This workflow diagram in Figure 38 illustrates the complete interaction flow of the IELTS Mastermind system, covering all planned features for both learner and admin roles. Users begin by registering and logging in through the authentication module. After successful login, learners are directed to the home page, where they can choose between exercises, mock tests, or evaluation tests. Each selected activity leads to a test-taking screen, followed by a results and feedback page that allows learners to review their performance. In parallel, admin users access administrative functions after logging in, including user management and practice content management, where they can add, edit, or remove user accounts and learning materials. This diagram clearly demonstrates the overall navigation structure of the system and highlights the separation of responsibilities between learners and administrators.

6.4.2 Implemented Features UI Workflow Diagram

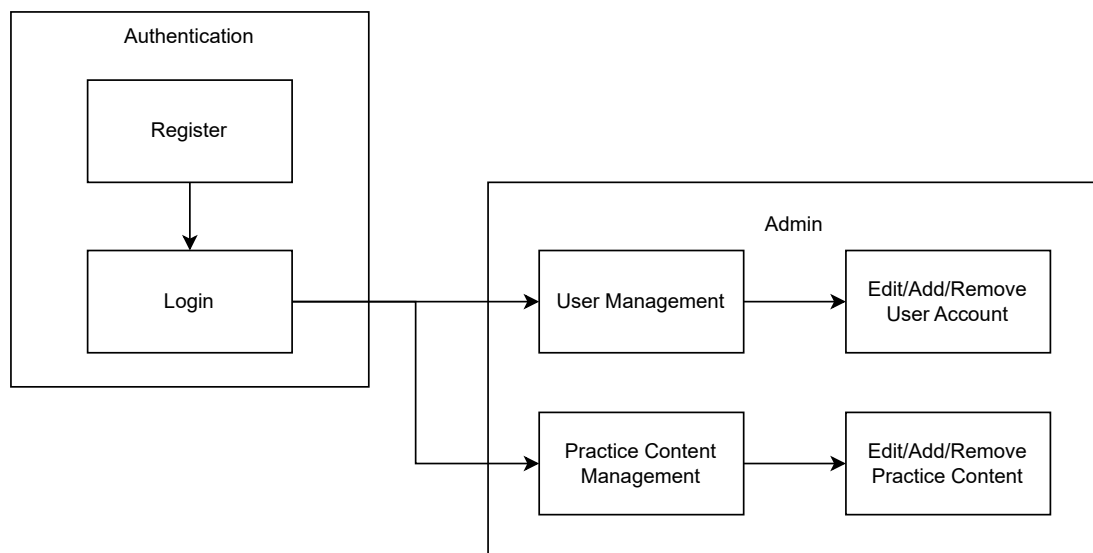
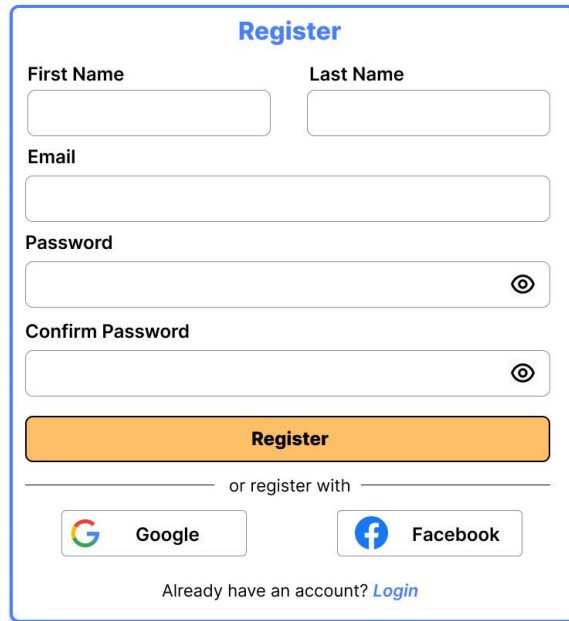


Figure 39: *Implemented Features UI Workflow Diagram*

This workflow diagram in Figure 39 represents the subset of features that have been implemented in the current version of the system. It focuses on the authentication process and core administrative functionalities. Users can register and log in to the system, after which admin users are able to access user management and practice content management modules. Within these modules, admins can perform essential operations such as adding, editing, or removing user accounts and practice content. Compared to the full system workflow, learner-specific features such as exercises, mock tests, evaluation tests, and result feedback are not yet included. This diagram highlights the current development scope while maintaining consistency with the overall system design.

6.4.3 Register



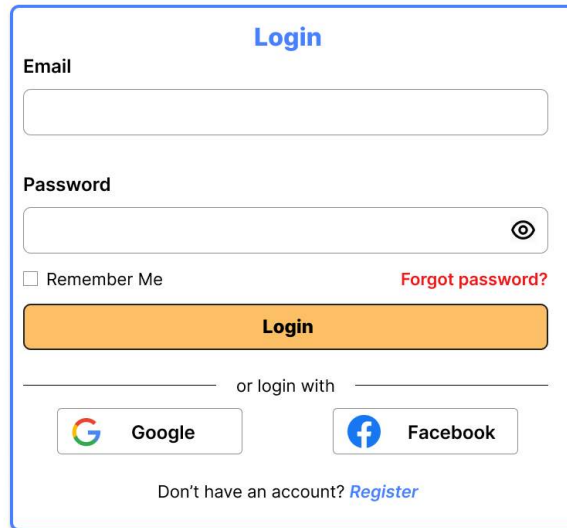
The Register form is titled "Register" in blue. It contains the following fields and elements:

- First Name**: A text input field.
- Last Name**: A text input field.
- Email**: A text input field.
- Password**: A text input field with a toggle eye icon on the right.
- Confirm Password**: A text input field with a toggle eye icon on the right.
- Register**: A large orange button.
- or register with**: A horizontal line with the text "or register with" in the center.
- Google**: A button with the Google logo.
- Facebook**: A button with the Facebook logo.
- Already have an account? Login**: A link in blue text.

Figure 40: Register Screen

The Register screen in Figure 40 allows new users to create an IELTS Mastermind account. It asks for basic information such as first name, last name, email, password, and confirm password. The password eye icon helps users check what they typed and avoid mistakes. The Register button is easy to see and is the main action on the page. The screen also provides Google and Facebook options for faster sign-up, and a Login link for users who already have an account.

6.4.4 Login



The login screen is a white rectangular box with a blue border. At the top center, the word "Login" is written in blue. Below it, there are two input fields: "Email" and "Password". The "Email" field is a simple white box. The "Password" field is a white box with a small eye icon on the right side. Below the "Password" field, there is a checkbox labeled "Remember Me" and a red link "Forgot password?". Below these, there is a large orange button labeled "Login". Below the button, there is a horizontal line with the text "or login with" in the center. Below this line, there are two buttons: "Google" with the Google logo and "Facebook" with the Facebook logo. At the bottom, there is a link "Don't have an account? Register" in blue.

Figure 41: *Login Screen*

The Login screen in Figure 41 allows users to securely access their IELTS Mastermind account. Users can log in using their email and password, with a password visibility icon to help avoid input errors. The Remember Me option is provided for convenience, allowing users to stay logged in on trusted devices. A Forgot password? link is included to support account recovery if users forget their credentials. The main Login button is clearly highlighted, and users can also choose to log in quickly using Google or Facebook. A link to the Register page is shown for new users who do not yet have an account.



6.4.5 Home

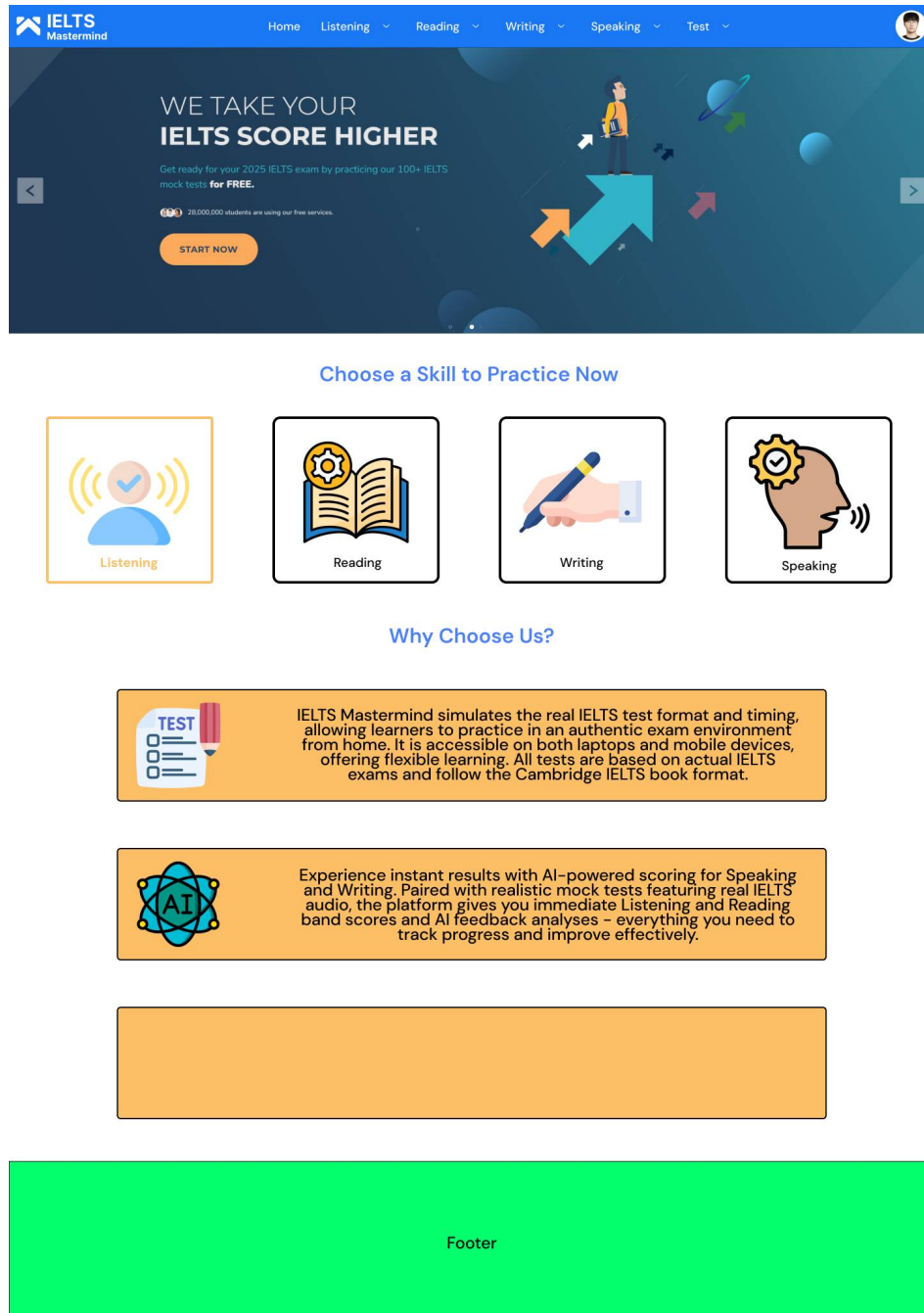


Figure 42: Home Screen



The Home screen in Figure 42 serves as the main entry point of IELTS Mastermind and provides users with an overview of the platform. At the top, a navigation bar allows users to access different sections such as Listening, Reading, Writing, Speaking, and Tests. A large banner highlights the main goal of the system and encourages users to start practicing. Below the banner, users can quickly choose a skill to practice through clearly displayed skill cards, making navigation simple and intuitive. The “Why Choose Us?” section briefly explains the key benefits of the platform, including exam-format practice and AI-supported feedback.

6.4.6 Browse

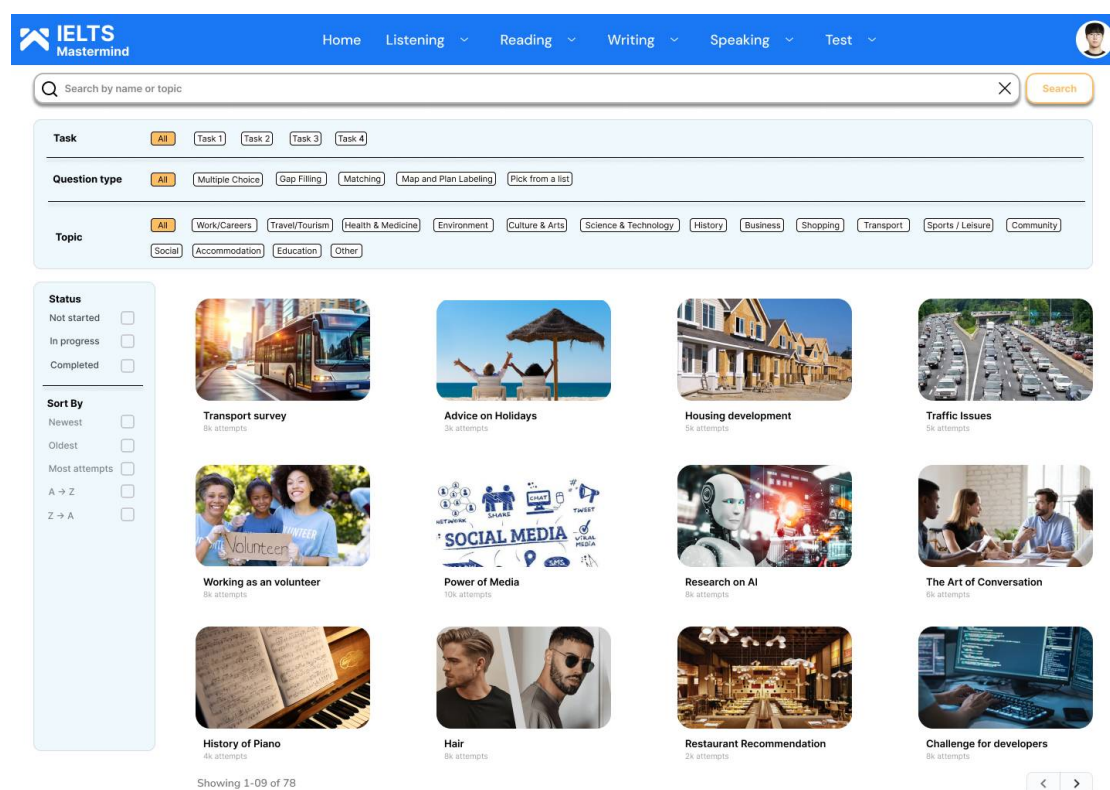


Figure 43: *Browse Screen*

The Browse screen in Figure 43 allows users to explore available practice content in a structured and flexible way. A search bar at the top enables users to quickly find content by keyword. Below it, filtering options such as skill type, question type, topic, and difficulty level help users narrow down results based on their learning needs. Practice items are displayed as clear content cards with titles and images, making them easy to recognize and select. A sidebar is provided to further refine the results, improving navigation when the content list is large.

6.4.7 Preview

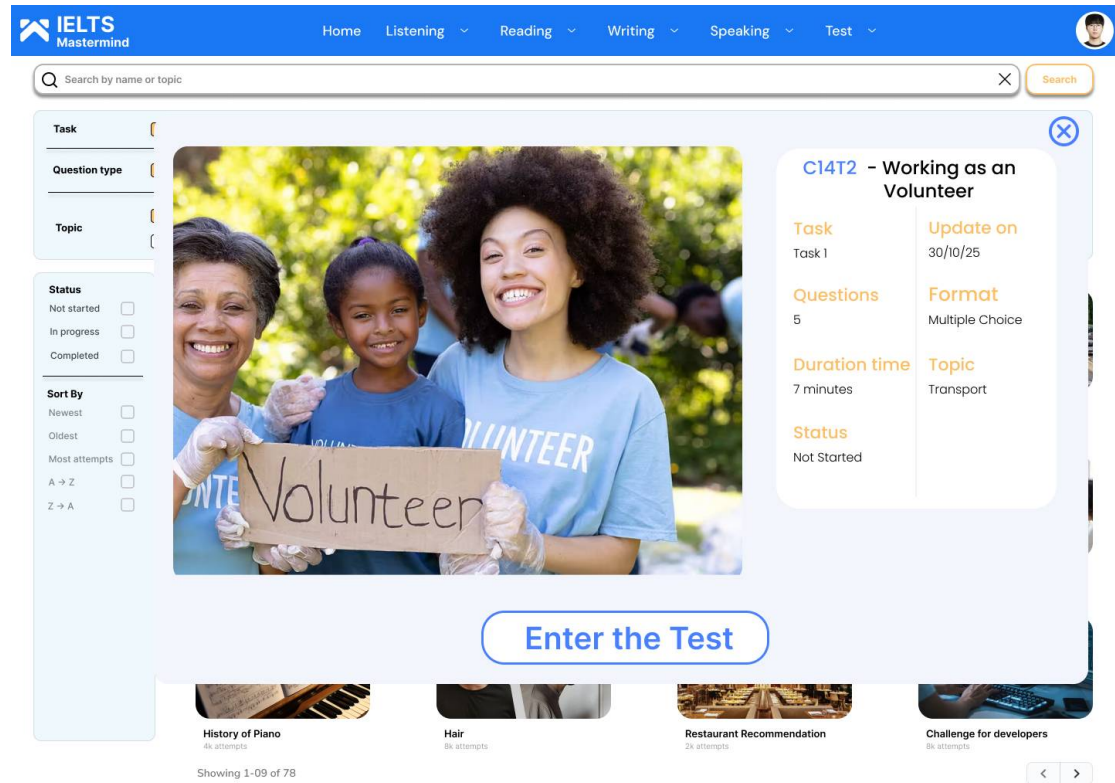


Figure 44: *Preview Screen*

The Preview screen in Figure 44 allows users to view key information about a practice task before starting it. The screen displays the task title, image, and important details such as question type, number of questions, topic, duration, and current status. This helps users understand what the practice involves and decide whether it matches their study goals. A clear “Enter the Test” button is provided to let users start the practice when they are ready.



6.4.8 Take

The screenshot displays the IELTS Mastermind 'Take' interface. At the top, a blue header bar contains the IELTS Mastermind logo on the left, the text 'Time Remaining 58:35' in the center, and a hamburger menu icon on the right. Below the header, the text 'Listening Task 1' is visible. A yellow instruction bar states 'Question 1-10 Write ONE WORD AND/OR A NUMBER for each answer.' The main content area is titled 'Volunteer Application' and contains a form titled 'Volunteer Programme – Applicant Enquiry' with the name 'Anna Miller'. The form is divided into three sections: 'Contact details' with fields for address, city area, postcode, and mobile phone; 'About the applicant' with a list of bullet points describing the applicant's interests and skills; and 'Availability' with fields for available days, start date, and work hours. At the bottom of the screen, there is a progress bar showing '00:51 / 6:32', a volume icon, and navigation controls (back, forward, and a circular arrow). A 'Practice' section with the text 'Done 0/10' is on the left, and a 'Submit >>' button is on the right. A row of question numbers (1-10) is located in the center of the bottom bar.

Figure 45: *Take Screen*

The Take screen in Figure 45 is where users complete a practice test in an exam-like environment. The screen displays the remaining time at the top to help users manage their time effectively, similar to the real IELTS exam. The main content area shows the practice material and questions clearly, allowing users to focus on answering without distractions. Navigation controls are provided to move between questions, and media controls are included when the task involves audio or other interactive elements. A Submit button is available to finish the test once all questions are completed.



6.4.9 Result

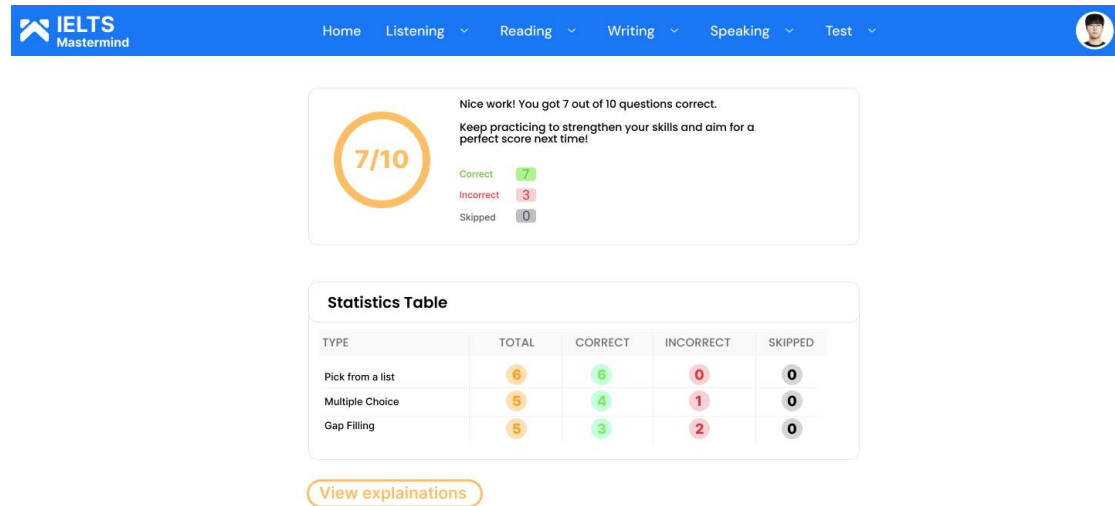


Figure 46: *Result Screen*

The Result screen in Figure 46 shows users their performance after completing a practice test. It displays the overall score and a short feedback message to help users quickly understand their results. A statistics table breaks down performance by question type, showing the number of correct, incorrect, and skipped answers. This helps users identify which types of questions they need to improve. A “View explanations” button is provided so users can review detailed answers and learn from their mistakes.



6.4.10 Profile

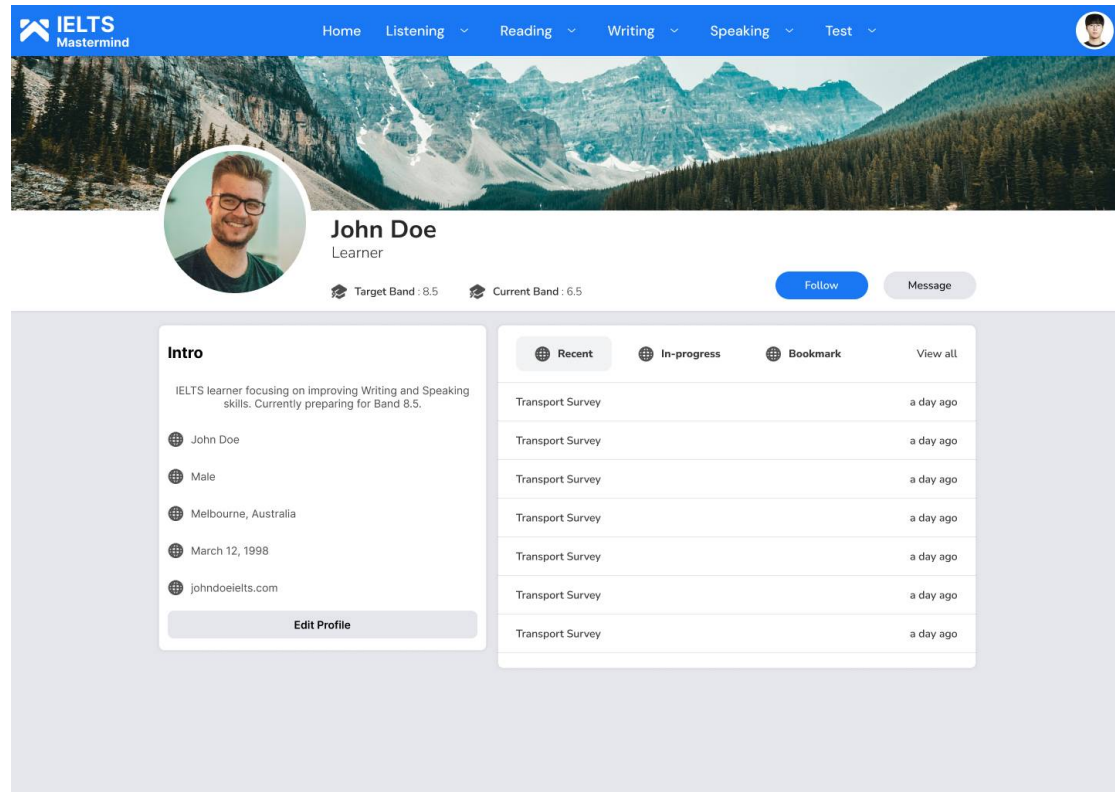
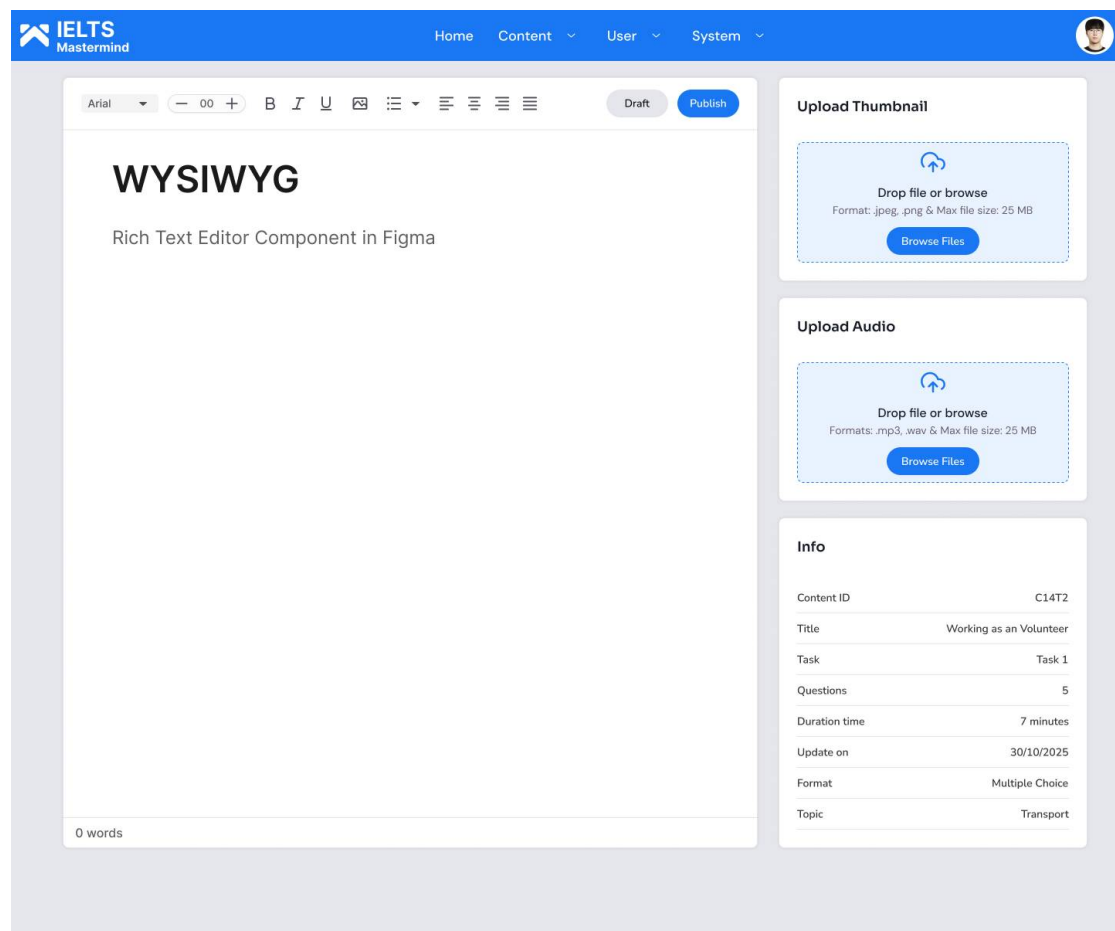


Figure 47: *Profile Screen*

The Profile screen in Figure 47 displays personal and learning-related information of a user in one place. It shows the user's profile picture, name, role, target IELTS band score, and current estimated band score. Basic personal details and an introduction section allow users to describe their learning goals. The screen also lists recent activities such as completed or bookmarked practice tasks, helping users track their learning history. An Edit Profile option is provided so users can update their information.

6.4.11 Content Management



Info	
Content ID	C14T2
Title	Working as a Volunteer
Task	Task 1
Questions	5
Duration time	7 minutes
Update on	30/10/2025
Format	Multiple Choice
Topic	Transport

Figure 48: *Content Management Screen*

The Content Management screen in Figure 48 is designed for administrators to create and manage practice content efficiently. It provides an editor that allows administrators to write and format content easily without technical knowledge. The screen also supports uploading related resources such as thumbnail and audio files. An information panel displays key details including content ID, task type, question count, duration, format, and topic, helping administrators verify content before publishing. Options such as Draft and Publish allow controlled content release.

6.4.12 User Management

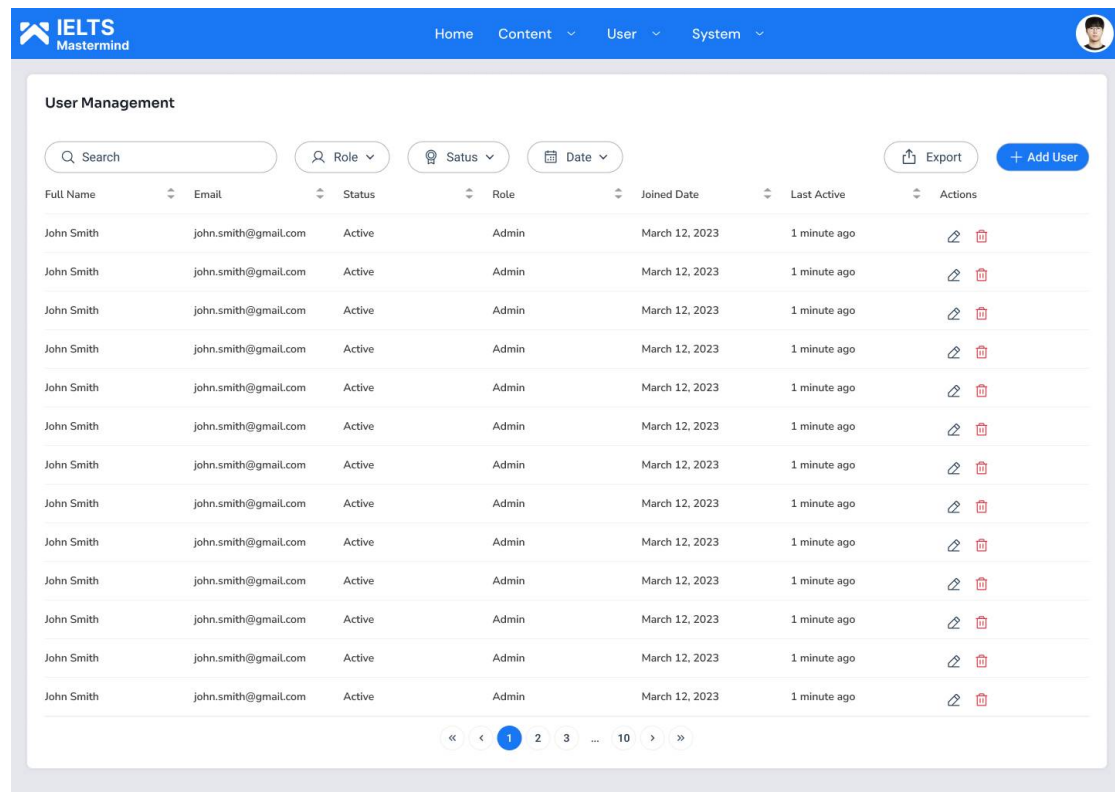


Figure 49: *User Management Screen*

The User Management screen in Figure 49 allows administrators to manage all user accounts in the system. It displays a list of users with key information such as name, email, account status, role, created date, and last activity. Search, filter, and sorting options help administrators quickly find specific users or manage large user lists. Action buttons are provided to add, edit, deactivate, or remove users, ensuring proper control over user access and roles.

7 System Implementation

This section presents the parts of IELTS Mastermind that have been implemented during the Specialized Project. Since the implemented screens closely follow the UI/UX designs presented in the previous section and do not differ significantly, they are shown here without additional explanation.

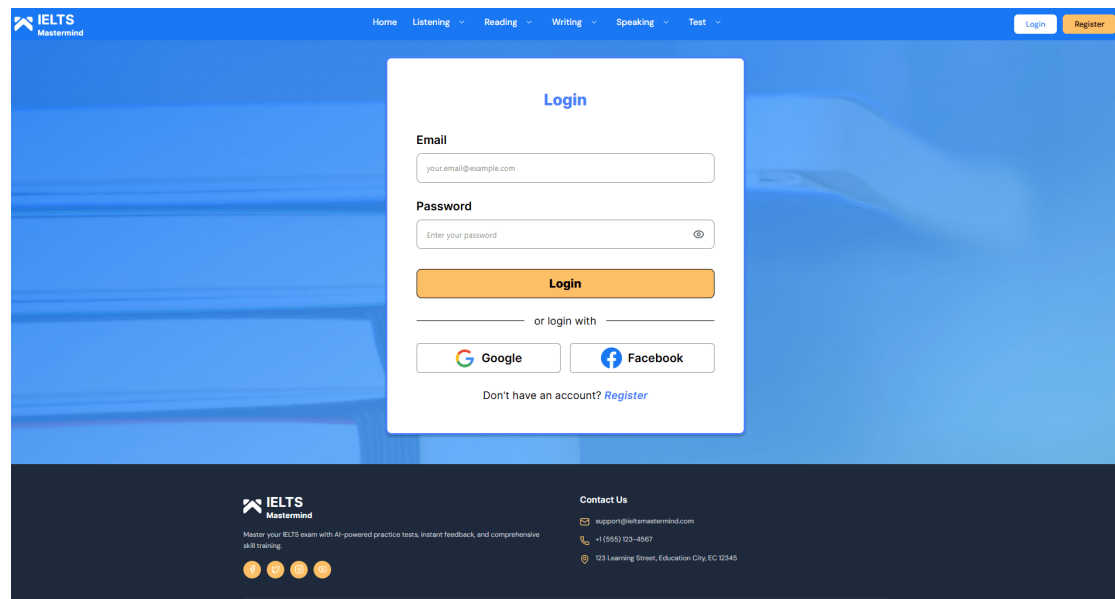


Figure 50: *Implemented Login Screen*



The image shows the 'Register' screen of the IELTS Mastermind website. The page has a blue header with the IELTS Mastermind logo and navigation links: Home, Listening, Reading, Writing, Speaking, and Test. There are 'Login' and 'Register' buttons in the top right corner. The main content area has a background image of a person reading a book. In the center, there is a white registration form titled 'Register'. The form includes fields for 'First Name', 'Last Name', 'Email', 'Password', and 'Confirm Password'. Below these fields is an orange 'Register' button. Underneath the button, it says 'or register with' followed by 'Google' and 'Facebook' login options. At the bottom of the form, it says 'Already have an account? [Login](#)'.

Figure 51: *Implemented Register Screen*

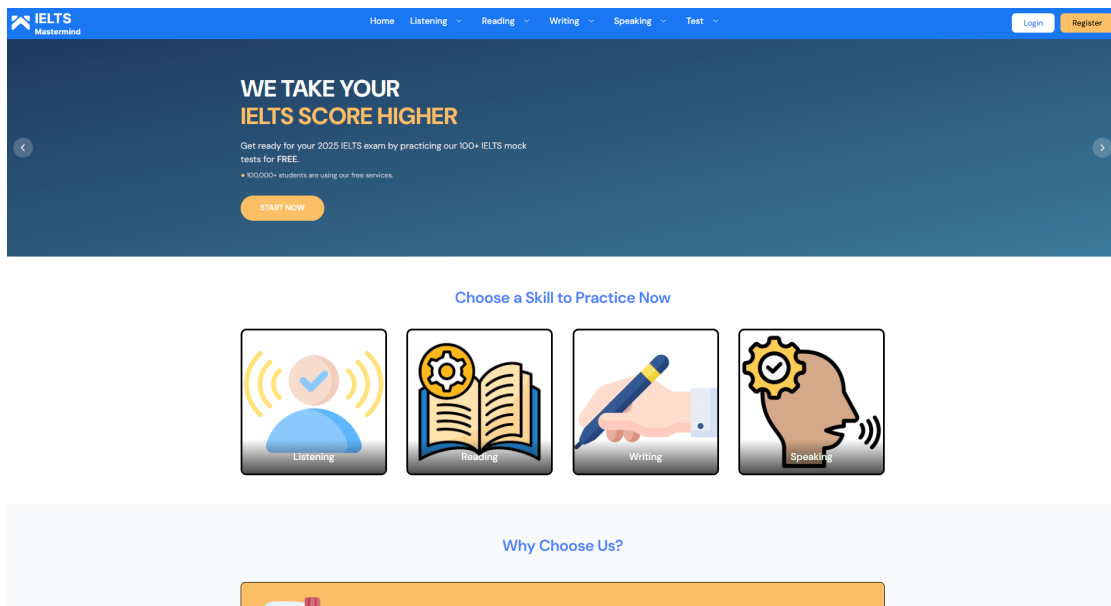


Figure 52: *Implemented Home Screen Part 1*

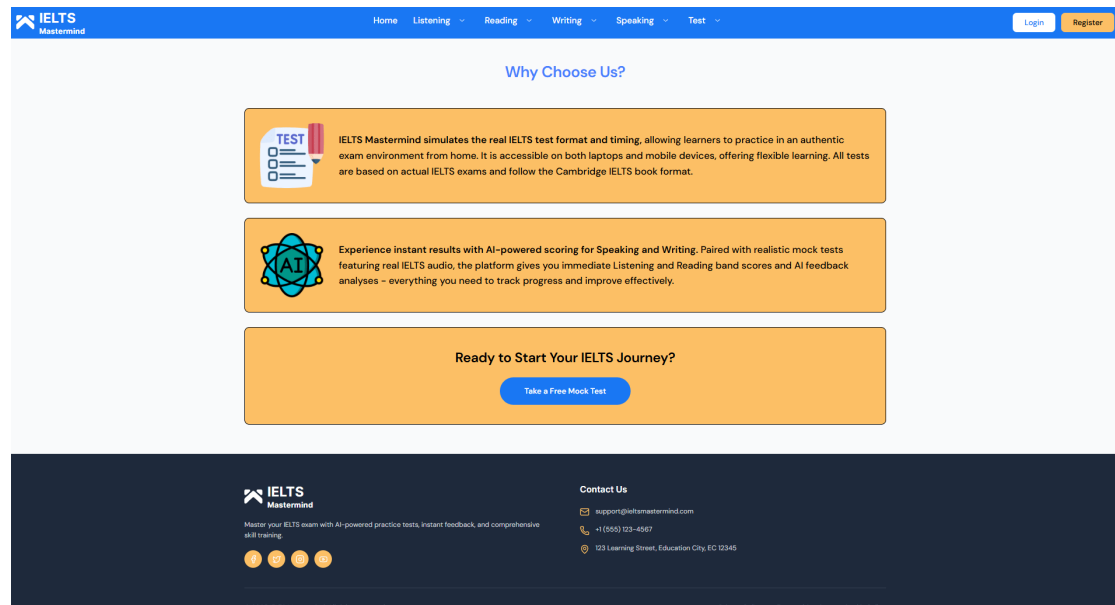


Figure 53: *Implemented Home Screen Part 2*

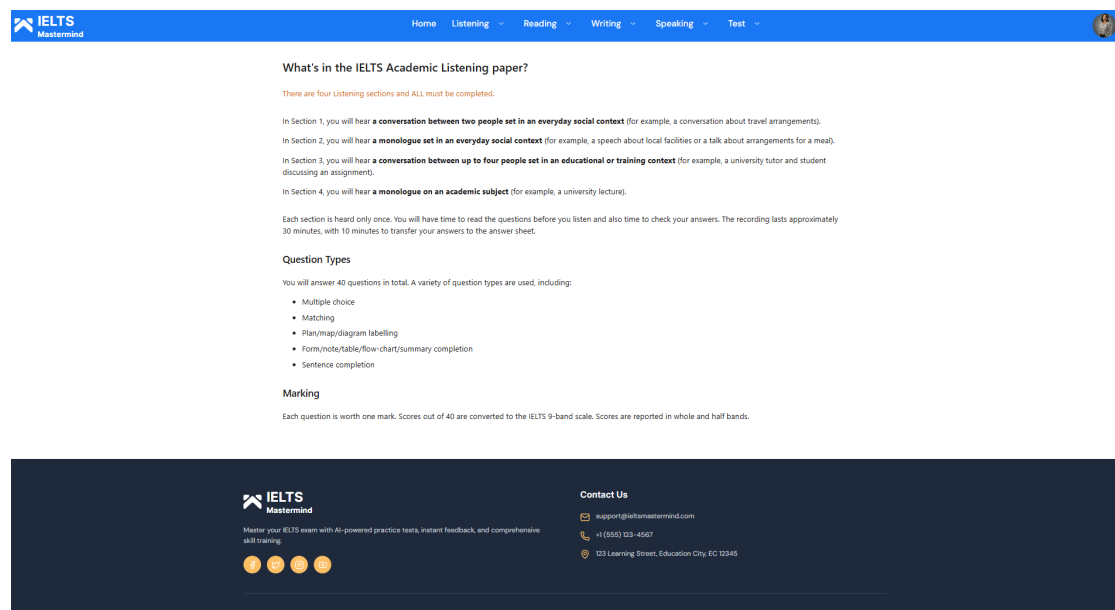


Figure 54: *Implemented Overview Screen*

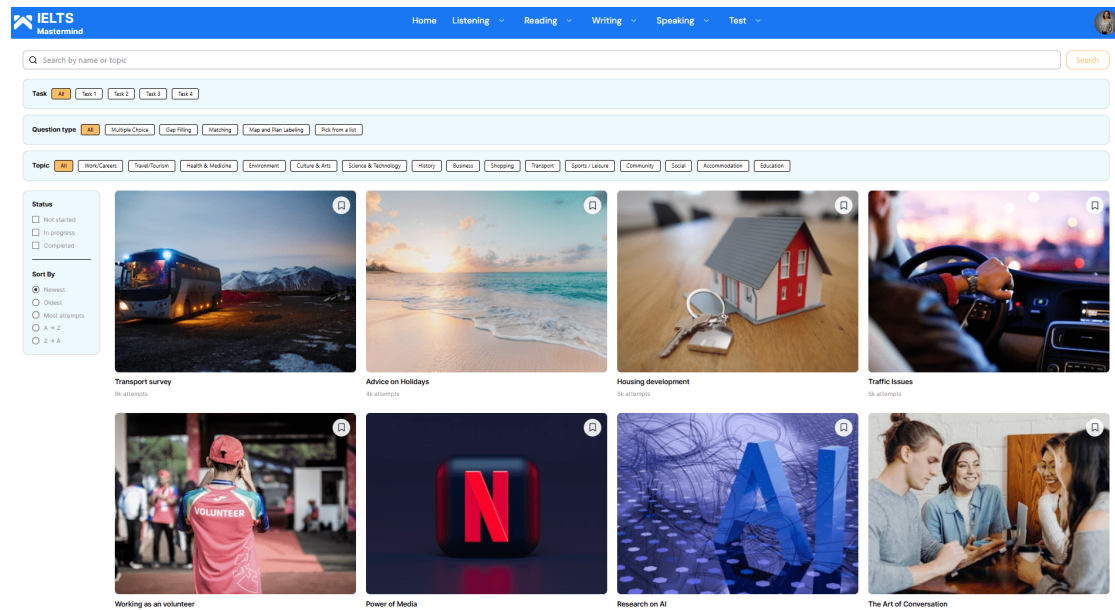


Figure 55: *Implemented Browse Screen*

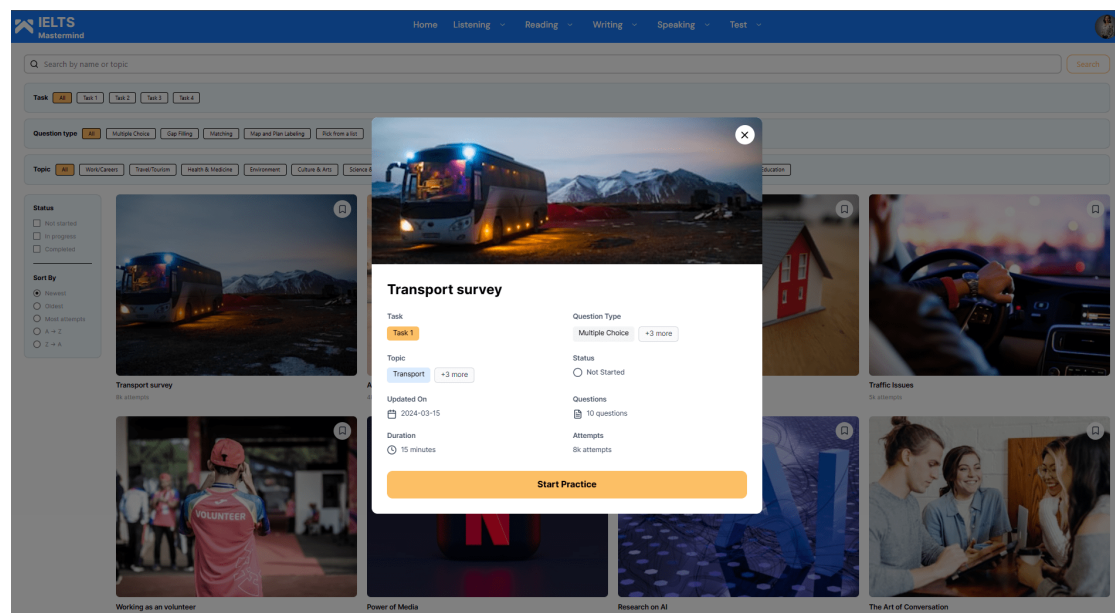


Figure 56: *Implemented Preview Screen*

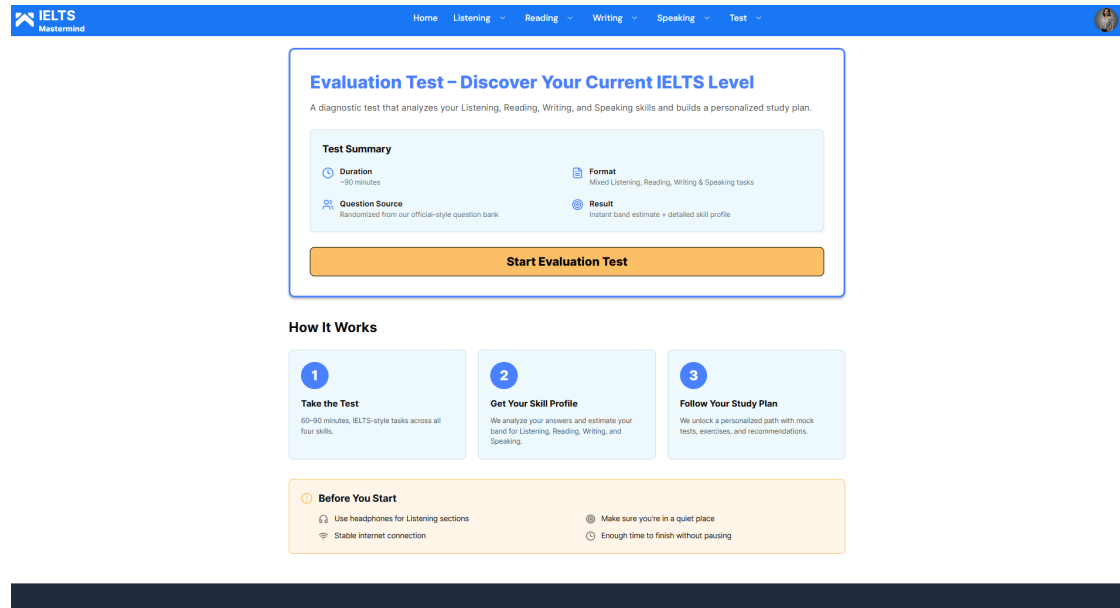


Figure 57: *Implemented Evaluation Test Screen*

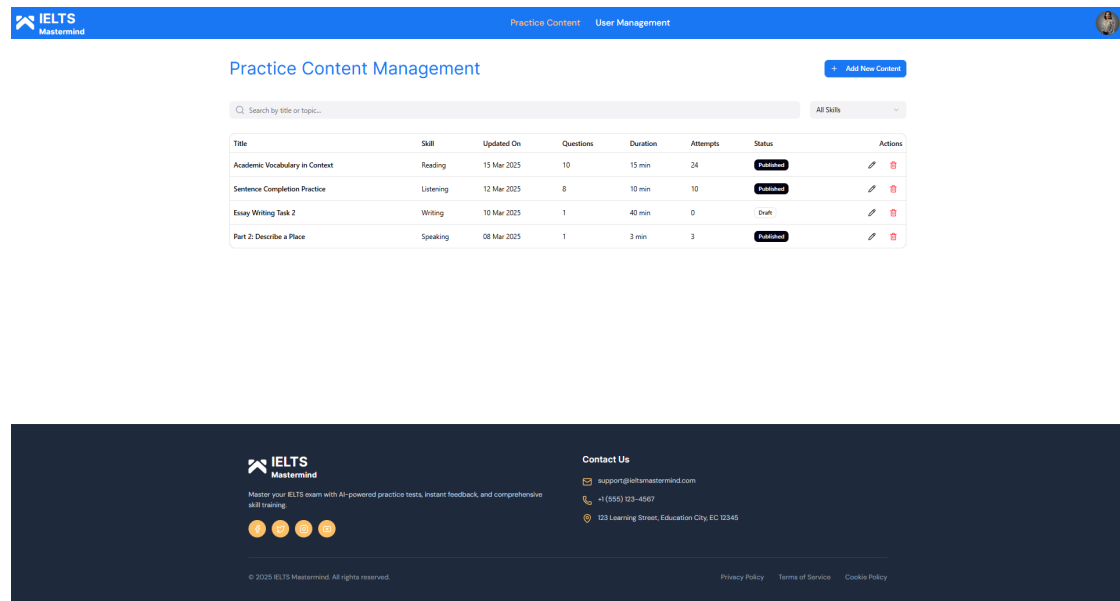


Figure 58: *Implemented Practice Content Management Screen*

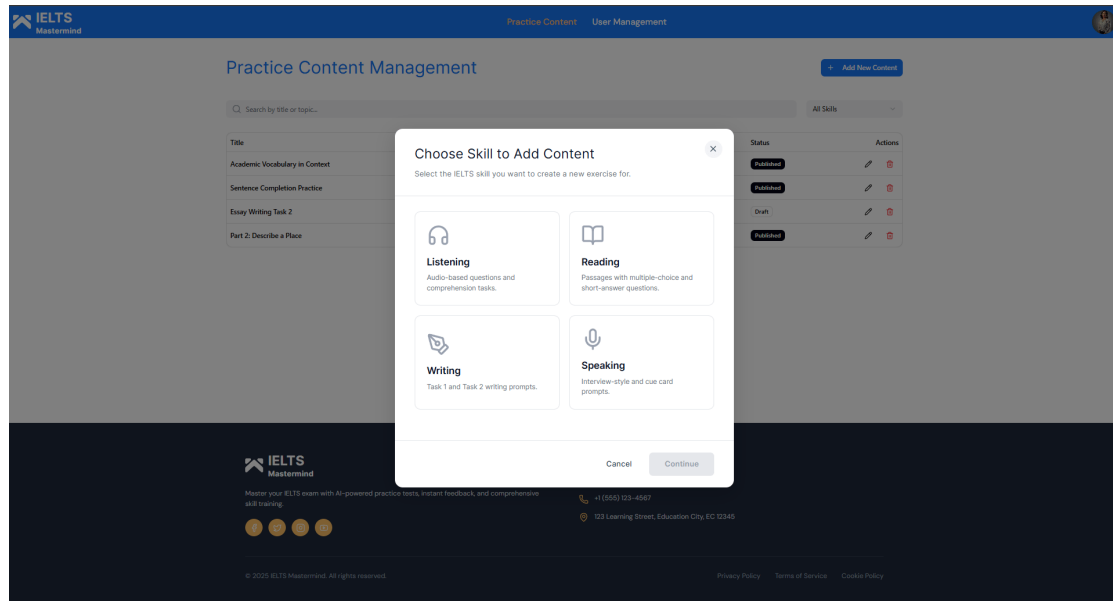


Figure 59: Implemented Add Content Screen

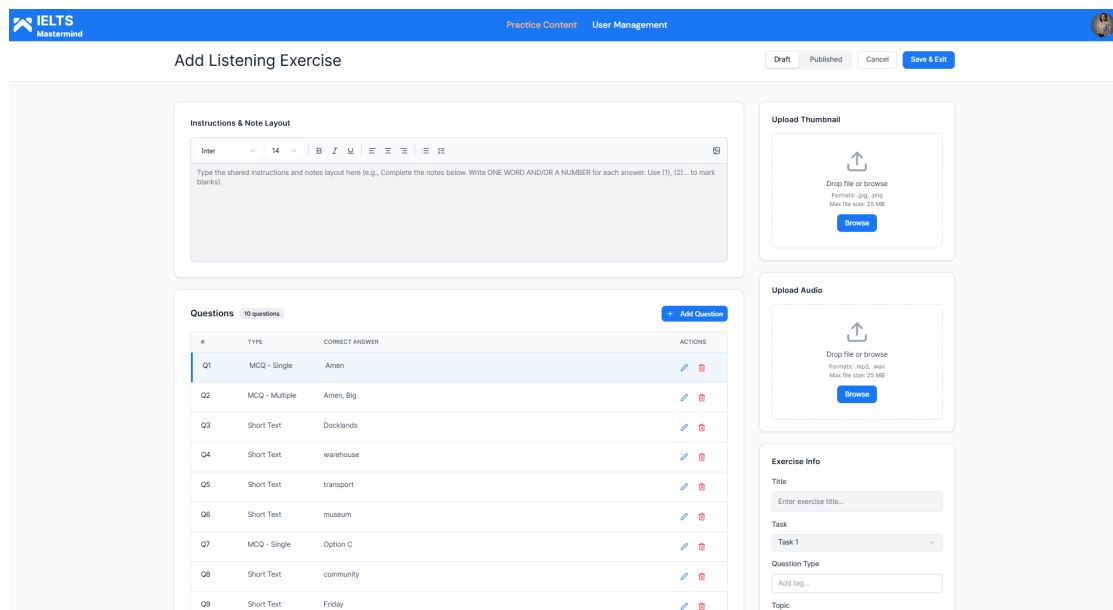


Figure 60: Implemented Add Listening Exercise Screen Part 1



Q9Short TextFriday

Q10Short Textwebsite

Answer & Scoring

Editing Question 1 - ✓ Saved

Question text

What is the first word in the audio?

Question type

Multiple Choice - Single Correct

CORRECT

OPTION TEXT

☒Amen

☐Option B

☐Option C

☐Option D

+ Add option

☐ Shuffle options

Explanation / Model answer (optional)

Provide an explanation or model answer that learners will see after submitting...

Save

Cancel

Topic

Add tag...

Updated On

12/17/2025

Questions

10

Duration (minutes)

5

Figure 61: Implemented Add Listening Exercise Screen Part 2

IELTS
Mastermind

Practice ContentUser Management

User Management

Manage user accounts and permissions

Add New User

Total Users

5

Active Users

4

Learners

4

Administrators

1

Search by name or email...

All Roles

All Status

User	Role	Status	Joined Date	Last Active	Tests Completed	Actions
John Smith @ john.smith@example.com	Learner	Active	1/15/2024	3/20/2024	12	
Sarah Johnson @ sarah.j@example.com	Learner	Active	2/1/2024	3/19/2024	8	
Michael Chen @ michael@example.com	Administrator	Active	12/10/2023	3/21/2024	0	
Emily Davis @ emily.d@example.com	Learner	Inactive	1/20/2024	2/15/2024	5	
David Wilson @ d.wilson@example.com	Learner	Active	3/1/2024	3/21/2024	3	

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Figure 62: Implemented User Management Screen

SPECIALIZED PROJECT - HK251

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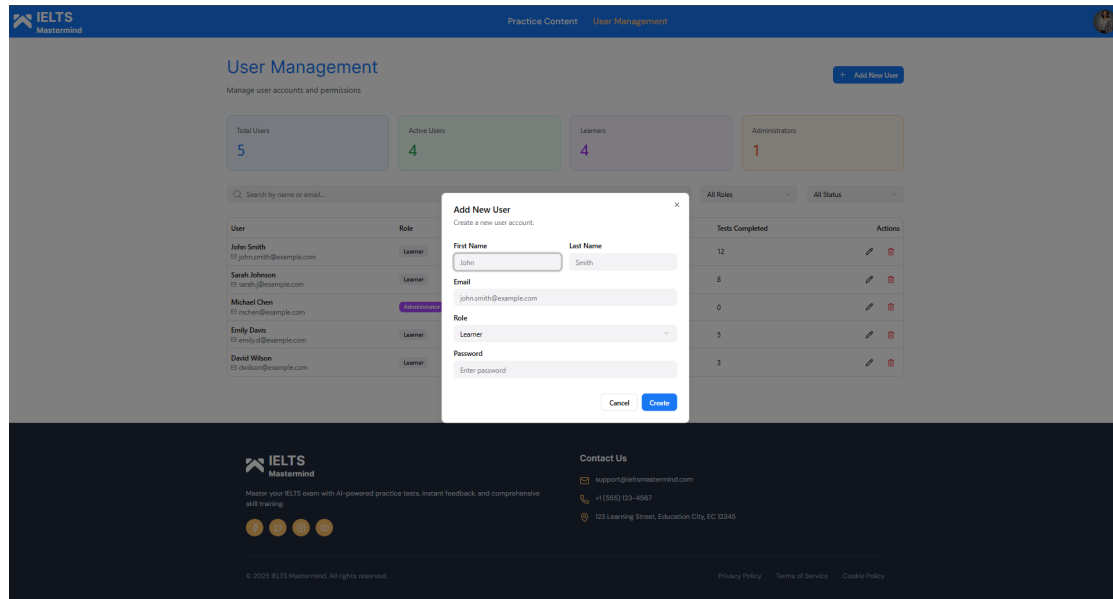


Figure 63: *Implemented Add User Screen*

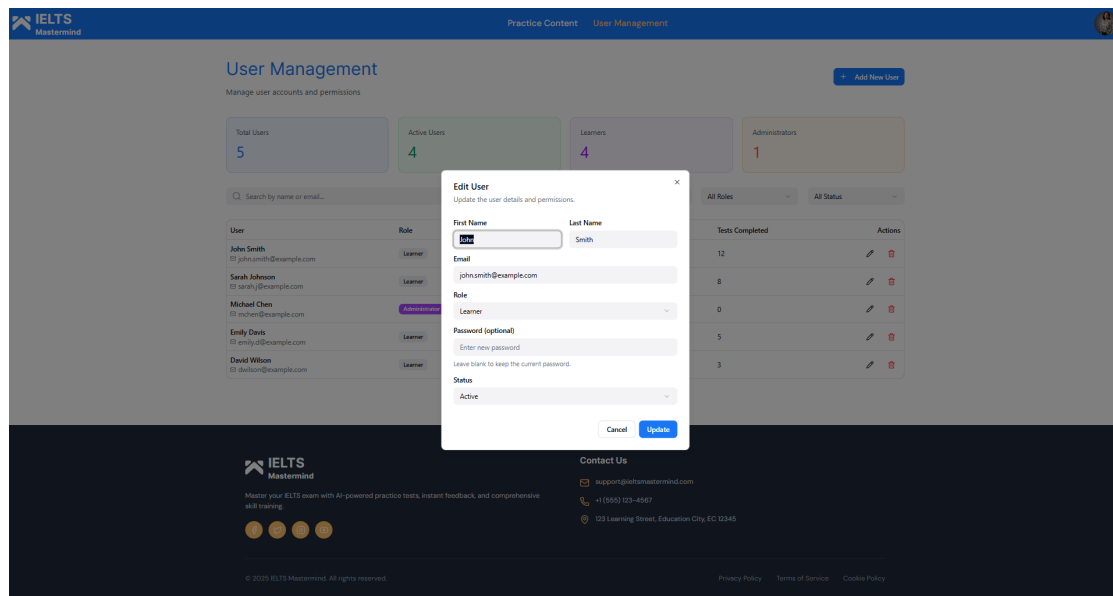


Figure 64: *Implemented Update User Screen*

8 Conclusion

8.1 Achieved Results

This Specialized Project presents IELTS Mastermind, an online IELTS learning platform designed to support learners in all four skills: Listening, Reading, Writing, and Speaking. The system aims to simulate a tutoring-like experience by providing exam-format practice, helping learners identify strengths and weaknesses, and supporting improvement through feedback and personalized learning support. The project was proposed to address common problems in IELTS self-study, such as limited personalized guidance, difficulty in self-assessment, and unbalanced support for Writing and Speaking compared to other skills.

To develop the system in a structured way, we completed the main stages of system analysis and design. We defined the functional requirements for different user groups, and modeled the system using use case diagrams, use case scenarios, activity diagrams, and sequence diagrams to clearly describe how key features work. We also identified and wrote measurable non-functional requirements, focusing on Responsiveness, Reliability, and Overall Cost, which fit the constraints of a student project and ensure the system can be tested and evaluated.

In addition, we designed the system architecture and prepared key technical models, including the architecture design and diagram, data flow diagrams, and deployment diagram to explain how the system is organized and deployed. We also specified the technologies used and produced detailed system design outputs such as module design, database design, including ERD and relational schema, API design, and UI/UX design for core screens. Overall, this project delivers a complete foundation for implementing IELTS Mastermind, and it can be extended in the future with additional features.

8.2 Strengths

Through the design and partial implementation of IELTS Mastermind, the system demonstrates several strengths:

- The system is designed to support all four IELTS skills (Listening, Reading, Writing, and Speaking), helping learners avoid unbalanced skill development, especially in Writing and Speaking.
- IELTS Mastermind provides exam-format practice with official IELTS timing, allowing learners to become familiar with real test conditions and improve time management.
- The system architecture, database design, and modular structure are well-organized and scalable, making the system easier to maintain and extend in future development phases.
- Modern technologies and design approaches are applied, including clear UI/UX design, RESTful APIs, and modular backend structure, which support future feature expansion.
- The integration plan for AI-based feedback and progress tracking provides a strong foundation for personalized learning support, even though these features are still at an early stage.



8.3 Limitations

Besides its strengths, IELTS Mastermind still has several limitations due to time and resource constraints of the Specialized Project:

- The current project mainly focuses on system analysis and design, and only core features are partially implemented; many advanced functions are not yet fully developed.
- The system has not been fully optimized for mobile devices and different screen sizes, which may affect user experience on smartphones and tablets.
- Non-functional aspects such as performance optimization, security hardening, and large-scale testing are limited and have not been fully evaluated.

8.4 Future Development Directions

Based on the current progress of this Specialized Project, the Graduation Project will continue and complete the IELTS Mastermind system with full implementation, feature expansion, and comprehensive evaluation. To ensure a clear and manageable development roadmap, the next stage can be organized into four main phases as follows:

Phase 1: Review and Planning (Weeks 1–2)

- Collect feedback after the Specialized Project and identify remaining issues (bugs, missing flows, UI inconsistencies).
- Update the Product Backlog by prioritizing unfinished features and new improvements.
- Refine system design details such as database adjustments, API specifications, and UI/UX improvements for the Graduation Project scope.

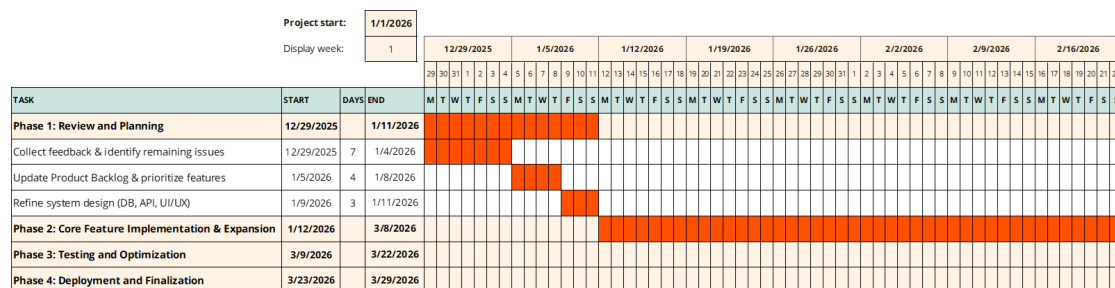


Figure 65: Gantt Chart for Phase 1

Phase 2: Core Feature Implementation and Expansion (Weeks 3–10)

This phase focuses on completing the main system:

- Practice Module: Complete and improve unfinished features related to practice content access, answer submission, and attempt management, and improve the overall practice workflow for all IELTS skills.



- Personalization Module: Complete and improve the evaluation process, skill profiling, and study plan generation to provide more accurate and adaptive learning paths for learners.
- Feedback Module: Complete and improve automated feedback and band score generation to deliver more detailed, consistent, and reliable performance analysis, especially for Writing and Speaking skills.
- Progress Tracking Module: Complete and improve progress dashboards, reports, and analytics to help learners clearly monitor their performance, time spent, and improvement over time.

In addition to completing and improving the current modules, the system may be extended with new modules if time permits.

- Tutor Module: Add a new tutor role to manage courses, lessons, learning materials, and practice content; allow tutors to assign tasks, provide feedback/grades, and track learner performance.
- Communication Module: Implement interaction features such as one-to-one messaging, group discussions, and forum-style topic boards to improve engagement and learner support.
- Motivation Module: Introduce gamification features (achievements, daily goals, weekly/monthly challenges, and leaderboards) to encourage consistent study habits.

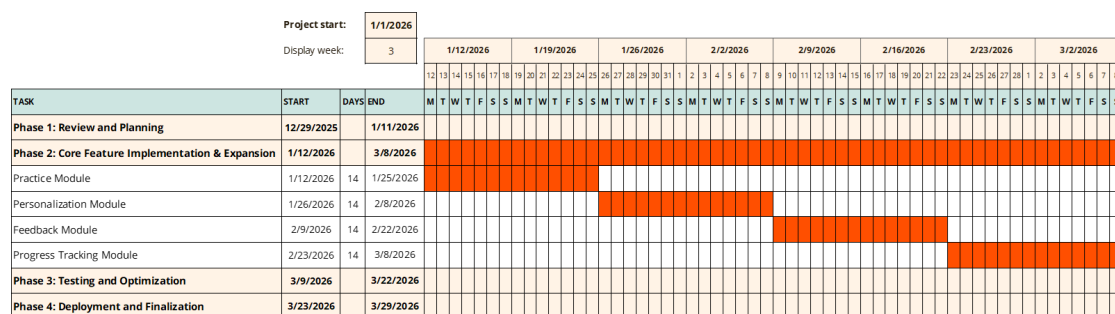


Figure 66: Gantt Chart for Phase 2

Phase 3: Testing and Optimization (Weeks 11–13)

- Conduct functional testing to verify all major features work correctly across different roles.
- Conduct non-functional testing and optimize system stability.
- Improve AI-related accuracy where applicable.

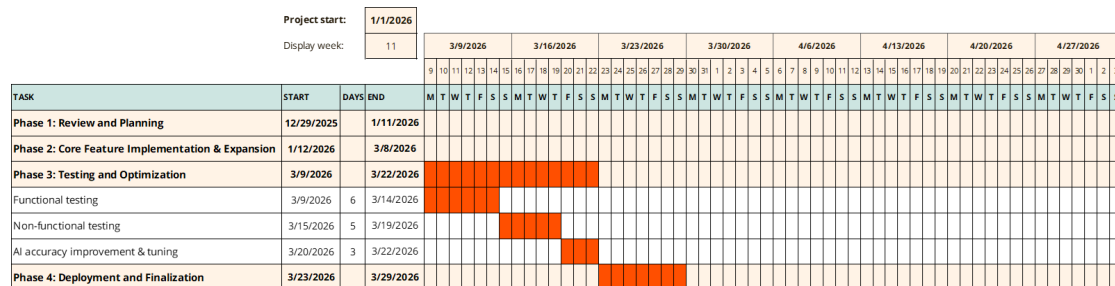
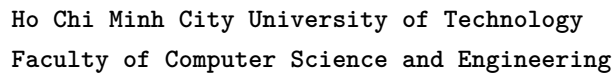


Figure 67: *Gantt Chart for Phase 3*

Phase 4: Deployment and Finalization (Weeks 14–16)

- Deploy the system to a real environment, configure domain and security settings.
- Finalize documentation, including the Final Report, User Guide, and technical deployment notes.
- Prepare the final demo and presentation materials for the Graduation Project defense.

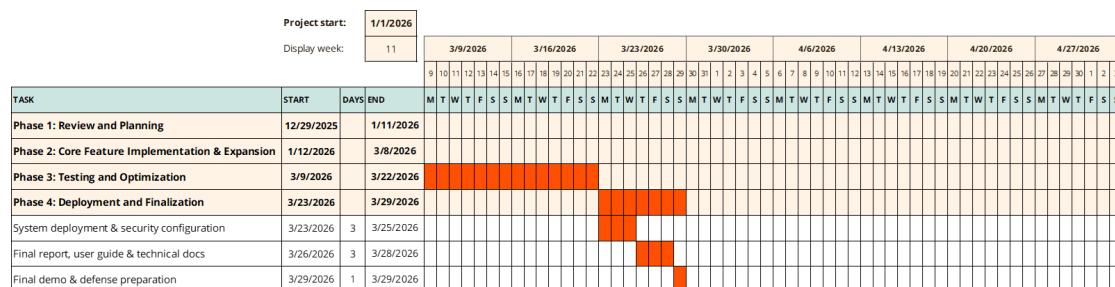


Figure 68: *Gantt Chart for Phase 4*



8.5 Members Contribution

No.	Fullname	Student ID	Tasks Assigned	Effort
1	Nguyễn Đoàn Hải Bằng	2252078	• Activity Diagrams & Sequence Diagrams (selected features) • UI/UX Design screens (Register, Login, Home, Browse, Preview, Take, Result, Profile, Admin screens) • System Implementation write-up (implemented parts + screenshots) • Conclusion (Achieved Results, Strengths, Limitations, Future Plan) + report formatting	100%
2	Nguyễn Quang Duy	2252120	• System Overview (related systems, objectives, stakeholders, users) • Architecture section (style selection, comparison, chosen architecture) • Data Flow Diagrams (Context + Level-0 + module DFDs) • Deployment Diagram discussion	100%
3	Huỳnh Mai Quốc Khang	2252293	• System Requirements (Functional + Non-functional requirements) • Use Case Diagrams & Use Case Scenarios (core modules) • Class Diagram • API Design (Auth, Profiles, Practice, Personalization, User/Role, Content Mgmt)	100%



References

- [1] K. E. Kendall and J. E. Kendall, *Systems Analysis and Design*, 9th ed., Pearson, 2013.
- [2] G. B. Shelly, T. J. Cashman, and H. J. Rosenblatt, *Systems Analysis and Design*, 6th ed., Course Technology–Thomson Learning, 2006.
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